
PETKHO PORTAL: SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA

A Capstone Project
presented to the Faculty of
College of Computer Studies
Camarines Sur Polytechnic Colleges

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Technology

By
Jey Kyle E. Peregrino
Mary Pons Rica E. Relatores
Anne Psylocke S. Urbina

December 2025

DEDICATION

Every difficult task needs our own hard work and the guidance of people we trust, especially those close to our hearts. We dedicate this work to everyone who has inspired and supported us. Our loving parents, friends, and classmates who pray for us and encourage us to stay positive helped us reach this achievement.

We also want to thank our client, Kho Veterinary Clinic, for their patience, along with the hardworking teachers and staff of Camarines Sur Polytechnic Colleges. Most of all, we thank our Almighty Father for giving us strength and the gift of life to experience this opportunity. With your support, prayers, and encouragement, we have come a long way—and we know we can go even farther.

-TEAM QUANTUM NEXUS

ACKNOWLEDGMENTS

This research would not have been possible without the guidance and help of the following individuals who generously gave their time and effort from the start until the completion of this study. The researchers would like to sincerely thank the following:

To **Rosel O. Oñesa, MIT**, the OIC-Dean of the College of Computer Studies, for his approval and support in allowing the researchers to conduct this study.

To **Jocelle B. Monreal, DIT**, the Capstone Subject Adviser and Project Adviser, for her time, patience, kindness, unwavering support, for always giving clear and helpful answers to the researchers' questions, and for her willingness to share her knowledge, which greatly contributed to the success of this study.

To **Philip Alger M. Serrano, MIT**, the Capstone Project Consultant, for his generosity and time in sharing his knowledge to guide the researchers throughout the project's development.

To **Ma. Allaine C. Agna, LPT**, the Capstone Project Grammarian, for her help and support in editing and improving the manuscript.

To the **Panelists: Kaela Marie N. Fortuno, MIT, Vincent B. Cortez, MIT, and Ian P. Benitez, DIT** for their valuable ideas, comments, and suggestions that helped the researchers improve the study.

To **Kho Veterinary Clinic**, for their cooperation as the client of the study, and for their time and understanding in meeting the needs of the researchers for the system's development.

To **Mr. Anthony L. Llabres**, for his continued assistance in recording the panel's comments, suggestions, and recommendations during the capstone defense.

To **our friends, classmates, and loved ones**, for the happy memories, support, and words of encouragement that lifted our spirits during hard times and helped us stay confident.

To **Mrs. Agnes S. Urbina, Mr. and Mrs. Dolfo B. Peregrino and Mrs. Maryann E. Relatores** for their unending financial, spiritual, and moral support, and for believing in the researchers' potential. Your sacrifices inspired us to work harder and finish our Capstone Project.

Lastly, thanks to our **Almighty Father**, our Lord and Savior, for giving us the wisdom, knowledge, strength, and guidance in facing challenges like this and for providing us with the determination to pursue and stay focused on our study.

-TEAM QUANTUM NEXUS

ABSTRACT

Title:	Petkho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada
Authors:	Jey Kyle E. Peregrino Mary Pons Rica E. Relatores Anne Psylocke S. Urbina
Number of Pages:	212
School:	Camarines Sur Polytechnic Colleges
Degree Conferred:	Bachelor of Science in Information
Keywords:	Local Based Veterinary System Automation of Pet Medical Health Record Invoice Automation

The PetKho Portal is a smart online system created to help Kho Veterinary Clinic Rinconada improve its daily work and provide better service to pet owners. The clinic used to rely on handwritten records, walk-in appointments, and manual updates, which often caused delays, missing information, and extra workload for the staff. This project was developed to solve these problems by organizing all important tasks in one easy-to-use platform. The system includes features such as online appointment scheduling, pet record management, inventory tracking, billing and payment recording, referral handling, and automatic notifications. It was developed using the Agile approach, which allowed the team to build the system step by step and improve it based on feedback. Different tests such as performance testing, security testing, use case testing, and user acceptance testing were done to make sure the system works smoothly and safely. The results showed that the PetKho Portal helped reduce errors, save time, and make information easier to access. Pet owners also found it more convenient to book appointments and receive updates. Overall, the PetKho Portal provides a simple, modern, and reliable solution that supports the clinic in giving better and more organized veterinary care.

TABLE OF CONTENTS

TITLE PAGE	i
APPROVAL PAGE	ii
PANEL OF EXAMINERS	iii
DEDICATION	iv
ACKNOWLEDGMENTS	v
ABSTRACT.....	vii
TABLE OF CONTENTS.....	viii
LIST OF TABLES	xii
LIST OF FIGURES	xiii
CHAPTER 1 INTRODUCTION	1
1.1 Project Context	1
1.2 Purpose and Description of the Project.....	3
1.3 Objectives of the Project.....	5
1.4 Significance of the Study	6
1.5 Objectives of the Project.....	6
1.6 Project Dictionary	7
Notes	11
CHAPTER 2 REVIEW OF RELATED LITERATURE AND SYSTEMS	14
2.1 Related Literature and Systems	14

2.1.1 Record Management	14
2.1.2 Monitoring System	17
2.1.3 Information System.....	19
2.1.4 Efficiency of Smart Veterinary Care	22
2.2 Synthesis of the State-of-the-Art	25
2.3 Gap Bridged by the Study.....	27
Notes	29
CHAPTER 3 TECHNICAL BACKGROUND.....	32
3.1 Overview of Current Technologies to be Used in the System.....	32
3.2 Resources	32
3.2.1 Hardware Specifications	33
3.2.2 Software Specifications	34
3.2.3 Program Specifications	36
3.2.4 Programming Environment.....	45
3.3 Testing Plan	47
3.3.1 Types of Testing	48
3.3.2 Testing Tools and Framework	49
Notes	53
CHAPTER 4 METHODOLOGY, RESULTS AND DISCUSSION	54
4.1 Planning	54
4.1.1 Convenient Sampling.....	55
4.1.2 Scaling	57
4.1.3 Statistical Test.....	61

4.1.4 Identified Problems.....	62
4.1.5 Proposed Solutions	64
4.1.6 Fishbone Diagram.....	66
4.1.7 Gantt Chart.....	67
4.2 Design	68
4.2.1 Entity Relationship Diagrams	68
4.2.2 Data Flow Diagram.....	70
4.3 Development.....	84
4.3.1 Admin Modules	84
4.3.2 Client Module	97
4.4 Testing	106
4.4.1 Overall Result	116
4.5 Deployment.....	118
Notes	121
CHAPTER 5 SUMMARY, FINDINGS, CONCLUSION AND RECOMMENDATION	122
5.1 Summary.....	122
5.2 Findings	123
5.3 Conclusions.....	125
5.4 Recommendations.....	126
BIBLIOGRAPHY	128
APPENDICES	135
APPENDIX A EVALUATION TOOL	136

APPENDIX B DOCUMENTATION	146
APPENDIX C SAMPLE INPUT/OUTPUT/REPORTS	147
APPENDIX D USER’S GUIDE.....	152
APPENDIX E COMMUNICATION LETTERS	168
APPENDIX F NON-DISCLOSURE AGREEMENT FORM	169
APPENDIX G JOINT AFFIDAVIT OF UNDERTAKING.....	181
APPENDIX H PROJECT TEAM ASSIGNMENT FORM.....	182
APPENDIX I FINAL PROJECT TITLE FORM	183
APPENDIX J CAPSTONE HEARING FORM (TD, POD, FOD)	184
APPENDIX K PANEL RSC (TD, POD, FOD)	187
APPENDIX L CONSULTATION LOGS FORM.....	194
APPENDIX M SECRETARY CERTIFICATE	198
APPENDIX N GRAMMARIAN CERTIFICATE	199
APPENDIX O CERTIFICATE OF UTILIZATION	200
APPENDIX P ACM FORMAT	201
APPENDIX S CERTIFICATE OF PLAGIARISM CHECK	208
CURRICULUM VITAE	209

LIST OF TABLES

Table 1: Hardware Specification.....	33
Table 2: Software Specifications	34
Table 3: Functional Requirements	36
Table 4: Front End	46
Table 5: Back End.....	47
Table 6: ISO-Aligned Questionnaire Testing Framework	50
Table 7: Respondents	56
Table 8: Performance Testing Scale	57
Table 9: Security Testing Scale	58
Table 10: Use Case Testing Scale.....	59
Table 11: User Acceptance Testing Scale.....	60
Table 12: Identified Problems.....	63
Table 13: Proposed Solutions	65
Table 14: Pet Owners Performance Testing	107
Table 15: Vet Owners/Resident Doctor Performance Testing.....	108
Table 16: IT Experts Performance Testing	109
Table 17: Pet Owners User Acceptance Testing.....	110
Table 18: Vet Owners/Resident Doctor User Acceptance Testing.....	112
Table 19: Pet Owners Use Case Testing.....	113
Table 20: Vet Owners/Resident Doctors Use Case Testing.....	114
Table 21: IT Experts Security Testing	115
Table 22: Test Case Result for Pet Owners	116
Table 23: Test Case Result for Vet Owners/Resident Doctors	117
Table 24: Test Case Result for IT Experts.....	118

LIST OF FIGURES

Figure 1: Flowchart.....	37
Figure 2: System Architecture	40
Figure 3: Security/Authentication Measures.....	43
Figure 4: Hierarchy Diagram	47
Figure 5: Fishbone Diagram	66
Figure 6: Gantt Chart	67
Figure 7: Entity Relationship Diagram	69
Figure 8: Context Diagram	71
Figure 9: Data Flow Diagram Level 0	73
Figure 10: DFD Level 1-Login	75
Figure 11: DFD Level 1-Appointment Management.....	76
Figure 12: DFD Level 1-Referral Management.....	77
Figure 13: DFD Level 1-Pet Record Management	78
Figure 14: DFD Level 1-Invoice Management.....	79
Figure 15: DFD Level 1-Adoption Management.....	80
Figure 16: DFD Level 1-Stock Management.....	81
Figure 17: DFD Level 1-Report Management.....	82
Figure 18: DFD Level 1-User, Staff, Branch Management.....	83
Figure 19: Landing Page.....	85
Figure 20: Login Page.....	86
Figure 21: Admin/Vet Owner Dashboard.....	87
Figure 22: Appointment Management	88
Figure 23: Pet Record Page.....	Error! Bookmark not defined.

Figure 24: Invoice Page	Error! Bookmark not defined.
Figure 25: Adoption Page	Error! Bookmark not defined.
Figure 26: Stock Management	Error! Bookmark not defined.
Figure 27: Report Management	Error! Bookmark not defined.
Figure 28: Notification Management	Error! Bookmark not defined.
Figure 29: User Management	Error! Bookmark not defined.
Figure 30: Veterinary Staff Management	Error! Bookmark not defined.
Figure 31: Branch Management	Error! Bookmark not defined.
Figure 32: User Login Page	Error! Bookmark not defined.
Figure 33: User Dashboard	Error! Bookmark not defined.
Figure 34: User Appointment Management Page	Error! Bookmark not defined.
Figure 35: User Pet Record Page	Error! Bookmark not defined.
Figure 36: User Pet Adoption Page	Error! Bookmark not defined.
Figure 37: User Invoice Page	Error! Bookmark not defined.
Figure 38 User Notification Page	Error! Bookmark not defined.
Figure 39: Network Model	Error! Bookmark not defined.
Figure 40: UML Deployment Diagram	Error! Bookmark not defined.

CHAPTER 1

INTRODUCTION

The chapter described the research study, which included its principal problem and the need to develop digital veterinary care management systems. The Kho Veterinary Clinic Rinconada had some challenges in tracking the health records of pets and scheduling an appointment with a client and communicating with them through the use of manual means to operate in their clinic. The Petkho Portal was one of the systems that employed state-of-the-art veterinarian care and monitoring to enhance operational procedures and provide quality care for patients.

1.1 Project Context

With the integration of digital technologies aimed at improving clinical efficiency and patient outcomes, the field of veterinary medicine rapidly evolved. There has been a marked shift in most parts of the world toward the adoption of digital veterinary systems, reflecting the wider trend of digital transformation in healthcare. In this regard, tools like EMRs, telemedicine platforms, cloud-based scheduling systems, and remote health monitoring gadgets have become common in veterinary practices across the globe. This can be attributed to increased expectations from pet owners for personalized and timely care, increased pet ownership rates, and evidence of how digital solutions enhance efficiency in veterinary services. Recent global market research projects that the veterinary software market will see impressive growth in coming years as clinics invest in systems that make the process easy and improve their relationships with clients.

Despite these advancements, many veterinary clinics, especially in developing or underserved areas continue to face persistent challenges due to their reliance on manual, paper-based systems. Smart cities and digital public health worked together to improve the lives of people and animals using technology. With tools like sensors and the Internet of Things, daily activities could be monitored to keep track of health and respond to emergencies [2]. Traditional methods of managing patient records, booking appointments, and communicating with clients often resulted in

disorganized data, inefficient workflows, and service delays. With the improvement in information technology, the process of providing institutional services was efficient and effective. Among institutions were veterinary clinics, which started to increasingly use information technology to improve the way they provided services. Some of the challenges facing such institutions included the tedious process of completing forms, scheduling of clients, and the lack of appropriate information regarding the various services available to clients. All these factors necessitated the design and development of a veterinary clinic management system, which was done through an iterative model and a structured design approach to effectively address the challenges confronting the institutions and improve the experience of the clients. Additionally, the clinics were unable to perform a timely follow-up, efficiently administer vaccinations, and retrieve patients' medical history as needed. Importantly, the lack of an automated approach made the administrative task more cumbersome, thus exposing the clinics to the possibility of human error, which led to a series of problems bottlenecks that reduced productivity. These global challenges were widely shared, especially among smaller veterinary clinics that lacked the resources to adopt large-scale commercial software.

At the local level, the problems being experienced globally were related to the operations of small veterinary institutions like Kho Veterinary Clinic Rinconada. The clinic operated through traditional methods of information handling, scheduling of appointments, as well as customer relations. However, the clinic realized the need to improve the operations of the clinic through the adoption of innovation, leading to the implementation of the Petkho Portal: A Smart Veterinary Care and Monitoring System. With the implementation of the innovation, the clinic was able to improve upon the operations of the clinic by creating a platform to conduct the operations of the clinic effectively, including information handling as well as the scheduling of appointments. With the innovation, the clinic was able to avoid the losses that could have resulted from the use of traditional methods of information handling, as the information of the pets was easily accessible

through the portal.

With respect to the customer experience as well, it should be noted that The Petkho Portal allowed for an improvement in this sphere to an extent by delivering notifications pertaining to upcoming appointments to the owners of the pets that were brought to the veterinary clinic for treatment. With the help of such modern technologies as cloud computing services and electronic communication systems, the veterinary animal healthcare services were not only improved internally with respect to their operations but were able to foster a collaborative approach with respect to animal health as well. With the help of this system's data analysis functions with respect to detecting and managing animal health issues at an early stage, animal health surveillance was able to be effectively utilized as part of managing biological risks and their related risks as well.

In essence, the Petkho Portal reflected a localized application of global best practices in digital veterinary care. By aligning with the trends and technologies that shaped the international veterinary landscape, it positioned Kho Veterinary Clinic Rinconada to deliver more efficient, accurate, and client-focused services. This initiative not only addressed the clinic's existing limitations but also marked a significant step toward modernizing veterinary healthcare delivery in the local community. The project demonstrated how digital solutions, when thoughtfully implemented, could transform traditional animal healthcare systems—bridging the gap between global innovation and local need.

1.2 Purpose and Description of the Project

The Petkho Portal: A Smart Veterinary Care and Monitoring System was intended to further enhance and modernize the activities carried out in Kho Veterinary Clinic Rinconada. In contrast with the conventional and often time-, error-, and data-intensive approach to animal health-related activities, the system offered for the first time in the country an "innovative electronic mode in the delivery and monitoring of veterinary services." What was unusual in this kind of system

was that it was "designed for the simultaneous use and updating in real-time of up to three or four branches," a feature that not only enhanced the continuity and quality of the animal health services offered in the different branches but also facilitated increased coordination and teamwork among employees in various branches. In this regard, the Petkho Portal was a very modern and innovative system in various aspects.

In order to address the situations of disorganization revealed in the Kho Veterinary Clinic Rinconada, as well as communication challenges, the project sought an innovative solution for replacing the conventional practices, which were rife with human errors, at the veterinary clinic. Petkho Portal, the developed solution, was designed with the intention of reducing human errors, which often resulted from conventional practice, while also enhancing communication with the pet owners regarding their health information. This ensured a better quality of service as the quality of care rose. Petkho Portal also ensured the accuracy of the information presented regarding a pet, which could otherwise be difficult due to the conventional approach.

The proposed project, Petkho Portal, has been designed in order to effectively address some of the nagging operational concerns that have been associated with the clinic, such as unprocessed patient information, unprocessed patient appointments, as well as lack of communication between the clinic personnel and pet owners, which may impact the level of service delivery in a negative way, considering that such issues may translate to a lack of proper documentation as far as the medical care of pets in the clinic goes, as well as a reactive approach towards service delivery compared to a responsive approach.

But in essence, the Petkho Portal aimed to create a rather stable and structured operational framework that would serve both the veterinary staff and clients alike. In this respect, the system enabled the clinic to raise its bar higher in attending to patients and reduce unnecessary administrative tasks caused by human error. It also made it easier for pet owners to stay tuned with current events about the health journey of their pets, thus attaining a more glued relationship

between the clinic and its clients. It was, therefore, proposed to upgrade the services provided by a veterinary at standards that are sustainable and scalable, harmonizing with the emerging demand for digital transformation in healthcare.

1.3 Objectives of the Project

The general goal of the undertaking in the project was the creation and implementation of Petkho Portal, a smart veterinary care and monitoring system. Such a system had the following goals: the optimization of the clinic's operations by enhancing the automation of patient record and appointment bookings as well as the optimization of communication with pet owners.

1. To identify the challenges in the existing veterinary clinic management processes, in terms of:

1.1 Patient record management

1.2 Appointment scheduling

1.3 Communication between the clinic and pet owners

2. To develop a component veterinary care and monitoring system that addresses the identified challenges by providing a centralized platform for:

2.1 Authentication and authorization

2.2 Appointment and pet Record Management

2.3 Referral Management

2.4 Inventory Management

2.5 Billing and Invoicing

2.6 Notification and Communication

3. To test the level of satisfaction of the management system based on ISO/IEC 25010 using the following:

3.1 Performance Testing

3.2 Security Testing

3.3 Use Case Testing

3.4 User Acceptance Testing

4. To deploy the developed system to Kho Veterinary Clinic Rinconada.

1.4 Significance of the Study

Recognizing the significance of the study, the objective of this research work is to create an automated system that will enhance the management, monitoring, and tracking of pet veterinary services in order to improve the delivery of services of the Kho Veterinary Clinic Rinconada. The results of the study were of significant benefits to the following::

Veterinary Staff: The system significantly streamlined various tasks like handling patients' information and requests. In other words, the system enhanced efficiency.

Pet Owners: The various components offered in the portal are helpful to pet owners; for example, they are able to book appointments and view health records for their pets and stay updated with information regarding vaccinations and further pet treatments.

Clinic's Owner: This person can take advantage of the ability offered by the system that can centralize the operations of the clinic, hence being able to monitor the performance of the clinic.

Future Researchers/Developers: This study provides valuable information regarding the integration of technology in the veterinary field; hence, it can be used as reference material in future studies focusing on the evolution of veterinary management systems or technological solutions in the field of digital healthcare. This study was significant in the evolution of the veterinary field; the study improved the level of service delivery, communication, and operational efficiency.

1.5 Scope and Limitations of the Project

The Petkho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada. The project developed, tested, and deployed the application to aid the operations

of the clinic by automating the necessary operations as well as the flow of information between the veterinarians and the owners of the pets. Some of the modules developed in the project application include authentication, authorization, the management of appointments, the handling of pets, the handling of referable patients, inventory control, as well as sending bill invoices to customers. These were meant to resolve the problems facing the clinic regarding the handling of patient records, appointments, as well as the sending of information to customers of the clinic. Finally, the project was completed, meaning the application was successfully deployed to the Kho Veterinary Clinic Rinconada to aid in operational usage.

Nevertheless, this well-thought-out system should be expected to have some limit at some point that may affect its efficiency to some extent and usability as well by future users. It should be noted that the functionality of the Petkho Portal would entirely depend on the availability of internet connection services in the particular area where this system would be utilized. Additionally, the precision of the fully automatic communication facilities would depend entirely upon the data provided by the users regarding their particular contact information. It should also be noted that this particular management and monitoring system would by no means replace the particular expertise that the veterinarian would have to employ in his operations.

1.6 Project Dictionary

In giving a clearer picture about the terms utilized within the context of the present research, the technical terms utilized within the present research have been conceptualized and operationally defined. These technical terms are considered the most vital in evaluating the functionalities and other significant features associated with the Petkho Portal: A Smart Veterinary Care and Monitoring System designed for the veterinary clinic named Kho.

Authentication. It is the process of verifying the identity of a user attempting to access a digital system through validated credentials such as usernames and passwords [14]. For the PetKho

Portal, this means that a user can access their dashboard only after their login information is validated, and incorrect credentials prevent entry into restricted modules.

Authorization. It refers to determining the specific actions or system features an authenticated user is allowed to access based on their role [15]. Within this system, authorization ensures that veterinarians, staff, and pet owners only reach the modules relevant to their responsibilities, maintaining operational security.

Appointment Scheduling System. It refers to the Portal's online scheduling module that allows pet owners to book appointments, receive reminders, and enables veterinarians to manage their daily schedules more efficiently [1]. The portal uses this system to organize and manage appointments between clients and service providers, improving workflow and reducing wait times.

Automated System. A system that performs tasks automatically through programmed rules, triggers, or schedules without requiring manual intervention [5]. The portal uses automation for sending reminders, generating invoices, updating records, and processing data to reduce staff workload and minimize human error.

Billing. It is the process of calculating, documenting, and communicating charges for services and products [16]. Within the PetKho Portal, billing computes service costs, generates invoices, issues receipts, tracks payments, and maintains financial records for the clinic.

Clinic Management Dashboard. A centralized interface that visually presents key data needed for organizational oversight and decision-making [8]. This dashboard provides administrators with real-time insights into appointments, transactions, and records, supporting faster and informed decisions.

Customization. It is the ability to modify or tailor a system to meet user-specific requirements [17]. The portal allows customization of interfaces, features, and functionalities to align with the operational needs of Kho Veterinary Clinic.

Data Security. The protection of digital information from unauthorized access, misuse, or breaches [4]. The system ensures data confidentiality, integrity, and availability, safeguarding pet records and financial information.

Digital Inventory. An electronic method for tracking and managing stock, supplies, and consumables [10]. The portal uses digital inventory to monitor veterinary medicines, prevent shortages, and maintain smooth clinic operations.

Financial Transparency. The clear visibility and accountability of financial transactions and records [18]. Through the portal, clients can view accurate billing, payment histories, and transaction logs, enhancing accountability.

Invoices. Official documents listing products or services rendered along with their corresponding costs [19]. The portal generates invoices automatically for every transaction, detailing treatments, services, and purchased items.

Online Appointment. A digital scheduling approach that allows clients to book services through online platforms [20]. Pet owners can request appointments remotely through the portal, offering a faster and more convenient alternative to walk-ins.

PDF Referral. A digital document used to endorse a client or patient to another service provider [21]. The portal generates PDF referrals for pets being transferred to external clinics, ensuring proper documentation.

Pet Health Records. A compilation of a pet's medical history, including treatments, vaccinations, and medications [6]. Within the portal, these records are updated by veterinarians and accessible to pet owners, supporting informed care decisions.

Real-time Data Updates. Instant updating and synchronization of data across a system [11]. The portal ensures that changes to medical records, appointments, and monitoring modules are reflected immediately to support timely decisions.

Receipt. A document confirming payment for goods or services [22]. The portal automatically generates detailed receipts after payment.

Referral Module. A system feature that facilitates referral processing between service providers [23]. The portal allows veterinarians to refer pets to other branches or external clinics, with automatic PDF generation for documentation.

System Administration. The process of managing and maintaining system operations, resources, and user controls [12]. The portal provides administrative tools to manage accounts, track data, and maintain system performance.

Notes

- [1] Rosmani, Arifah Fasha and Mokhtar, Muhammad Harith. 2023. An Online Scheduling Platform for Veterinary Appointments. *Jurnal Intelek* 18, 2 (Aug. 2023), 16–26. <https://doi.org/10.24191/ji.v18i2.22092>
- [2] Benis, Arriel; Haghi, Mostafa; Deserno, Thomas M.; and Tamburis, Oscar. 2023, May. *One Digital Health Intervention for Monitoring Human and Animal Welfare in Smart Cities: Viewpoint and Use Case*. *JMIR Medical Informatics* 11 (May 2023), e43871. <https://doi.org/10.2196/43871>
- [3] Katrin Beyer. 2023. Unlocking the potential of ICT innovation in veterinary healthcare: The pathway to improve practices and business model. *Procedia Computer Science* (2023), 4775–4784.
- [4] Péter B. Bök and Dario Micucci. 2024. The future of human and animal digital health platforms. *Journal of Reliable Intelligent Environments* (September 2024). <https://doi.org/10.1007/s40860-024-00232-0>
- [5] Mohammad Dayoub. 2023. Applications of smart agriculture to improve animal production: Opportunities, challenges, solutions, and benefits. *Preprints* (2023). <https://doi.org/10.20944/preprints202310.1229.v1>
- [6] Holger Görge. 2023. Solutions and prospects for digital documentation of treatment data on-farm. *Smart Agricultural Technology*. Elsevier B.V.
- [7] Wendy Hauser. n.d. Slow down to speed up: How to create practice efficiency. *Peak Veterinary Consulting*.
- [8] Dedi Iskandar. n.d. Development of a technology-based animal electronic medical record information system for animal health services.
- [9] Nurul Badriah Jaffar. 2021. The development of veterinary clinic management system using structured approach. *Applied Information Technology and Computer Science* 2, 2 (2021),

- 1555–1567. <https://doi.org/10.30880/aitcs.2021.02.02.100>
- [10] Marinette Babila and Hui Xiao. 2020. The principal as a curriculum-instructional leader: A strategy for curriculum implementation in Cameroon secondary schools. *International Journal of Education and Research* 8, 4 (2020).
- [11] Harshad D. Patil. n.d. Smart surveillance and animal care system using IoT and deep learning.
- [12] Sasmita Swain. 2024. Smart livestock management: Integrating IoT for cattle health diagnosis and disease prediction through machine learning. *Indonesian Journal of Electrical Engineering and Computer Science* 34, 2 (May 2024), 1192–1203. <https://doi.org/10.11591/ijeecs.v34.i2.pp1192-1203>.
- [13] Research Exhibition in Mathematics and Computer Sciences (REMACS 5.0). 2023.
- [14] Microsoft. 2025. What is Authentication? Microsoft Security. <https://www.microsoft.com/en-my/security/business/security-101/what-is-authentication>
- [15] Maciej Kosinski. 2025. Authentication vs. Authorization: What's the Difference? IBM Think. <https://www.ibm.com/think/topics/authentication-vs-authorization>
- [16] Asher Ashfaq. 2024. What is Medical Billing? Grants for Medical. <https://www.grantsformedical.com/what-is-medical-billing.html>
- [17] Roger S. Pressman and Bruce R. Maxim. 2020. *Software Engineering: A Practitioner's Approach*. McGraw-Hill.
- [18] Lee Graham and Michael Lang. 2025. *Financial Management Systems in Healthcare Organizations*. Springer.
- [19] Angela Martinez and Paul Simmons. 2023. Electronic Invoicing Systems in Healthcare. *Journal of Digital Finance* 9, 1. <https://doi.org/10.1234/jdf.v9i1.2023>
- [20] Arifah F. Rosmani and Mohd H. Mokhtar. 2023. Veterinary Online Appointment System:

Enhancing Efficiency. CS101, Universiti Teknologi MARA.

- [21] David Lin, Sarah Chen, and Rafael Torres. 2024. Veterinary Referral Document Management: Concepts and Practices. *Journal of Information Systems and Informatics* 8, 2. <http://journal-isi.org/index.php/isi/article/download/863/421>
- [22] Mark D. Evans. 2026. Service Management System Billing Chapter. *Studylib*. <https://studylib.net/doc/9400191/the-medical-billing-system>
- [23] Laura K. Thompson and Daniel R. Vega. 2025. Digital Referral Modules in Veterinary Practice. *Journal of Veterinary Clinic Digital Systems* 5, 1.
- [24] Kevin C. A. Amarille, Lina P. Reyes, and Mark D. Cruz. 2024. Web-Based Integrated Information System of Phoenix Veterinary Clinic. *Asian Scientific Journals*.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presented a comprehensive review of related literature and existing systems that served as the theoretical and practical foundations for the development of the study. The insights gathered from various sources provided direction and support for the researchers in shaping the framework and methodology of the project. Moreover, the review enhanced the researchers' understanding of essential concepts and technologies, helping them explore innovative solutions and best practices for the successful implementation of the proposed study.

2.1 Related Literature and Systems

This chapter provided a review of related literature and systems from both foreign and local studies. It served as a guide for the researchers in the development of the system, offering insights and ideas on how to implement the proposed study effectively. The studies presented in this section highlighted systems with similar objectives, features, and outcomes, which helped in drawing comparisons and identifying best practices relevant to the subject matter.

2.1.1 *Record Management*

This research mainly focused on the dynamic field of veterinary nursing. Specifically, it aimed to examine how the application of technology could be most effective in enhancing the quality of patient care. Furthermore, it offered an exhaustive overview of the work performed by veterinary nurses. Moreover, the document delves deeper into how veterinary nurses work in conjunction with other practitioners in addressing various disciplinary areas. It also emphasizes the obstacles associated with the quality of patient service. This research employed the black box testing method. Black box technical testing was a test of the input or output functionality of the software. In this project, several types of Blackbox Testing will be performed to make sure the system works well and meets the needs of its users[21]. Black Box Testing is a software testing method in which the functionalities of software applications are tested without having knowledge

of internal code structure, implementation details and internal paths. Black Box Testing mainly focuses on input and output of software applications and it is entirely based on software requirements and specifications. It is also known as Behavioral Testing[22]. It delved into technological advancements, including improvements in medical and surgical technology, electronic health records (EHR), and telemedicine. The findings had indicated that the use of technological devices played an important and transformative role in ensuring increased accuracy and improved surgical and communication processes among teams. Challenges including issues around training, finance, and use of information to protect against cyber threats existed as other issues observed during the study. This highlighted that the use and embracing of technological devices in the processes used by veterinary nursing practitioners had an increased capability to improve and enhance the standard and core role played by the practitioners in the ever-changing domain of animal and pet healthcare [1].

The study results revealed a number of weaknesses that exist in inventory management, whereby approximately 59.5% of the facilities do not have standard operating processes when it comes to warehouse management. Also, none of the facilities were found to make use of bin cards as well as software applications in their inventory management process. However, in approximately 83.8% of the facilities, there were no disposal guidelines, and disposed medications were not disposed of properly. Private veterinary wholesalers had relatively good performance when it comes to recommended storage facilities (86.25%), while government clinics had a poor storage performance rating in almost half of the facilities (48.3%). Based on the study, a number of challenges were revealed, including inadequate infrastructures, inadequate personnel, lack of access to affordable medications, inadequate regulatory control, and budget constraints. However, a comprehensive approach that involves all the stakeholders in addressing these challenges and improving the overall performance of the sector has been proposed as a recommendation [17].

As a result of technological advancements such as the internet and wireless networks, many businesses, including veterinary clinics, implemented these technologies in their day-to-day operations, and this trend grew in popularity in recent years. This project proposed that veterinary clinics adopt an online, web-based appointment system to make the appointment process more efficient and straightforward. The research had two objectives: designing and developing a web-based concept for an online appointment booking system. The appointment system was subjected to a usability test to determine whether the proposed system simplified appointment scheduling for users. This study employed the waterfall model, which consisted of six phases, including requirement, analysis, design, coding, testing, and implementation [4].

This article sought to address certain perspectives on the changing dynamics of animal healthcare and how innovation is changing veterinary practice. It helped to understand and appreciate the significance of introducing innovation to improve effectiveness. At the same time, this author sought to address some critical questions about the types of innovation to incorporate into vet clinics, what level of satisfaction people have with such services, and types of organizational challenges that people faced. In addressing these questions, this author used various methods such as literature reviews, reasoning approaches, and 102 qualitative interviews with pet owners. This was highly applicable to those involved directly within animal care units and to those interested in understanding these perspectives. The findings helped to improve our understanding and see innovation that can improve effectiveness within animal care units [5].

This study focuses on the development of Rabdash, a mobile application designed to digitize rabies data collection for Davao City's CVO and private veterinarians. It was developed using the Scrum Agile method and features real-time reporting, user management, and security in the handling of data. As a result, it promotes easy transitioning from manual to digital tracking of data, which may improve rabies surveillance and management. The usability testing yielded an average usability score of 80.63, but several usability issues were noted, along with poor internet

access in remote areas. Thus, the recommendations include improving the UI, allowing offline capability, integration of analytics tools, user training, and enabling public participation to add to the efforts of rabies control [6].

2.1.2 Monitoring System

The rise of the pet care industry has also emphasized the importance of providing facilities to those who own or plan to own pets in an easy and reliable manner. The focal point of this article is to present how an integrated platform was developed regarding providing facilities of adopting pets, seeking guidance regarding their care, and consulting professional vets in one go. The overall idea of the study is to highlight how technology has the potential to serve as a great assistance in achieving efficiency in matters of providing facilities to pets.

In this study, the use of diversity metrics as a black-box alternative to white-box coverage criteria. These metrics are designed to function without requiring internal model access or execution results. It adapt and evaluate three diversity metrics to assess their ability to measure input diversity effectively. Using four datasets and five DNN models, it examine the correlation between these metrics and their fault detection performance, and compare them to leading white-box coverage methods. Additionally, it introduce a novel approach for estimating fault detection in DNNs. This experimental results demonstrate that diversity in image features within test inputs serves as a more dependable indicator for DNN testing than traditional coverage metrics. Notably, one black-box diversity metric significantly outperformed existing methods in fault detection efficiency and computational speed, reinforcing doubts about the adequacy of current coverage criteria for generating effective natural test inputs [2].

This is a project aimed at developing a modern system of cattle health monitoring. This system is developed with modern technology that can be utilized for tracking important health parameters of cows. As we all know, it is of utmost importance that the lifespan of cattle is increased, along with the quality of dairy production. It is a fact that farmers face a problem in

tracking the health of cows on a larger scale due to a lack of time. This system is mainly developed for the sustainability of cattle ranching. By using advanced technology such as skin temperature sensors, heart rate sensors, and motion sensors, this system can track critical health parameters such as heart rates, activities, heat stress, and sleeping patterns. With machine learning algorithms, this system can also demonstrate the cattle's health. A mobile application will also provide users with timely visualizations to facilitate prompt actions in case of developing health concerns. This innovation will change the face of cattle management by improving health concerns, resource management, and increasing profitability for livestock farmers. Monitoring of cattle health will act as a preventative system in detecting health concerns in cattle because of the major challenges in monitoring animal health effectively in the industry; generally, practices such as visual observations of cattle health are inaccurate [15].

This thesis aims to explore how new technologies affect veterinary science, with special emphasis on selected application scenarios related to veterinary health informatics. The thesis analyzes a weight loss program for cats with higher effectiveness compared to traditional approaches but with reservations on cost from pet owners. Pet health insurance statistics related to various breeds have also been analyzed to develop education on respective disease prevalence, with higher occurrences among purebred breeds of cats. In another application, predictive analytics based on pet insurance statistics have provided insights on future incidence of periodontal disease and skin tumor occurrence among Feline breeds, and there is a need to exercise caution while interpreting such statistics. Besides, attitudes toward telemedicine and other related ICT solutions among veterinary practitioners have specifically looked at challenges arising when consultations have to be done remotely, such as charging fees for these services, especially with reference to the COVID-19 pandemic. The thesis specifically aims to highlight scenarios like affordable and user-friendly technologies, breed-specific care, and telemedicine that have a transformative effect on

practice, while cautioning against a lack of scientific understanding and further research with reference to its application [7].

Animal welfare in a pasture environment cannot be monitored effectively despite its numerous natural benefits. For instance, these are beneficial in detecting diseases or calving, and drones are required to manage animal herds. However, VH makes use of sound and mild electric shock. However, there are certain animal welfare issues associated with VH, mainly those with a low learning ability. Therefore, more research has to go into these systems before they are finalized in order to promote animal welfare [8].

The research seeks to investigate the impact of styles of communication in veterinarian-client-patient relationships on clinical choices made in healthcare settings. Among the expectations identified in focus groups conducted by researchers were those of the pet owners; they have the expectation of being understood, having information that suits their needs, as well as having the ability to participate in the clinical choices made. Veterinarians, however, have their own challenges like time constraints, many clients in their practices, as well as those who do not communicate using their own type of language. The researchers have indicated that this is essential in enhancing trust in the healthcare settings [10].

2.1.3 Information System

In this respect, the present study heavily relied on Healthray 2023, which presented basic characteristics of clinical management systems that acted like signposts in the formulation of the survey and assessment framework for the study. Usability, functionality, performance, and security were seen as important quality factors that influenced the support provided to daily operations like scheduling an appointment, keeping client records, and communicating with clients by the veterinary and clinical information system. Systems with intuitive navigation, clear interface elements, and ease of operation had been shown to improve user acceptance by reducing errors and cognitive load during real-world tasks. Healthray's recommendations reinforced the understanding

that platforms facilitating precise features, workflows, and system performance improved decision-making and user experience, reducing the user's workload, whether a pet owner or employee themselves. Load time, performance, and data processing all improved user experiences, and user trust was enhanced, as was information security through features such as authenticity, access control, and security communication. Thus, there is agreement that digital clinical platforms have to incorporate elements like usability, functional suitability, performance efficiency, reliability, and system security to effectively serve users. This study introduced another digital system called 'a,' which is an open-source platform that facilitates real-time interactions with various machine learning classifier types and veterinary electronic health records. In addressing some of the challenges that arise when trying to integrate system applications with EHR system limitations, 'Anna' has proven to work as an independent system. Based on this and through its application in terms of three different diagnostic types of dog illnesses, 'Anna' has proven that it is capable of rapid roll-out options, application environment types, and increases access to various types of ML applications used within veterinary practice. [11].

While many industries are moving in an environmentally responsible direction, healthcare is in a paradox: even though the aim of hospitals is to make people healthier, they highly contribute to health issues through environmental burdens. Human medicine has taken some steps toward sustainability by minimizing waste and carbon emissions through the identification and institution of sustainable practices. This study considers the findings of AgencyAnalytics (2024) on system responsiveness and performance, which guided the criteria for evaluating the clinic portal. In clinical information systems, performance and reliability are key indicators of quality, especially in settings where fast access to patient data and uninterrupted operations are critical. Research on digital platforms shows that quick loading, stable processing, and consistent responsiveness greatly affect user satisfaction and workflow efficiency, as delays can disrupt tasks and reduce confidence in the system. According to AgencyAnalytics, systems should maintain low latency, smooth

transitions between functions, and stable operation even under heavy data loads or high user activity. These qualities help both the clinician and the clients by maintaining smooth interaction sessions, reducing waiting periods, and providing reliability. Overall, the literature focuses on the need for strong performance and reliability in assisting the digital clinical applications to enable the users to perform their day-to-day activities in an efficient manner [24].

In contrast, veterinary medicine lacks such a comprehensive evaluation. This study was intended to replicate the practice of a review method in the area of human healthcare to evaluate the current information related to environmental sustainability in veterinary clinics. There are only three peer-review opinion articles from the UK pertaining to the area of environmental sustainability in the eight databases searched for information. Not one was related to companion animal practice, the main practice area in the United States for the veterinarians. Information from the article suggests that veterinarians could integrate environmentally conscious practice within the veterinary field by adopting "green practices," learned from other industries in the areas of energy use, water management, waste reduction, purchasing, transportation, and behavioral changes at different levels. A great need exists for veterinarians to practice sustainability for animal and environmental well-being. [12].

Veterinary professionals and small animal veterinary workers (SAVWs) in particular are instrumental in the prevention of zoonotic diseases; however, they experienced challenges during the Covid-19 disease outbreak worldwide. Based on a survey of 1204 small animal veterinary workers in the United States of America, it was noted that factors such as reducing the number of work hours, PPE use, and other factors such as caregiving influence their capacity to prepare for and respond to the Covid-19 outbreak in their respective regions in the United States of America. The importance of managing their schedules in order to prepare and improve their capacity to respond to the Covid-19 outbreak in their respective regions in addition to addressing their caregiving challenges was also emphasized in order to prepare for and respond to similar outbreaks

related to zoonotic diseases in various regions in the world in the near future through a One Health approach [13].

2.1.4 Efficiency of Smart Veterinary Care

The article shows the importance of having a strong vocational identity while comparing different professions in this aspect. The article emphasizes that the development of vocational identity is an essential aspect in the training process in VET, as it plays an essential role in career development. It is influenced by both the chosen profession and the experience gained in an apprenticeship. It has been investigated that having a strong vocational identity positively effects motivation in the long term and work performance. It plays an important role in the acquisition of work responsibility together with the importance of quality [14].

The development of IT has affected wide areas in life today, including veterinary health care. An Animal Health Center (Puskesmas) is a facility that conducts veterinary services through professional veterinarians and paramedics. However, compliance with animal welfare remains low because the government's attempt at implementation is less than adequate and supported by not fully implemented regulations. Traditional methods of recording data, such as patient cards, possess shortcomings, including limited accessibility, risk of loss, potential human error, and environmental impact. An animal medical record information system solves all the above-mentioned problems by ensuring access to efficient data management, high-quality health services, and routine care. In the development of this system, the Waterfall Method was applied. [9].

More than half of Norway's upper secondary students choose a vocational education and training pathway. However, these students are more likely to be at risk of marginalization compared to students who follow other educational routes. Evidently, students face many barriers when it comes to vocational training and education in Norway as some of them are not likely to secure apprenticeships after their two-year course in VET. Additionally, students are likely to be at a higher risk of marginalization based on some of their family-related influences. This article seeks

to examine how local VET addresses risks through systems theory developed by Luhmann. The systems theory considers VET as consisting of separate blocks or systems of understanding and addressing marginalization risks. Using a case approach with key stakeholders through a series of interviews is likely to yield meaningful findings as depicted in this case. The findings indicate that there are institutional logics which potentially act as barriers to integrating education and work. Nevertheless, there is a likelihood of bridging these gaps through communication and understanding to form a cohesive whole for addressing marginalization risks. [3]

The improvement of animal health and welfare in livestock systems relies heavily on effective and reliable assessment tools. The current project develops a novel data-driven approach based on animal-based indicators for the assessment of health and welfare conditions on Swiss farms housing common livestock species, including cattle, sheep, goats, pigs, and poultry. The methodology contained a systematic literature review to identify animal-based indicators, field comparisons using Welfare Quality® scores, and applied machine-learning models in interconnected agricultural and animal health databases to identify at-risk farms. Data across species were variable, with dairy cows and pigs having more data, and some indicators still need full validation. Features such as participation in federal programs and farm management behavior proved to have high predictive accuracy. Conclusions: Combined data-based metrics with on-site assessments give most valid results, and innovative farming technologies might be promising to reduce the need for on-site assessments in the future. [16].

The article under discussion, focusing on the topic of remote vital sensing as a practice in veterinary medicine, shows a phenomenon that involves the sensing or measurement of vital parameters from a living creature without conducting an unduly invasive method. Further, the article focuses on a number of technologies that have been applied in vital sensing within veterinary medicine, including infrared thermography, rPPG, radars, wearable sensor approaches, and computer vision along with machine learning. Some of the applications, benefits, and demerits of

these approaches are addressed, showing their potential use within a number of animal species, focusing on the future prospect of leveraging vital sensing within the realm of veterinary medicine. Further, some of the drawbacks, challenges, and reliability attributed to the application of vital sensing in the medical sector are addressed, along with future prospective applications including the focus that will be given to the combined approach with artificial intelligence and vital sensing [18].

This is due to the fact that the latest technological trends have increased the number of pet owners worldwide as pets serve as a source of stress-relieving measures and have been turned into a hobby for some individuals, making them spend as much as possible on their pets. During the COVID-19 outbreak, pets, especially those that had close interaction with infected individuals, have been pointed out as carriers of the virus. On the other hand, 36% of the surveyed individuals stated that they did not own any pets. Due to the movement restrictions put in place by some governments to help control the COVID-19 outbreak in 2020, some individuals did not have the ability to access and attend to their pets as usual. This project seeks to design and develop an online Veterinary Clinic Management System to improve client booking in pet clinics throughout the globe [19].

Over the years, Electronic Medical Records (EMRs) have undergone extensive development and are now playing a more prominent role in transforming healthcare delivery. Although it has to be emphasized that while the general motivation behind using EMRs remains aligned to delivering an overall upgrade in the level of medical care available to all, it is also important to examine their influence in specific individual physician-patient relations as well. The present study aims to examine their use in accordance with their application in veterinary medicine in Philippine society by emphasizing confidentiality in the duties of persons in this profession, such as diagnosing diseases in animals, treating wounds or performing surgery while in professional service roles. The importance of their use in delivering effective care will be emphasized through

reference to how they facilitate the management of their duties in delivering professional care to animals while providing owners with important tools such as tracking appointments and providing information to store safely in regards to their pets in order to care for them properly [20].

The application of testing in this regard, as illustrated in the study, incorporates a set of black box testing in order to effectively evaluate the efficiency of the management system. More specifically, in this regard, performance testing is incorporated in order to evaluate the efficiency of the system in effectively performing a number of tasks, assignments, and responses, as well as the ability and efficiency of a seamless response devoid of considering and understanding code mechanisms. With regard to security testing, as illustrated in this study, the case, in this regard, incorporates a potential and efficient response, as well as ability, in order to accept external forces and pressure in considering and enhancing data security. With regard to this, in this study, a series of efficient responses, as illustrated in this study, incorporates use cases in order to evaluate a series of authentic, genuine, and efficient responses, as well as scenarios, in order to evaluate appointment schedules, as well as responses compatible and congruent to this regard in a veterinary clinic.

This particular chapter covered a critical analysis of existing literature and systems related to the suggested proposal within the field of veterinary care. Both local and international information related to the subject has been reviewed and considered in the analysis, aiming to create a solid theory of existing related work for the development of a thorough framework of research. Already, there is a clear development in terms of record management systems, monitoring systems, and information technologies related to veterinary care. At the same time, there is a need for the development of solutions related to issues of managing information, regulation, and utilization of available technologies. As far as the related information has been reviewed, issues of information records, on-line systems, and system performance have been thoroughly emphasized.

2.2 Synthesis of the State-of-the-Art

The current landscape of veterinary healthcare technology suggests a remarkable evolution

With the integration of digital systems, advanced monitoring instruments, and innovative management platforms aimed at improving quality and efficiency in animal care. A considerable amount of recent research emphasizes the adoption of EHRs and web-based management systems that help overcome several long-standing concerns regarding data handling, accessibility of patient records, and appointment scheduling. These digital solutions have also helped streamline clinical operations and improve communication between veterinarians and pet owners. However, emerging studies also point to several contemporaneous concerns on cybersecurity risks, user training needs, and some cases of low system and data reliability.

Concurrently, the veterinary profession is increasingly adopting advanced health monitoring technologies such as wearable technologies, remote devices, and various aspects of artificial intelligence that incorporate various aspects of machine learning to monitor the health status of an animal through the use of various health indicators. This technology is highly advantageous in the management of health status through the construction of specific decision models. However, challenges such as the lack of internet connection, the usability of devices for health monitoring, as well as the physiological uniqueness of various animals, are factors that are yet to be fully resolved.

Additionally, the evolution of the use of mobile applications, as well as agile software development techniques, has also fueled the pace of innovation in veterinary informatics. For instance, the creation of mobile applications for diseases, e.g., rabies, shows the use of real-time data collection, as well as user management, in enhancing public health results. The use of machine learning classifiers with veterinary EHR shows the use of artificial intelligence in veterinary practice, as well as the limited need for software evolution.

Despite the above improvements, various studies emphasize the existing challenges which hinder the fruition of the actual potential of the technology. These include, but are not limited to, infrastructure, regulatory, access, as well as professional needs. There is also the emerging need for

sustainability in the veterinary field, an issue similar to the human health initiative. Another factor highlighted as instrumental in creating a better workforce that can efficiently utilize the emerging technology, with a high level of animal welfare, is the provision of vocationally inspired education.

For the viability and usability of veterinary management systems to be guaranteed, extensive evaluation procedures, such as those considered through black box testing, have to be conducted. These include aspects concerning performance under different circumstances, security mechanisms to safeguard critical information, and basic functional requirements concerning aspects to be met concerning those utilising these systems. It is a reaffirmation of just how important it is to develop systems which, apart from their cutting-edge nature, also have security and usability aspects guaranteed for their viability in actual clinical settings.

In summary, based on various literature reviews, veterinary technology is advancing greatly through the various inventions introduced by digital innovation, various devices for health monitoring, and various related information management technology. Even though there are various challenges relating to security, integration, and usage, existing literature reviews have provided a solid foundation that can be used to deliver efficient, reliable, and efficient veterinary technology. In conclusion, the various inventions being introduced in the technology sector are playing a very crucial role in enhancing the efficiency of the offered services, thereby contributing to the improvement of pet health.

2.3 Gap Bridged by the Study

From the review of related literature and systems, the results indicate that even though many studies have been conducted related to the inclusion of technology in veterinary health care, most existing systems are limited to only one functionality, such as appointment scheduling, maintaining electronic medical records, or health monitoring. Most of the systems exist independently, leading to fragmented processes lacking in efficiency and continuity within veterinary operations. Secondly, there is still a platform that faces hitches pertaining to data

unavailability, poor mechanisms for monitoring, inability to ensure interoperability, and lack of communication between clients and veterinarians. Therefore, an innovative, more integrated, and adaptive system is required to address these inadequacies.

This gap will be bridged by the study that comes with a vast web-based application in the field of veterinary care and monitoring that encompasses all the key features. This application can help run the clinic with considerable ease with more communication shared among pet owners and the vets. This application can benefit smaller veterinary clinics because not only is the application easy to use but also affordable. Associated with the application are the reliability, performance, and security measures that make the application more efficient with the innovations that are bridged in the study with the purpose of strengthening the veterinary services with the promotion of better pet welfare

Notes

- [1] Olufunke Afuolabi. 2023. Exploring the impact of emotional intelligence on veterinary nursing: Enhancing patient care and well-being.
- [2] Zahra Aghababaeyan. 2023. Black-box testing of deep neural networks through test case diversity. *IEEE Transactions on Software Engineering* 49, 5 (May 2023), 3182–3204.
<https://doi.org/10.1109/TSE.2023.3243522>
- [3] Anna Cecilia Ragnarsdóttir. 2023. Bridging gaps in vocational education and training systems in Norway. *Journal of Vocational Education and Training* (2023).
<https://doi.org/10.1080/13636820.2023.2255992>
- [4] Rosmani Arifah Fasha and Mokhtar Muhammad Harith. 2023. An online scheduling platform for veterinary appointments. *Jurnal Intelek* 18, 2 (Aug. 2023).
<https://doi.org/10.24191/ji.v18i2.22092>
- [5] Katrin Beyer. 2023. Customer experiences drive the need of innovation in veterinary practices. *Procedia Computer Science* (2023), 3094–3103.
- [6] Gabrielle R. Cabrera. A centralized mobile app for dog rabies vaccination reporting by private and city veterinarians in Davao City.
- [7] Barak N. Hadar. *Emerging Technology and Health Informatics in Veterinary Medicine: Technology-Enhanced Weight-Loss Program, Disease Patterns and Predictions from Insurance Data, and Perceptions of Telemedicine in the COVID-19 era*.
- [8] Anders Herlin. 2021. Animal welfare implications of digital tools for monitoring and management of cattle and sheep on pasture. *Animals*. MDPI AG.
- [9] Dedi Iskandar. *DEVELOPMENT OF A TECHNOLOGY-BASED ANIMAL ELECTRONIC MEDICAL RECORD INFORMATION SYSTEM FOR ANIMAL HEALTH SERVICES*.
- [10] Nina Janke. 2021. Pet owners' and veterinarians' perceptions of information exchange and

-
- clinical decision-making in companion animal practice. *PLoS ONE*. 16, 2 February (Feb. 2021). DOI:<https://doi.org/10.1371/journal.pone.0245632>.
- [11] Chao-Yu Kong. *Enhancing AI Accessibility in Veterinary Medicine: Linking Classifiers and Electronic Health Records*.
- [12] Maria K. Koytcheva. 2021. A Systematic Review of Environmental Sustainability in Veterinary Practice. *Topics in Companion Animal Medicine*. 44, (Aug. 2021). DOI:<https://doi.org/10.1016/j.tcam.2021.100550>.
- [13] Diego R. Marquez. 2023. ASSESSING US SMALL ANIMAL VETERINARY CLINIC ADAPTATIONS AND THEIR IMPACTS ON WORKFORCE COVID-19 PREPAREDNESS AND RESPONSE. *Health Security*. 21, 6 (2023). DOI:<https://doi.org/10.1089/hs.2023.0041>.
- [14] Christian Nägele. 2024. I ECER VETNET Proceedings 2024 Nicosia. VII, (2024). DOI:<https://doi.org/10.5281/zenodo.13624225>.
- [15] Shubham D. Pasalkar. *Agrisense Integrated Health Surveillance for Cattle*.
- [16] Davide Vecchio. *Development of a data-driven method for assessing health and welfare in the most common livestock species in Switzerland: The Smart Animal Health project*.
- [17] Amanuel Wondie Mekonen. 2024. Assessment of veterinary pharmaceutical warehouse management practices and its associated challenges in four selected zones and Bahir Dar city of Amhara regional state, Ethiopia. *Frontiers in Veterinary Science*. 11, (2024). DOI:<https://doi.org/10.3389/fvets.2024.1336660>.
- [18] Xin Zhao. . 2025. Remote Vital Sensing in Clinical Veterinary Medicine: A Comprehensive Review of Recent Advances, Accomplishments, Challenges, and Future Perspectives. *Animals*. Multidisciplinary Digital Publishing Institute (MDPI).
- [19] Research Exhibition in Mathematics and Computer Sciences (REMACS 5.0). 2023.
- [20] Sherwin S. Chua, John A. Alis, and Mark Joseph Oria., "WEB-Based Veterinary Clinic with

-
- e-Mail and SMS Notification," *2024 9th International Conference on Business and Industrial Research (ICBIR)*, Bangkok, Thailand, 2024, pp. 1160-1165, doi: 10.1109/ICBIR61386.2024.10875747.
- [21] Putri Kartika Ayuningtyas 2023. Performance And Functional Testing With The Black Box Testing Method. *International Journal of Progressive Sciences and Technologies*. 39, 2 (2023). <https://doi.org/10.52155/ijpsat.v39.2.5471>
- [22] What is Black Box Testing? Techniques, Types & Example: 2023. .
- [23] Healthray. (2023). 8 essential features of clinical management systems. Retrieved from <https://healthray.com/blog/clinic-management-systems/8-essential-features-of-clinical-management-systems/>
- [24] AgencyAnalytics. (2024). *Page load time KPI: Definition and importance*. Retrieved from <https://agencyanalytics.com/kpi-definitions/page-load-time>

CHAPTER 3

TECHNICAL BACKGROUND

This chapter has provided a detailed overview of the various technologies, resources, and methodologies used in developing this system. The researchers identified various technologies used in developing this system through both front-end and back-end development to provide a safe user interface to users.

3.1 Overview of Current Technologies to be Used in the System

The Kho Veterinary Clinic was previously making use of a manual system where things were slow and not very accurate. Access was also a problem for some people, considering the manual system. To deal better and more effectively, the PetKho Portal was introduced, leading to the development and use of modern web application technology and others in achieving access and efficient working for the Kho Veterinary Clinic. Overall, better and more efficient systems have been put in place to create a more efficient experience for users and clients everywhere.

3.2 Resources

The development of the PetKho Portal is a result of carefully selected hardware and software resources. Cloud-based servers are to be used to host the software, facilitating scalability, reliability, and maintenance. The portal would also support users' devices, such as a computer, tablet, or smartphone, to enable them to access the PetKho Portal. The software components are based on an assortment of open-source platforms, helping to cut expenses while receiving maximum support. The ability to proficiently work on a combination of PHP, JavaScript, CodeIgniter, and MySQL helps to accelerate development based on operational clinic needs. In addition, the cloud-based infrastructure ensures secure data storage and high system availability, while the chosen technologies enhance performance and future scalability. This overall setup makes the system dependable and easy to improve as the clinic continues to grow. It also allows clinic staff to manage records and services more efficiently with less technical difficulty.

3.2.1 Hardware Specifications

Table 1: Hardware Specification

HARDWARE	SPECIFICATIONS
Computer Type	Laptop
Processor	Intel Core i5 or Higher
Storage	256GB Solid State Drive
Memory	8GB RAM or Higher
Power Supply	65W or Higher
Input Devices	Keyboard and Mouse
Output Devices	Monitor

Hardware Specifications (Table 1): The hardware specifications show an appropriate laptop configuration that can support the development and other general computing processes. The use of a laptop as opposed to a computer or a desk shows a level of flexibility and convenience. The flexibility has always been an integral part of computer and communication systems. In the era of evolving threats, new concepts are introduced that seek to upgrade the existing security measures. These concepts are likely to provide enhanced protection mechanisms that can effectively support the safety of the stored information as well as the information that is in the process of transmission. Through this research paper, the various concepts that can provide an upgrade to information security are presented. These concepts are geared towards preventing malicious software from gaining unauthorized access to stored information and ensuring that appropriate security software performs the expected functions with integrity and appropriate standards [4]. The processor, although initially cited as an Intel Core i5 or better processor, is cited within the narrative as having been an AMD Ryzen 7 3700U processor, which is a quad-core processor with the capacity to multitask through the use of a multithreading technology.

The 8GB RAM is sufficient enough for running existing Operating Systems and development environment software. In some instances where more power is required, such as running virtual machines and data processing, an upgrade from 8GB RAM to 16GB RAM can be

ideal. The 256GB Solid State Drive has faster read and write times compared to conventional hard drives and is effective in booting the Operating System and accessing frequently used programs.

The Radeon Vega Mobile Graphics, an integrated graphics processor, is not suited for graphics-intensive tasks such as 3D modeling or heavy games but runs well in a standard development environment or for the purpose of executing multimedia content. Basic input devices are provided in the system in the form of a keyboard and a mouse. The keyboard and mouse are standard input devices required in a system but are critical in determining the system's usability or its potential for productivity within the system environment. A monitor forms the primary output device in the system, which could be the in-built display within the laptop casing or an external display screen of the system providing the necessary interface to the user in the form of a visual display. The power supply provided in the system is a 65W power supply. This power supply is enough to power the laptop while in normal usage. This ensures that the laptop is energy efficient while providing the required power to the system while in operation. Finally, the laptop runs on an operating system called Windows 11. This operating system provides the needed environment for developers as well as end-users in software development applications while providing the needed security for the system while in normal usage. The hardware provided in the system is cost-effective while still being powerful enough to carry out the needed operations in software.

3.2.2 Software Specifications

Table 2: Software Specifications

SOFTWARE TOOLS	SPECIFICATIONS
Operating System	Windows 10 or 11, MacOS 10.9 or later
Programming Language	Visual Studio Code, PHP v8.1.0
Scripting Language	HTML5, CSS3, Bootstrap, JavaScript
Framework	CodeIgniter
Database and Server	XAMPP 8.1, MySQL, phpMyAdmin 8.1
Browser	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari

Software Specifications (Table 2): The software tools listed play an important role in the development of the system as well as its deployment and subsequent use. It is compatible with both Windows 10/11 and macOS Version 10.9 and later versions. It uses the Visual Studio Code as an editor for coding purposes due to its lightweight UI and support for multiple programming languages through its extensive array of extensions. The main backend language is PHP Version 8.2.12, as it is relatively popular due to its support for server-side platforms.

For the management of the database and the server, It makes use of the XAMPP 8.1 package, which comprises Apache and MySQL and phpMyAdmin. It currently runs on the outdated MySQL 5.0.37 version, offering a serious security risk and missing many handy features available in later versions. An upgrade to the latest version of MySQL will be advisable. It supports popular browsers such as Chrome, Firefox, Edge, and Safari. During development, developers can make use of XAMPP for easy internet-based deployment. In this setup, development and testing become flexible and accessible.

Besides, with support for various versions of standard web browser applications, the system is also allowed to maintain very high compatibility across different user environments. This accommodates a wide range of devices, from desktop computers, laptops, and tablets to smartphones, ensuring that not too many display or usability flaws would result due to different platforms and browser versions. On top of this, users could access the PetKho Portal in convenience and comfort, regardless of their preferred device or environment.

Also, regular updating of tools, libraries, and frameworks used during development greatly improves the security, reliability, and feasibility of the system. The system stays up-to-date with the latest technology, eliminates the possibility of technology incompatibility after some time, and provides the best possible performance and functionality for the system. Above all, these best programming practices put together guarantee the development of an environment with the best possible efficiency for the development of the PetKho Portal.

Furthermore, the system follows standard web development practices such as modular coding, proper version control, and environment-based configuration to ensure maintainability and scalability. The use of structured coding conventions allows developers to easily understand, modify, and extend system functionalities as future requirements arise.

3.2.3 Program Specifications

Table 3: Functional Requirements

FUNCTIONAL REQUIREMENT	DESCRIPTION
Customer / Pet Owner Management	They can register, log in, book and manage appointments, access and download pet records, and submit adoption applications.
Appointment & Referral Management	The system lets pet owners book appointments, and vets manage schedules and referrals.
Pet Records Management	Vets and assistants can manage and view detailed pet records, including medical history, treatments, and vaccinations.
Invoice & Billing Management	The system lets sub-branch staff create invoices and main vets approve them.
Stock & Inventory Management	Sub-branch staff manage inventory, while main vets monitor stock requests.
Adoption Process Management	The system facilitates the pet adoption process. Pet owners can submit adoption applications, which can then be reviewed and managed by clinic staff.
Reporting & Analytics	The system generates reports for each role, while the Veterinary Owner accesses all operational and financial reports.
User, Staff & Branch Management	The Veterinary Owner (Admin) has full control over managing user accounts, staff details, and information for all branches (Main, Pili, Nabua).
Notification System	The system sends automated notifications for appointments (reminders, confirmations), referral status updates, and other relevant communications to users and staff.

Table 3 presents the core functional requirements of the system, highlighting the key features and processes that support veterinary operations, user management, and service delivery across all roles and branches.

Flowchart

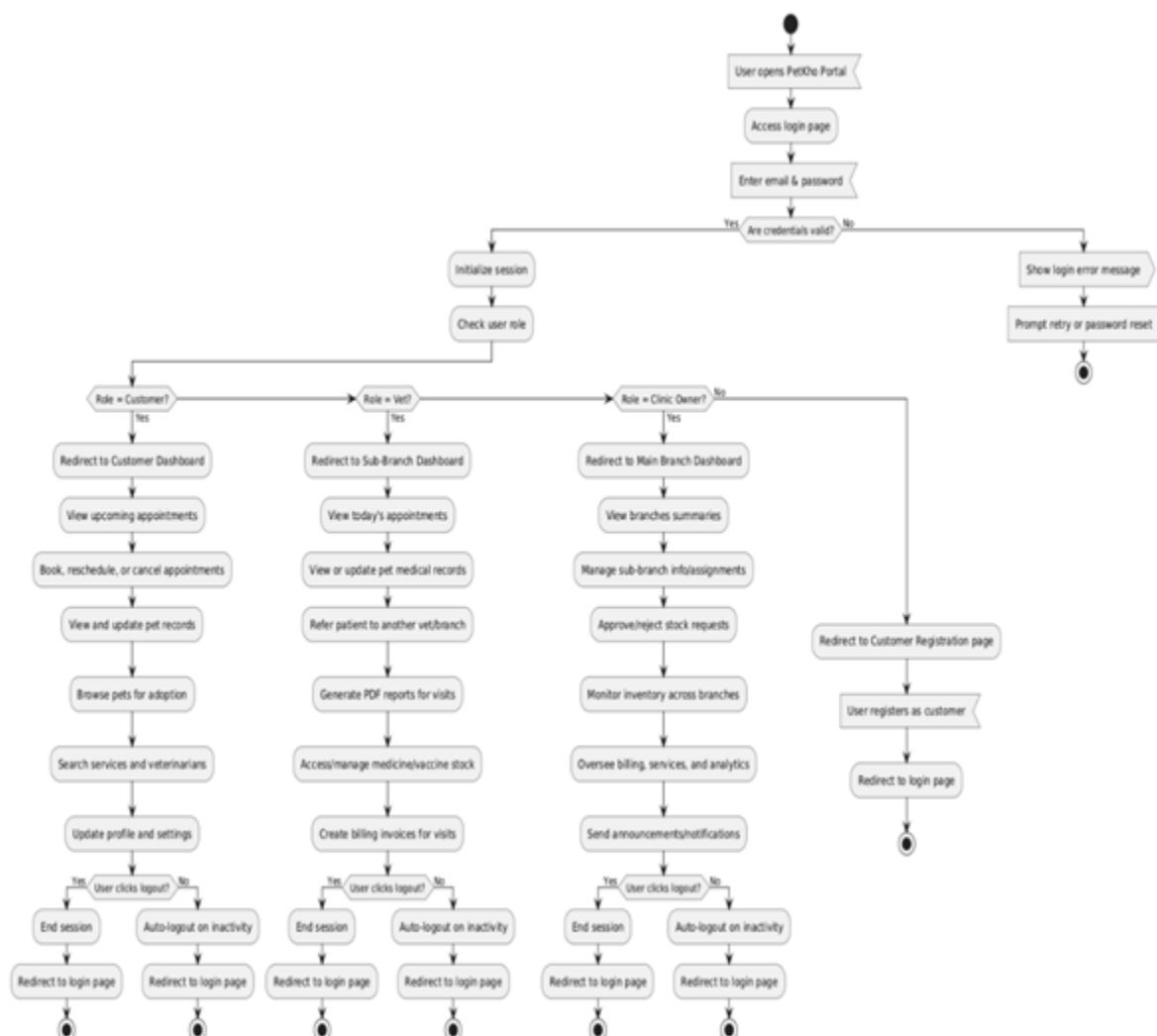


Figure 1: Flowchart

The PetKho Portal is a veterinary management system based on roles, focusing extensively on access and functionality through a safe and well-structured process for logging in. It serves three basic roles: Customers, Veterinarians, and Clinic Owners. The logging process starts with a safe interface where users enter their details and undergo verification based on data format and

completion. The system then processes the details to find matching credentials, paralleling a hashing process to ensure safety during verification, and provides adequate error responses for failed attempts, although account locking could also occur to prevent brute force.

After the authentication is successful, a secure session is activated that stores important information regarding the user, their roles, user ID, and the time the user logged into the system. The user is redirected to their own dashboard based on their roles after gaining access to the system. The customers are allowed to access various facilities such as booking, pet information, adoption, and account management. Their interface is easy to use and friendly for a user with little knowledge regarding technology.

Veterinarians have access to a sub-branch-veterinarians-focused dashboard that caters to operations such as managing appointments, noting patient notes, creating referrals, managing medicine inventories, as well as creating invoices to clients. At the same time, the Clinic Owners have access to the main branch dashboard that allows them to manage activities such as stock approval, inventory management, as well as managing stock referrals and services across multiple branches.

If the system does not find an assigned role for a user or if the user is new, it redirects to a registration form that allows only public registrations for the Customer role. The rest of the roles, like Veterinarian and Clinic Owner, are managed internally to keep the integrity and control of the system. With this username and password, users can log in to the platform once they have signed up. Session management entails a logout button and automatic timeout after periods of inactivity that prevent unauthorized access due to unattended sessions.

Security is an integral part of the login system that includes best practices from the industry, such as encrypted HTTPS communication, password hashing with bcrypt or Argon2, secure cookies, and session rotation. In addition, all login activities are logged for auditing

purposes, while also serving threat detection. Role-based access ensures that users interact only with features relevant to their role, hence simplifying navigation while reducing risk.

It has been designed to be scalable and extensible, while the module structure shall allow addition of new roles or features easily without recoding. Future enhancements will add two-factor authentication, biometric login capability, social media authentication, and Single Sign-On to help future application expansions. These will reinforce the security in the system and safeguard user interests. Overall, this future-ready architecture will ensure that the PetKho Portal remains in step with advances in technology and the evolving needs of the veterinary field.

Moreover, the login and role-based access mechanism contributes significantly to the overall usability and performance of the PetKho Portal. By loading only role-relevant modules and data upon successful authentication, the system optimizes resource usage and reduces unnecessary processing overhead. This approach not only enhances system responsiveness but also improves user experience by presenting a clean and focused interface tailored to each role. As a result, users are able to perform their tasks more efficiently, reinforcing the system's reliability as a comprehensive and secure veterinary management solution.

In summary, the system is designed to be scalable and flexible, with a module structure that allows new roles or features to be added easily without the need for major recoding. Future enhancements may include two-factor authentication, biometric login, social media authentication, and Single Sign-On to support system growth and improve user security. Overall, this future-ready architecture ensures that the PetKho Portal can adapt to technological changes and continue to meet the evolving needs of the veterinary field. In addition, these improvements will help protect user data and build greater trust among clinic staff and clients. The architecture also supports smoother system updates with minimal disruption to daily clinic operations. This ensures long-term reliability and continuous improvement of the portal. Ultimately, the system is built to provide a secure, stable, and user-friendly platform for future use.

System Architecture Diagram

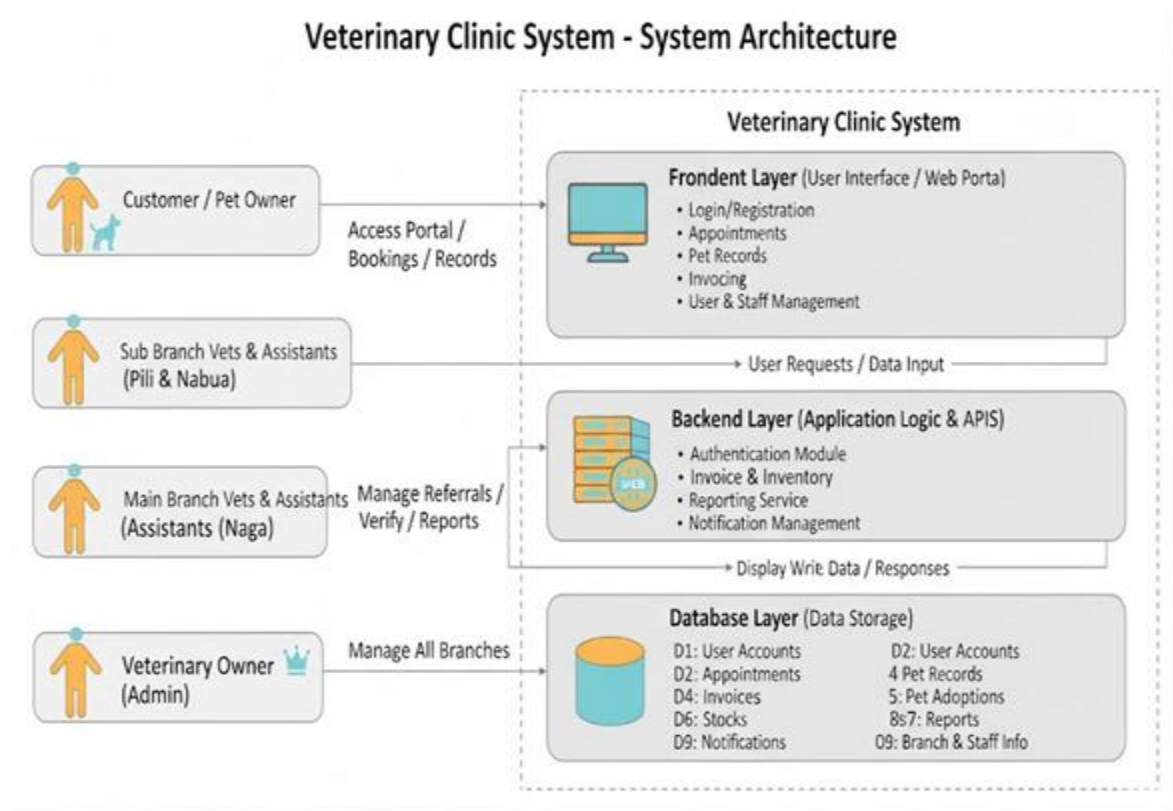


Figure 2: System Architecture

The Security and Authentication Architecture of the PetKho Portal consists of multiple layers, which is intended to maintain the confidentiality, integrity, and limited access to veterinary, client, and operational data. However, this security feature blocks access to the system by any unauthorized person while simultaneously delivering reliable service to all the legitimate users, including the Pet Owners, Vet Assistants, Vet Residents, and the Veterinary Owner. Application of security controls at different layers by the system drastically reduces the probability of data theft, abuse of rights, and unauthorized operations in the system.

Starting with the user layer and the user presentation layer, the access to the PetKHO Portal uses access via secure web interfaces. The user login and registration details are sent to the servers by users using the HTTPS protocol for the encryption of the information being sent, so the data is

safe. There is also the use of client-side validations to prevent the input of incorrect information into the system, as well as the use of timeout to reduce the risk of unauthorized access due to user sessions.

The access control layer is where an application connects to an API Gateway; here, we have the initial security control in the backend of the system. It builds up all incoming requests and also attempts to eliminate any improper or suspicious request attempts to this backend system. Moreover, it also controls brute-force attack attempts by limiting how fast an unauthorized entity could send login attempts or other requests to this backend system. Only approved and properly formatted requests are received by authentication services.

A centralized Authentication Service is in charge of validating the information provided by the users. This information is then checked against a database consisting of the securely stored information pertaining to the user accounts. Use of the hashing technique in passwords guarantees the security of the user information even in the unlikely event the database is compromised in some manner. Once the user information has been validated by the authentication service, a secure session is created for the user. Once the user has been validated, the user role is recognized by the system, based on which the access control is provided for the entire portal.

Amidst this scenario, the system utilizes a Role-Based Access Control (RBAC) system in which authorization occurs through a specific role. This indicates that the users are restricted to the availability of the modules or the data based on their roles within the system. For example, the Pet Owner can only carry out their individual or client-related responsibilities, the Vet Assistant or Vet Residents can have their operations/medical-related roles fulfilled within the system, while the Veterinary Owner has the admin privileges in the system. Essentially, this defined system prevents the elevation of privilege in the system in a meaningful manner.

The access control of all available application modules of the PetKHO portal is restricted

or restricted based on authentication of users regarding their roles to ensure that only certain users are allowed to access and execute certain operations. For security reasons, certain data of similar systems is maintained secure in certain databases that allow only restricted reading and writing to maintain integrity and prevent unauthorized access and changes. Moreover, some audit trails are maintained to trace some of the most critical operations like changes and attempts to gain certain access and administer certain functions of this system and, at the same time, security alerts are also sent to administrators regarding any suspicious or critical activities that have occurred or are taking place. These security measures combined help to form a supporting basis for security that allows the system to maintain a secure and reliable environment.

In addition, the system incorporates a comprehensive set of defensive mechanisms at both the infrastructure and data layers to further reinforce overall security, reliability, and system integrity. These measures include secure server configurations with updated security patches, firewall protection, and controlled network access to minimize exposure to potential threats. Database access is strictly managed through role-based privileges, ensuring that only authorized users can view, modify, or manage sensitive records. Regular and automated data backups are performed to prevent data loss and to allow quick recovery in case of system failure, human error, or unforeseen incidents. The system is also designed to protect against common web vulnerabilities such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF) by using input validation, prepared statements, token-based request verification, and secure coding practices. These techniques help ensure that user inputs are properly checked and that malicious commands cannot be executed within the system. Error handling and logging mechanisms are carefully implemented to prevent the exposure of sensitive system details, such as server paths or database structures, while still capturing enough information for administrators to diagnose issues, track system behavior, and perform effective troubleshooting. Logs are securely stored and monitored to identify suspicious activities and support ongoing system maintenance. Through these combined

safeguards, the PetKho Portal not only provides strong protection against external attacks and internal misuse but also ensures long-term system stability, data accuracy, and operational continuity. By maintaining a secure and well-monitored environment, the system builds confidence among users, protects sensitive information.

Security/Authentication Measures

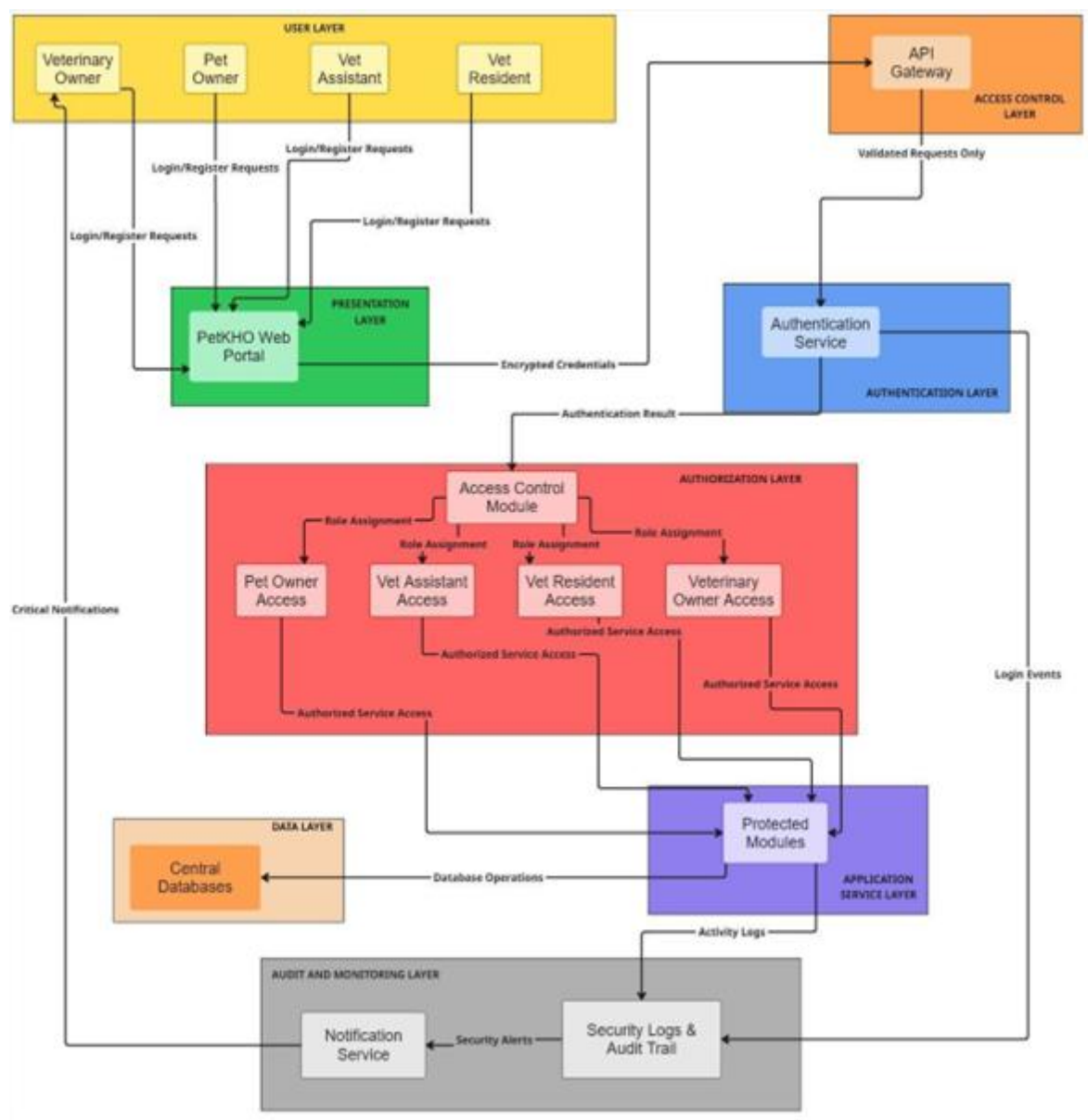


Figure 3: Security/Authentication Measures

The Security and Authentication Architecture of the PetKHO Portal consists of multiple layers, which is intended to maintain the confidentiality, integrity, and limited access to veterinary, client, and operational data. However, this security feature blocks access to the system by any unauthorized person while simultaneously delivering reliable service to all the legitimate users, including the Pet Owners, Vet Assistants, Vet Residents, and the Veterinary Owner. Application of security controls at different layers by the system drastically reduces the probability of data theft, abuse of rights, and unauthorized operations in the system.

Starting with the user layer and the user presentation layer, the access to the PetKHO Portal uses access via secure web interfaces. The user login and registration details are sent to the servers by users using the HTTPS protocol for the encryption of the information being sent, so the data is safe. There is also the use of client-side validations to prevent the input of incorrect information into the system, as well as the use of timeout to reduce the risk of unauthorized access due to user sessions.

The access control layer is where an application connects to an API Gateway; here, we have the initial security control in the backend of the system. It builds up all incoming requests and also attempts to eliminate any improper or suspicious request attempts to this backend system. Moreover, it also controls brute-force attack attempts by limiting how fast an unauthorized entity could send login attempts or other requests to this backend system. Only approved and properly formatted requests are received by authentication services.

A centralized Authentication Service is in charge of validating the information provided by the users. This information is then checked against a database consisting of the securely stored information pertaining to the user accounts. Use of the hashing technique in passwords guarantees the security of the user information even in the unlikely event the database is compromised in some manner. Once the user information has been validated by the authentication service, a secure session is created for the user. Once the user has been validated, the user role is recognized by the

system, based on which the access control is provided for the entire portal.

Amidst this scenario, the system utilizes a Role-Based Access Control (RBAC) system in which authorization occurs through a specific role. This indicates that the users are restricted to the availability of the modules or the data based on their roles within the system. For example, the Pet Owner can only carry out their individual or client-related responsibilities, the Vet Assistant or Vet Residents can have their operations/medical-related roles fulfilled within the system, while the Veterinary Owner has the admin privileges in the system. Essentially, this defined system prevents the elevation of privilege in the system in a meaningful manner.

The access control of all available application modules of the PetKHO portal is restricted or restricted based on authentication of users regarding their roles to ensure that only certain users are allowed to access and execute certain operations. For security reasons, certain data of similar systems is maintained secure in certain databases that allow only restricted reading and writing to maintain integrity and prevent unauthorized access and changes. Moreover, some audit trails are maintained to trace some of the most critical operations like changes and attempts to gain certain access and administer certain functions of this system and, at the same time, security alerts are also sent to administrators regarding any suspicious or critical activities that have occurred or are taking place. These security measures combined help to form a supporting basis for security that allows the system to maintain a secure and reliable environment. Regular monitoring and timely updates of security protocols ensure that potential vulnerabilities are addressed promptly. This proactive approach helps maintain user trust and protects sensitive veterinary and client data from emerging threats.

3.2.4 Programming Environment

This section presents a comprehensive discussion regarding the programming language and the existing technology used to develop the front end as well as the back end of the PetKho Portal. The technology used was evaluated based on a set of factors that include the proficiency of

the developers, community support, its capability to scale, and the ease with Which technology or programming language is integrable. Through the application of the technology or the programming language used, the developers ensured that the PetKho Portal is based on a system design that is articulated, secure, reliable, and maintainable. Nonetheless, the application of technology or programming language enabled the developers to guarantee that the PetKho Portal is adaptive to emerging technology as well as the developments that are forthcoming in the future.

Front End

Table 4: Front End

TECHNOLOGY CATEGORY	SPECIFICATION CATEGORY	VERSION
Programming Language	HTML, CSS, JavaScript	ES2023
Framework	CodeIgniter	15.5.4
CSS Framework	Bootstrap	5.3

Table 4 illustrates some of the technologies utilized in crafting the front end of this system. Note that this user interface has been developed using a mix of PHP, HTML, and CSS version 8.4, which has provided a dynamic, flexible, and user-friendly environment. Further, to promote enhanced functionality and neat and well-organized code, this project utilizes a mix of the CodeIgniter Framework version 4 to encourage easy integration of both the front end and back end of this application. Lastly, this application employs Bootstrap version 5.3 to ensure a modern look and user-friendly feel in all pages of this application.

Additionally, the system incorporates Bootstrap version 5.3, a powerful front-end toolkit that ensures a modern, professional, and visually appealing interface. Bootstrap provides pre-designed components, responsive grid systems, and interactive elements that improve user experience and minimize development effort while maintaining consistency across all pages. Its built-in responsiveness ensures that the application adapts seamlessly to different screen sizes and devices, making navigation and interaction easier for users of all technical backgrounds.

Back End

Table 5: Back End

TECHNOLOGY CATEGORY	SPECIFICATION CATEGORY	VERSION
Programming Language	JavaScript	ES2023
Framework	CodeIgniter	15.5.4
Database	MySQL	9.2.0

Table 5 highlights the technologies necessary in the creation of the backend in the system's development process. The backend process in the system handles essential system operations and has been developed with the use of PHP version 8.2.12, which has been crucial in ensuring a solid foundation in system development, thus enhancing efficiency in system functionality.

Another crucial aspect in system development was the use of the CodeIgniter Framework, version 4, which has been necessary in ensuring a lightweight yet powerful approach in managing backend system operations. Its MVC (Model-View-Controller) architecture promotes organized coding, easier maintenance, and faster development of new features within the PetKHO Portal.

3.3 Testing Plan

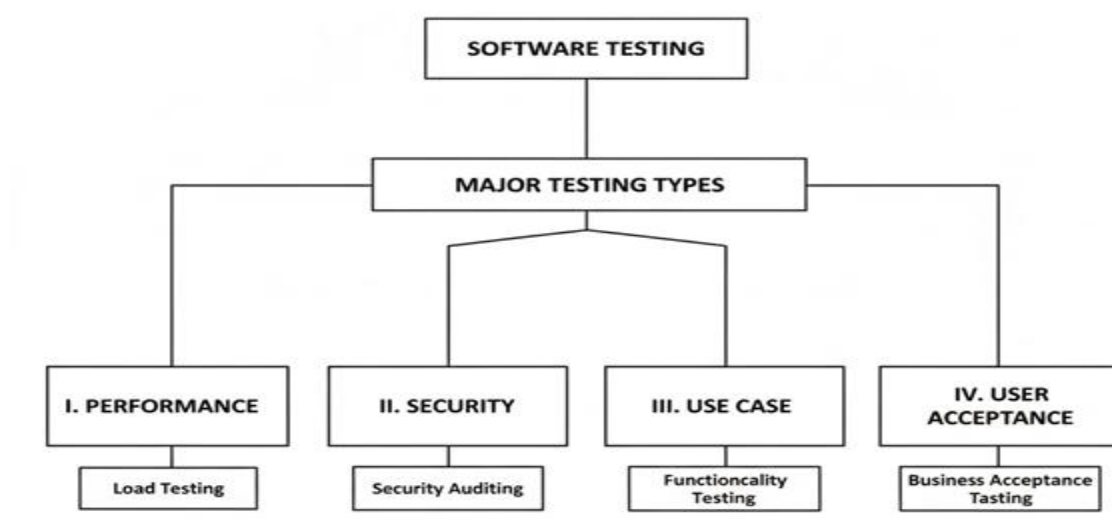


Figure 4: Hierarchy Diagram

3.3.1 Types of Testing

Additionally, MySQL 9.2.0 is highly significant to the management of backend functionality and thus increase efficiency in system development as it was crucial to managing system databases in the backend process.

In this project, various types of Blackbox Testing will be performed in order for the system to be in a good working condition and suit the needs of its users. First is UAT, which checks if the system meets the needs of the users. This involves key feature testing using alpha testing, which identifies early bugs, and the validation of important business rules to ensure that the system should act as expected. Still, there is system-level UAT testing, which considers the whole software as a single unit and confirms if it is ready to go into real usage. These testing processes collectively help ensure that the system is reliable, functional, and provides a smooth experience for all users before deployment.

The second is Security Testing. This testing is to defend the system from various kinds of threats and unauthorized access. It involves the scanning of the system for known vulnerabilities and performing an access control test so that only authorized users can gain access to sensitive information. Further, automated system security testing will also be performed to model various attacks and verify the system's defenses with no need for human effort.

Next is Performance Testing, which checks how well the system handles different amounts of work. Load testing is done to see if the system remains fast and stable under normal user traffic, while stress testing pushes it beyond normal limits to find out where it might fail. System performance testing uses tools to automatically repeat these tests to find any bottlenecks and ensure the system can handle expected and unexpected loads.

Finally, Use Case Testing will be carried out by simulating real user scenarios. This includes scenario testing that covers full workflows and happy path testing which focuses on the

most common, error-free user actions. The system use case validation automates these tests to ensure that key processes run smoothly for all types of users.

Any issues found during these tests will inform development priorities for bug fixes and feature enhancements.

3.3.2 Testing Tools and Framework

In this project, the testing shall not be done through automated software or programming-based testing tools, but used with a systematic and standard-based evaluation using a questionnaire tool. This method is very applicable for assessing the functionality, usability, performance, and security of the system from the client's perspective without the need to interface directly with the source code. The questionnaire shall be designed based on internationally accepted standards, mainly ISO/IEC 25010: System and Software Quality Requirements and Evaluation, ISO/IEC 9126, and ISO/IEC 27001, which define the key quality attributes in software systems. This approach tends to provide more detailed feedback from a stakeholder while reflecting realistic perceptions about the quality of the system.

The questionnaire will be divided into four major Black Box Testing types: User Acceptance Testing, Security Testing, Performance Testing, and Use Case Testing. The questions under each part of the questionnaire will focus on particular Black Box Testing domains and will be framed to assess the extent to which the system complies with the desired quality with respect to these Black Box Testing domains. Ask the system to perform well with respect to these, so it behaves well, provides security to the end users, and the system's required functionality relates to the needs of the end users. In such cases, the evaluation of the end users is completely dependent on Black Box Testing.

In the study cited above, data-driven strategies were used to reveal the hidden patterns lying within the chaotic raw data, leading to the derivation of the PDE models with the help of such

chaotic observations. Neural networks were used to figure out the characteristics of the system, such as the breaking of symmetry itself [3].

The table below represents a detailed outline of the various testing styles; the important areas tested through the test questionnaire and the exact ISO standard used as a basis of designing the evaluation criteria. This table acts as a starting point for designing the evaluation form for the clients, so the parameters precisely conform to internationally accepted software quality norms. By using the guidelines from these iso standards, the evaluation process is natural, precise, and comprehensive, so the quality of the entire software could be measured. This ensures the PetKho Portal conforms to world quality standards of dependability, usability, and user satisfaction during the entire test phase.

Table 6: ISO-Aligned Questionnaire Testing Framework

TESTING TYPE	FOCUS AREAS IN QUESTIONNAIRE	ISO STANDARD REFERENCE
User Acceptance Testing	System usability, task completion, compliance with requirements	ISO/IEC 25010
Security Testing	Data protection, user authentication, access restrictions	ISO/IEC 27001
Performance Testing	Response time, system stability under usage, reliability	ISO/IEC 25010
Use Case Testing	Workflow accuracy, user scenario handling, expected output	ISO/IEC 25010

The User Acceptance Testing portion of the questionnaire focus on whether the system is easy to use, meets the specific needs of the user, and complies with functional and business requirements. It is generally done by an end user or the client itself to verify/accept the software system before the software application is moved to the production environment. UAT is done in the final phase of testing after functional, integration, and system testing is performed [6]. Areas such as navigation clarity, successful execution of tasks, as well as user satisfaction, are assessed in this section. The process in this section has been influenced by the ISO 25010 standard, which highlights areas such as usability, functionality, as well as user experience. The characterization of

the ISO/IEC 25010 standard, as a quality model, was modified to define the platform's design, as well as its architecture/user interface, as discussed in [7]. The feedback from this section is very crucial for validating whether the system matches the day-to-day workflow and expectations for its target users.

The Security Testing questions are based on the standards of ISO/IEC 27001 and ISO/IEC 15408. This is essential in determining the level of security of the system in the context of the confidentiality, integrity, and availability of the data in the system. This set of questions examines whether the system can be protected from unauthorized access and whether the users are adequately authenticated in the system, thus greatly guarding against data breaches in the future.

Performance testing, according to ISO/IEC 9126, includes such details as the response time of the system, the reliability of the system under normal and heavy usage, and the performance consistency. The questionnaire needs the client to evaluate the loading time of pages or features, if there are slowdowns, or if the system crashes while loading or after loading the program. Traditional performance testing considers the input to the program as the single factor acting from the outside. This method completely neglects the performance factors of the program's configuration. This study aims to generate interest in the performance testing of configurable software systems by providing the answers to three questions regarding performance testing [2]. These insights can identify how scalable and efficient a system is. They can ensure that a system can operate under realistic usage without compromising the user experience.

Lastly, Use Case Testing is concerned with the assessment of whether or not a system can support a given workflow or common tasks with a high level of accuracy. The design phase of the system's processes is introduced by the concept of a use case [5]. This involves ensuring if the use cases result in the expected results when followed step-by-step, and if the error handling is clear and user-friendly. The questions themselves are aligned with the ISO/IEC 25010's functional correctness and completeness of processes, considering the real-world functionality of the system

to deliver the desired results while performing tasks.

By following such a system while conducting the process, it becomes ensured that a proper testing of the system takes place in a standard way, without which, there might be some complexities while getting a proper involvement from the clients regarding such a system as such, while ensuring that the end product should remain functional, usable, secure, and reliable in a real-world.

Notes

- [1] Finland Oy Justus Jaskari, W. *Ensuring Customer Value through Comprehensive Reporting.*
- [2] Xiaojun Han. 2021. A Study of Performance Testing in Configurable Software Systems. *Journal of Software Engineering and Applications.* 14, 09 (2021). DOI:<https://doi.org/10.4236/jsea.2021.149028>.
- [3] Dawid W. Sroczynski 2025. From disorganized data to emergent dynamic models: Questionnaires to partial differential equations. *PNAS Nexus.* 4, 2 (Feb. 2025). DOI:<https://doi.org/10.1093/pnasnexus/pgaf018>.
- [4] Stanimir Stoyanov. 2025. Survey on Hardware Components Providing Information Security. *AIP Conference Proceedings* (Mar. 2025).
- [5] Khairul Fahmi Syah and Sukirman Sukirman. 2023. Sistem Peminjaman Buku Perpustakaan dengan Identifikasi Kode Quick Response (QR). *Edumatic: Jurnal Pendidikan Informatika.* 7, 1 (2023). DOI:<https://doi.org/10.29408/edumatic.v7i1.13372>.
- [6] What is User Acceptance Testing: 2023. .
- [7] Ana Maria Nogueira da Cruz 2020. An open web-based platform for enhancing the visibility of Brazilian research. *Proceedings - 2020 International Conference on Computational Science and Computational Intelligence, CSCI 2020* (2020).

CHAPTER 4

METHODOLOGY, RESULTS AND DISCUSSION

This chapter explains the process of development of the proposed Petkho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada and assesses the viability of the efficient computerization process. To achieve this, the Agile method that focuses on flexibility, teamwork, and iteration is used to outline the whole process of the system development from the design to review to ensure that the design was effective based on the operation of the clinical activities.

4.1 Planning

This phase focuses on outlining the objectives, scope, and requirements of the project, the cornerstone to successful development of the Petkho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada. In this phase, the project developers reviewed the operations of the clinic to identify improvement opportunities in the processes through an assessment of its operations and records management systems. The research is guided by the agile design thinking methodology for the purpose of meeting the requirements of the end user, where the quality attributes workshop method has been acknowledged in the prototyping phase of the design in a bid to ensure the input of the different stakeholders to a guarantee of quality attribute control in the project[6]. By using interviews and observations, the essential information was collected which helped define the Pre-Questions involved in the functional and non-functional Pre-requirements of the system, so as to ensure the features were a perfect fit for the actual needs of the clinic. The goals, objectives, resources, and criteria were also set during the planning phase, which ensured the right path was set out for the entire designing process. By providing a plan, the time involved in the entire undertaking was kept organized as well. This structured approach during the

planning phase laid a strong foundation for the subsequent design and development stages, ensuring that the PetKho Portal would effectively address the clinic's operational challenges.

4.1.1 Convenient Sampling

Convenient sampling is commonly classified as one of the non-probability sampling methods. With this type of sampling, the researcher selects the sample on the basis of their convenience and accessibility of the respondents. Within the framework of the PetKho Portal system, the researcher was able to utilize the application of the convenient sampling technique to ensure the collection of clients, veterinary officers, and administrators who were accessible to the research. Although the application of the convenience sampling technique is advantageous due to the low cost of the research as well as the simplicity of the collection of the relevant information, the technique is restricted in the sense that the results obtained from the application of the technique are not sufficient to represent the results of the potential users of the system.

4.1.2 Respondents

Table 8 shows the distribution of the respondents who participated in the evaluation of the PetKho Portal system. These are a diverse group of users who participate in the daily operations and services involving Kho Veterinary Clinic, which ensued that the views of both clients and professionals would be represented. The participants were mainly composed of pet owners from the different branches of the clinic, such as 19 from the Naga Main Branch, 11 from the Pili Sub Branch, and 25 from the Nabua Sub Branch, and the researchers could gather the feedback of clients who frequently interact with veterinary services in various locations. Veterinary staff, assistants, and professionals, represented by two from each branch, together with the veterinary owner/director responding to the functionality of the system, its integration to the workflow, and its operational efficiency, were also included. Additionally, five IT experts were used to assess the technology aspect of the system, including user interface, security, reliability, and overall performance. The sampling method used was a convenient sampling technique, where respondents were picked based on their accessibility to participate in the research, making the research

convenient to carry out. This sampling technique is essential in the evaluation of a system because the researchers were able to get feedback immediately from users, who are the clients, the vet staff, the management, as well as the overall feedback from the IT team, ensuring that the evaluation was well-rounded.

Table 7: Respondents

RESPONDENTS	FREQUENCY
Naga Main Branch Pet Owners	19
Pili Sub Branch Pet Owners	11
Nabua Sub Branch Pet Owners	25
Naga Main Branch Vet and Assistant	2
Pili Sub Branch Vet and Assistant	2
Nabua Sub Branch Vet and Assistant	2
Vet Owner	1
IT Experts	5
Total	67

In resolving the operational problems raised by the Kho Veterinary Clinic, the researcher developed the PetKho Portal, which is intended to be used as a web-based application in the delivery of veterinary care services including the monitoring of these services. Based on the ISO/IEC 25010 software quality standards, the following areas of quality were considered: functional suitability, performance efficiency, usability, security, and maintainability. In the first test, the use case test, the system was tested to ascertain whether it meets the needs and expectations of the users in an operating clinic setup. The system was further tested to see whether it protects critical information and prevents unauthorized use in the second test, referred to as the security test. The performance test was carried out to test the performance of the system in handling normal operating activities. In the use case test, it was determined that all the necessary activities being

carried out in the system, including scheduling appointments, keeping pet records, and producing bills, were functioning properly.

4.1.2 Scaling

Scaling is the process where something is measured, ranked, or categorized through a specified criterion. It is normally used to illustrate performance, quality, and effectiveness in an orderly manner.

Likert Scale

Each number has a meaning: 1 – Poor / Strongly Disagree, 2 – Fair / Disagree, 3 – Good / Agree, and 4 – Excellent / Strongly Agree. This scale makes it easy for testers or users to provide feedback in a structured way, depicting areas that are working effectively as opposed to areas that need improvement.

Table 8 below presents the Performance Testing Scale that will be employed based on a Likert scale with values from 1 to 4. Performance testing tests the performance of the systems by checking how efficiently it works, how fast it runs, whether it supports heavy use without slowing down or giving errors to the users.

Table 8: Performance Testing Scale

RATING	SCALE	DESCRIPTION
1.00 - 1.75	Strongly Disagree	System is very slow, frequent delays
1.76 - 2.50	Disagree	Noticeable slowdowns affect user experience
2.51 - 3.25	Agree	Acceptable speed, minor delays
3.26 - 4.00	Strongly Agree	Very fast, smooth, no delays even under heavy load requirements.

This scale will make it simple to assess the efficiency of the system in operation given different conditions. A score of 1 implies that the system takes a long time, making it frustrating to use, thus creating significant delays in operation when in use by the clients. A score of 2 implies that the system works, although some delays are realized during its operation in the system environment. A score of 3 implies that the system works reasonably, with some minor delays that are insignificant in system operation, thus creating a better user interface experience. A score of 4 implies that the system operates at a high pace, enabling a smooth system operation even when the system has been subjected to significant system usage by clients in the system environment. Table 10 shows the Security Testing Scale using a 1–4 Likert scale. Security testing evaluates how well the system protects data and prevents threats, such as unauthorized access, hacking, or data loss. The scale allows testers to rate the system’s security based on what they observe during testing, making it easier to identify strengths and weaknesses and ensure the system is safe and reliable for users.

Table 9: Security Testing Scale

RATING	SCALE	DESCRIPTION
1.00 - 1.75	Strongly Disagree	The system is not safe and has serious security issues.
1.76 - 2.50	Disagree	The system has some weaknesses that need fixing.
2.51 - 3.25	Agree	The system is mostly safe with minor risks.
3.26 - 4.00	Strongly Agree	The system is fully secure and protects data well.

A score of 1 would mean that it was an insecure system that could pose some threats to users’ information and personal details, meaning that it would be unsafe for use by everyone. A score of 2 would indicate that it had some security measures to ensure that users' information was

protected and that it would be safe to use to some extent. However, it would also indicate that there were some security concerns that had to be worked on or rectified to ensure that users would not have some problems with it. A score of 3 would indicate that it was a secure system that had some small security concerns that had nothing significantly to do with its safety. A score of 4 would indicate that it was a highly secure system with the capacity to ensure that users' information remains secure and that it would also ensure that users have a safe experience when interacting with it or when it is used for operations in their daily lives.

Table 10 demonstrates the Use Case Testing Scale that utilizes a Likert Scale ranging from 1-4. Use case testing normally aims at ascertaining whether all the features in the system are functioning as anticipated and whether the users are able to carry out the desired functions smoothly and easily. Through the use of the Likert Scale, it is possible to easily assess the level of support that the system offers to the normal operations of the users.

Table 10: Use Case Testing Scale

RATING	SCALE	DESCRIPTION
1.00 - 1.75	Strongly Disagree	Many tasks fail; the system does not follow expected steps.
1.76 - 2.50	Disagree	Some tasks fail; workflow is inconsistent.
2.51 - 3.25	Agree	Most tasks work; there are minor issues in the steps.
3.26 - 4.00	Strongly Agree	All tasks work smoothly; the system follows expected steps perfectly.

A score of 1 indicates that a lot of the features or steps are not working, making it very frustrating and difficult for the user to efficiently execute their tasks or functions. A score of 2 shows a mix-up in the tasks executed by the system. This is based on the assumption that the system

is not efficient and that the mix-up could confuse the user and restrict the flow of their activities or tasks. A score of 3 reflects that most tasks are working as they ought to, except for a few minor glitches that do not affect the performance and flow of activities or tasks executed by the user. A score of 4 reflects that all the features or the processes are working as they ought to, with the system fully satisfying the needs of the user by executing tasks or activities very easily and efficiently.

Table 11 presents the User Acceptance Testing Scale using the Likert Scale with a rating of 1-4. UAT testing checks if the system meets the expectations of the intended user group in terms of its usage or utility. The scale here helps the user to rate their satisfaction level with the system, indicating the potential upgrades required for the system or the areas in the system that are functioning exceptionally well.

Table 11: User Acceptance Testing Scale

RATING	SCALE	DESCRIPTION
1.00 - 1.75	Strongly Disagree	Users are unhappy; the system does not meet their needs.
1.76 - 2.50	Disagree	Users see major problems; system needs many improvements.
2.51 - 3.25	Agree	Users are mostly satisfied; minor fixes needed.
3.26 - 4.00	Strongly Agree	Users are fully satisfied; system meets all needs and expectations.

A score of 1 means users are dissatisfied, and the system does not meet their needs. It is difficult to use. A score of 2 means there is a major problem, and many improvements need to be incorporated for the system to function well and meet the needs and requirements of customers/users. A score of 3 means users are satisfied, and small improvements need to be incorporated for the system to function well and meet the needs and requirements of

users/customers. A score of 4 means users/customers are satisfied, and the system meets their requirements and expectations. It is working well and is effective in serving users/customers regardless of the purpose for which it was created.

4.1.3 Statistical Test

In evaluating the acquired data, the researcher provided appropriate analysis and interpretation by utilizing several tools that would measure the efficiency of the overall system. These tools were utilized by the researcher to measure the accuracy, consistency, and reliability of the acquired results from the evaluation of the proposed system. A systematic approach that could measure the acquired qualitative results and overall opinions provided by the customers was obtained from the applications of the provided computations. Through the applications of the provided tools and computations, the researcher managed to evaluate the overall level of the new developed system in comparison to the current. The formulas used in this study are as follows:

The Weighted Mean was utilized when the various respondent groups had different sample sizes, such as the Naga Main, Pili Sub, and Nabua Sub branches. This statistical technique was seen to present a more accurate view of the general evaluation, especially with regard to giving much importance to those with larger samples.

Using this process, it was possible to multiply each group's mean score by their corresponding number of respondents to obtain a balanced form of a general average. This process avoids any form of bias that would have occurred had all the groups received equal value regardless of their number of participants.

$$\bar{X}_w = \frac{\sum(w_i x_i)}{\sum w_i}$$

Where:

X_w = Weighted Mean

w_i = Number of respondents per group (weight)

x_i = Mean score of each group

4.1.4 Identified Problems

It is to be noted that prior to preparing the project report for “PetKho Portal: A Smart System for Veterinary Care and Monitoring of Kho Veterinary Clinic Rinconada,” an extensive research regarding the current operations of Kho veterinary clinic was done by the research team members who gathered extensive information through an in-depth interview with veterinarians, employees, and administrative personnel involved with the operations of Kho veterinary clinic. Additionally, an extensive observation regarding the operations of the veterinary clinic was done by the research members. It was noted that some challenges were observed by the research members during the operations regarding maintaining accurate data regarding clients’ pets, managing appointments efficiently, tracking the subsequent treatments for the animals provided by the veterinary clinic, and maintaining proper relationships with the clients who brought their animals to the veterinary clinic for further treatments. These findings highlighted the need for a more organized and automated system to reduce human error and improve data accuracy. The research also indicated that a role-based management system could streamline workflow and enhance communication among staff members. Consequently, the PetKho Portal was designed to address these challenges by integrating appointment scheduling, pet records, and client management into a single, efficient platform.

These identified challenges have shaped the purpose and objectives of the PetKho Portal and have informed the design and development of the most important and significant features incorporated in the system. The main purpose was to simplify day-to-day running, reduce errors, and improve communication between employees and clients to enhance the quality of the services delivered. Table 12 shows a list of the overall challenges identified by the veterinary clinic, their descriptions, and thus a clear understanding of the areas the PetKho Portal addresses to resolve these shortcomings and improve their performance. By analyzing these challenges, the development team was able to prioritize features that would deliver the greatest impact. The system

was designed to automate repetitive tasks such as appointment tracking and pet record management. It also includes role-based access to ensure that staff members can efficiently perform their duties without compromising data security. Overall, these measures collectively enhance operational efficiency, reduce human error, and improve client satisfaction within the veterinary clinic.

Table 12: Identified Problems

CHALLENGE	DESCRIPTION
Manual Patient Record Management	Records are kept manually, leading to errors and difficulty in tracking patient
Inefficient Appointment Scheduling	Lack of a centralized system causes scheduling conflicts and long wait times.
Poor Communication with Pet Owners	Poor communication leads to missed updates and low client engagement.
Uncoordinated Referral Process	Referrals to specialists or other clinics are handled inconsistently, causing delays.
Lack of Inventory and Supply Tracking	Medical supplies are not tracked efficiently.
Manual Billing and Invoicing Processes	Done manually, which is time-consuming, prone to errors, and delays payment
Absence of Centralized Platform	Without proper security, data may be accessed by unauthorized users.
Limited Data Security and Access Control	Without proper authentication and authorization, sensitive data may be exposed to unauthorized access.

The problems outlined in Table 12 clearly confirm that there is room for improvement where the management of the clinic is involved. The problems and weaknesses of the traditional approach and inefficient management of the clinic through manual management indicate that there is a need to adopt an effective approach and management techniques in order to address the problems. For example, the problems indicate that manual management generates redundant

process problems. It is incompetent in managing data and has problems of coordination among team members. Manual management is likely to generate human errors, which can affect the efficiency and productivity of an organization. In order to address the problems and weaknesses of the traditional and manual approach and management of the Petkho clinic, there is a need to develop an effective portal known as Petkho Portal. The development of Petkho Portal will help Petkho clinic address and resolve existing problems, hence streamlining clinic operations using effective communication, enhancing the quality of service, and ensuring adequate decision-making, hence providing an effective and reliable experience towards enhancing efficiency and effectiveness in the service delivery of Petkho clinic.

4.1.5 Proposed Solutions

The Kho Veterinary Clinic Rinconada faced several problems since they used traditional methods of documentation. This implies that they were not effective in their operations since documentation is an essential aspect of an organization. The organization was handling all operations manually. The operations included handling patients, scheduling, and payment. They were treating all operations manually. This implies that there were several problems with how they were running their operations.

PetKho Portal was able to resolve these problems through automation of key business processes, facilitating effective interactions among employees as well as customers, while also promoting the maintenance of accurate as well as safe information. By incorporating features of appointment tracking, invoicing, records handling, as well as treatment tracking, the system was able to simplify operations to ensure high quality services. The PetKho Portal facilitated improvements to operation, bills, and access to records. Additionally, role-based access control ensured that staff members could perform their responsibilities efficiently while protecting sensitive client and patient data. This comprehensive approach not only enhanced workflow efficiency but also increased client satisfaction and trust in the veterinary clinic's services. As a

result, client satisfaction improved significantly, and trust in the veterinary clinic's services was reinforced, demonstrating that the PetKho Portal is a reliable, secure, and practical solution for modern veterinary operations.

Table 13: Proposed Solutions

CHALLENGES	PROPOSED SOLUTIONS
Manual Patient Record Management	The PetKho Portal offers a digital pet record system for secure, accessible, and organized management of patient information.
Inefficient Appointment Scheduling	An automated scheduling module enables online booking, sends reminders, and prevents overbooking for efficient appointment management.
Poor Communication with Pet Owners	A communication module automates reminders and confirmations while enabling the clinic to send announcements and health alerts.
Uncoordinated Referral Process	A digital referral system enables electronic forms, referral tracking, and secure record sharing for timely and coordinated pet care.
Lack of Inventory and Supply Tracking	An inventory management system tracks supplies in real time, sends low-stock alerts, streamlines reordering, and optimizes inventory to reduce waste.
Manual Billing and Invoicing Processes	An automated billing system generates invoices, enables online payments, sends overdue reminders, and tracks revenue to reduce errors and delays.
Absence of Centralized Platform	A centralized system integrates all key functions with role-based access control, a user-friendly interface, and a customizable dashboard for efficient clinic management.
Limited Data Security and Access Control	Robust security measures, including multi-factor authentication, role-based access, data encryption, and regular audits.

This proposed solution offered in the table below addresses the current operational challenges facing Kho Veterinary Clinic Rinconada with the implementation of the centralized automated digital solution. Each solution works on improving a specific area or process in the overall operations of the clinic, ranging from handling patient data records to communication with clients and billing. This Petkho Digital Solution Portals platform aims at improving the overall speed, quality, and accuracy in the operations of the clinic while minimizing human error. Moreover, the entire process is centralized on a single application that fulfills all the needs, enabling the free flow of data from one module or process to the next. Also, the proposed solution guarantees the highest level of security with measures such as data encryption. Thus, the proposed solution works on providing a better organized, efficient, responsive, user-friendly solution in the overall process that not only improves the quality of work among the staff but also the quality response or experience offered to the clients or customers.

4.1.6 Fishbone Diagram

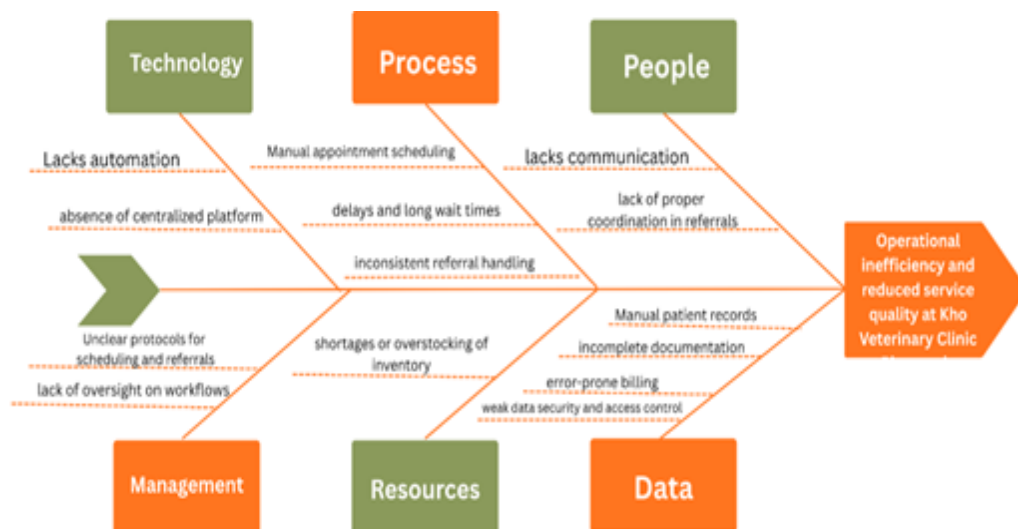


Figure 5: Fishbone Diagram

The figure above illustrates the Fishbone Diagram that identifies the root causes of operational inefficiency and reduced service quality at Kho Veterinary Clinic Rinconada. The

diagram categorizes the main issues into six key areas: Technology, Process, People, Management, Resources, and Data.

Challenges exist in the different key areas of the clinic. For the key area of technology, the clinic lacks the means of automating activities as well as a centralized system; thus, information scheduling is manual. This causes referrals to take longer. Another key area is the issues affecting the clinic's efficiency due to communication and workflow issues among the staff. This is considered in the key area of the clinic's management. Then comes the key area of the resource issues affecting the clinic in terms of its inventory. This is then considered in the key area of the data issues affecting the clinic.

4.1.7 Gantt Chart

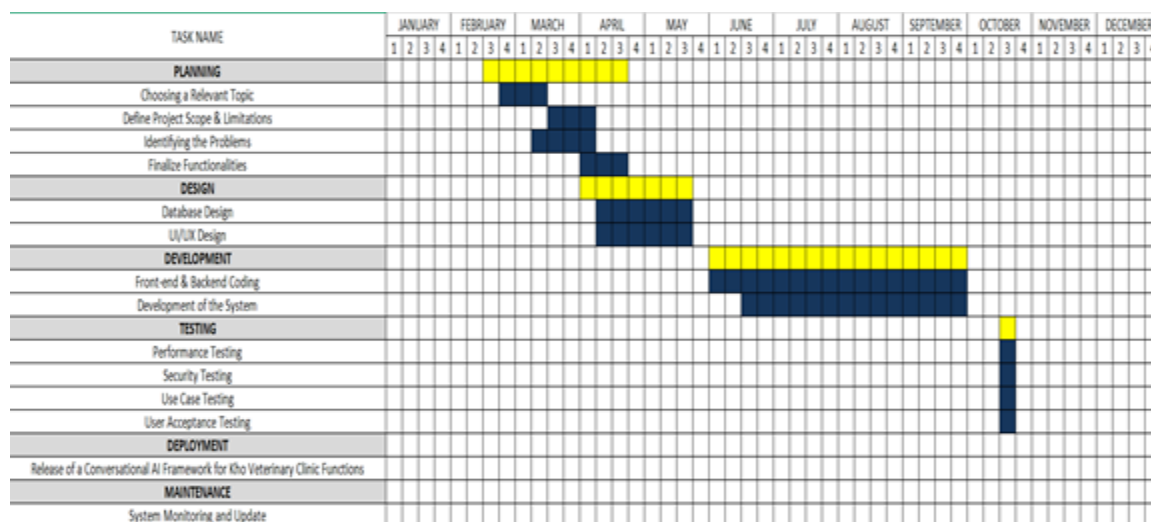


Figure 6: Gantt Chart

The plan of the PetKho Portal starts with the Planning Phase in January. At the beginning of the project, the theme is chosen and the problems identified. The design phase begins in April and ends in May. Considering the database and UI/UX design is crucial at this point as the system should have flexibility in terms of services. Moving on to the development phase, which starts in May and ends in September. Testing starts in October, ensuring functionality, reliability, and usability.

4.2 Design

In the project's design phase, the need to change focus from problem identification to solution development arose, toward a solution that would best address the needs of Kho Veterinary Clinic Rinconada. The overall structure of the solution, its major features, as well as user interface issues, are integrated to address issues related to clinic operations, where operations are manual, disorganized, and fragmented. Agile's approach to ensuring that proper distribution of the design effort is given, based on priorities, state, and requirements, is essential to improving the overall design solution to a project[2]. The integration of clinic operations, where patient management, appointment, referral, inventory, as well as communication, would operate as an individual solution, is ensured through the design phase to a great extent. This phase also emphasized iterative feedback and prototyping, allowing stakeholders to review design elements early and ensuring that the final system would be both practical and aligned with the clinic's operational needs.

4.2.1 Entity Relationship Diagram

Figure 7 depicts an Entity Relationship Diagram of Veterinary Management System that explains and defines entities and their logical structure and relationships with each other. This diagram forms an important basis as it explains how data is stored and efficiently maintained across all modules of Veterinary Management System. Such an ER diagram defines and illustrates entities like users, pets, appointments, adoptions, reports, etc., and their relationships with each other.

In addition, the diagram highlights how these entities interact with one another, illustrating one-to-one, one-to-many, and many-to-many relationships that exist in the system. For example, it shows how a single user can have multiple pets, schedule multiple appointments, and generate various reports, while each pet can also be linked to adoption records and medical histories. This structured representation helps developers, database administrators, and system analysts understand how information flows within the application, ensuring data consistency, reducing redundancy, and supporting efficient data retrieval. Furthermore, the ERD serves as a critical tool

during both the design and implementation phases of the Veterinary Management System. It provides a clear framework for creating database schemas, defining keys, constraints, and indexes, and ensuring that each module can interact seamlessly with others.

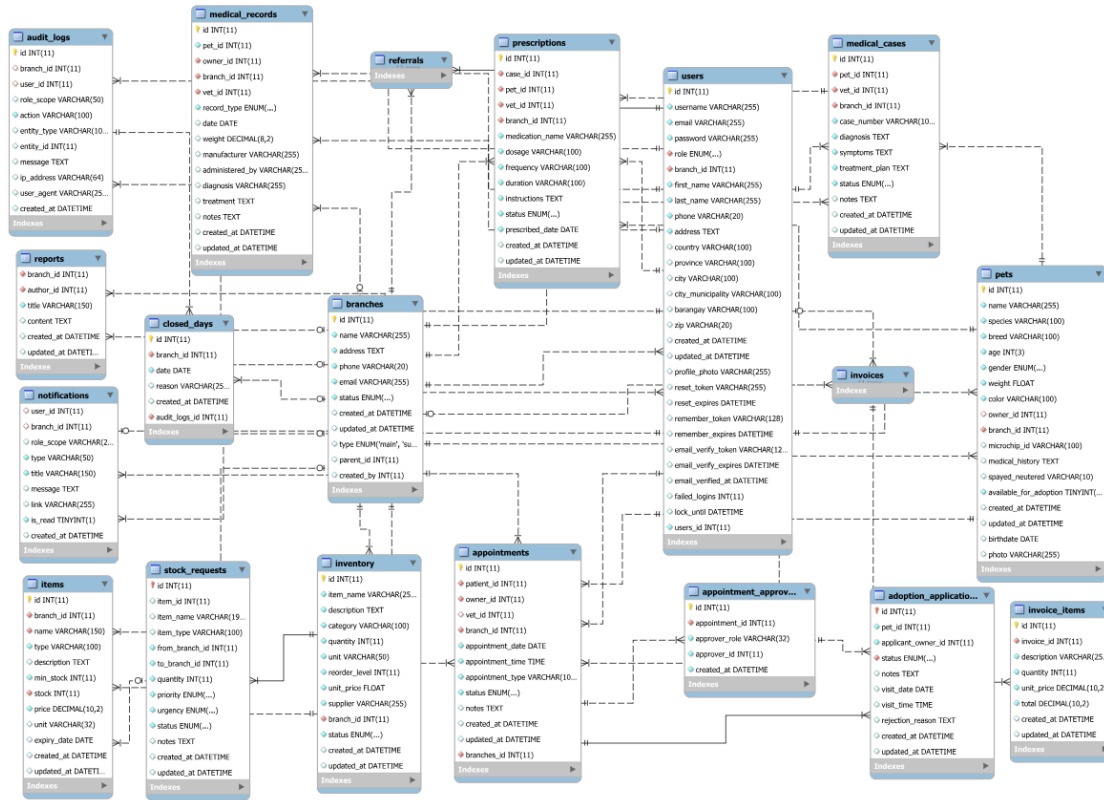


Figure 7: Entity Relationship Diagram

As seen from the above diagram, the Veterinary Owner fully controls and monitors each and every branch, which creates effective synchronization of the main and branches. Each branch operates independently on their own set of data, such as pet info, appointment info, employees, inventory, etc. This information is interconnected in the form of a single database. For example, the adoption module illustrates the capability of the system to deal with the whole adoption scenario through entities such as adoption_pets, adoption_applications, adoption_inquiries, as they are interlinked and store the necessary information regarding the status of the available pets to be adopted, the information regarding the applications, as well as the inquiries. Similarly, the

appointment module also interlinks the entities appointment, appointment_followup, and appointment_reschedules.

In addition, the pet record module includes various medical-related entities such as medical_records, health_conditions, and vaccinations, to ensure proper maintenance of each pet's medical conditions for consultation at later stages. The inventory module includes various entities such as inventory_items, branch_inventory, and stock_requests, to allow for the proper administration of medical-related stock used within each branch of the facility for the convenience of the staff, veterinarians, and the customers. In the same regard, supporting entities such as *reports* and *notifications* improve the functionality of the Veterinary Management System's administration by allowing for thorough notifications of each event within the facility. All these entities highlight the role of each entity within the ERD of the Veterinary Management System.

4.2.2 Data Flow Diagram

Figure 8 shows a Context Diagram, which is also referred to as a Level 0 Data Flow Diagram, and this diagram generally offers a high-level overview of the Veterinary Management System and how this system interacts with external entities. It shows us how information can flow from this system to different external entities in terms of both input and output processes. The context diagram establishes a general overview of what a system can actually accomplish. It can thus be said to represent a general overview of how different users in this system can communicate.

This diagram is important because it helps stakeholders easily understand the scope and functionality of the system without focusing on internal technical details. By presenting the system as a single process, it highlights the main data exchanges and responsibilities of the Veterinary Management System, making it useful for planning, analysis, and communication. The context diagram also serves as a foundation for more detailed diagrams, such as Level 1 and Level 2 DFDs,

ensuring that all system processes remain aligned with the overall objectives and user interactions of the application.

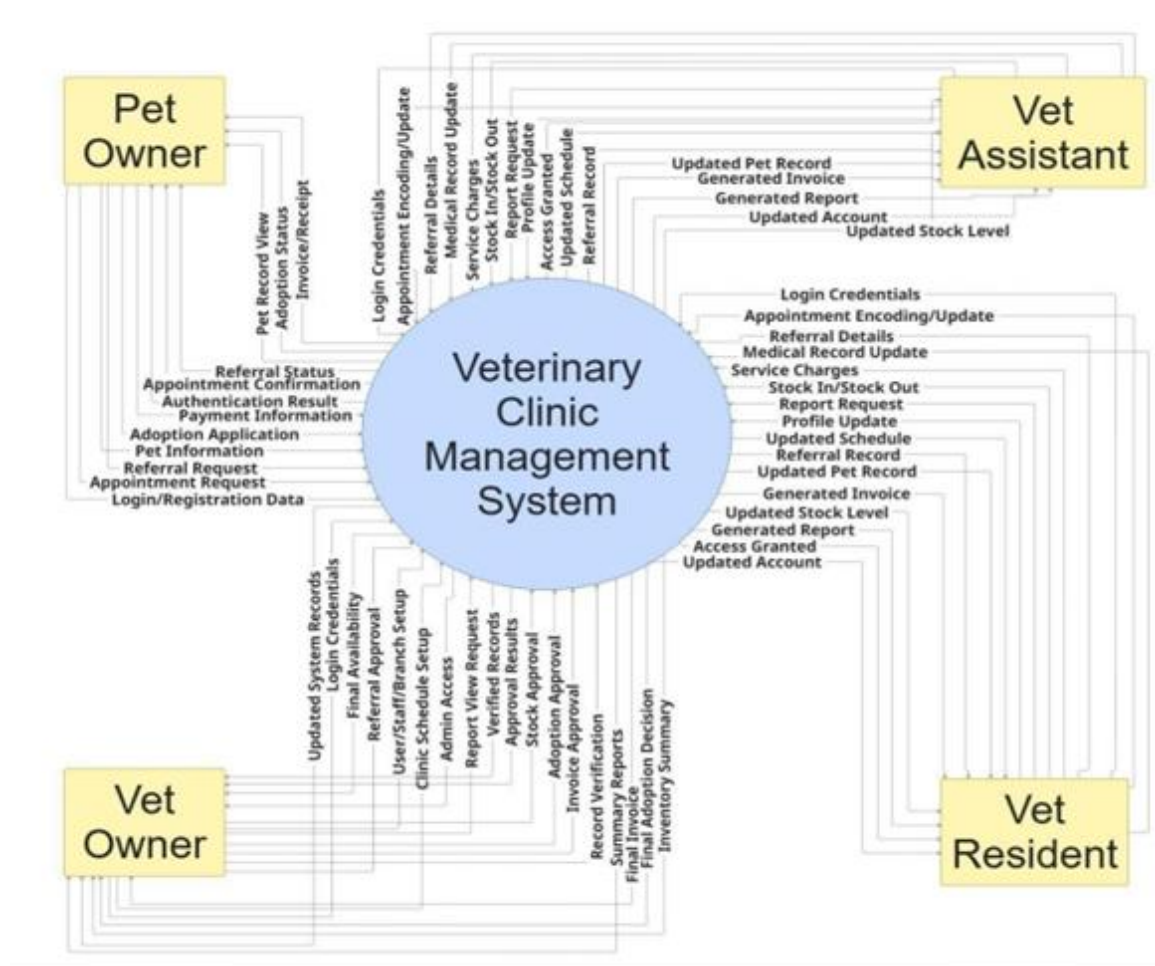


Figure 8: Context Diagram

It acts as the central hub that interlinks the four major entities, which are the Veterinary Owner, Main Branch Vets & Assistants (Naga), Sub Branch Vets & Assistants (Pili & Nabua), and the Customer or Pet Owner. The Veterinary Owner plays a crucial role in managing the whole clinic's operation by maintaining the staff, monitoring branch activities, and analyzing detailed reports to enable all branches to function efficiently and maintain the same service level. The Main Branch Vets & Assistants handle a wide range of operational processes, including creating appointments, creating adoptions, creating invoices for update, updating pet records, and approving referrals or stock requests necessary for clinic operations. Similarly, the Sub Branch Vets &

Assistants are responsible for handling the adopted pets, updating the status of adoptions, maintaining appointment records and schedules, reviewing the performance of a branch, and preparing reports that need to be reviewed by the main branch and the owner.

On the other hand, the Customer or Pet Owner interacts with the system by registering or logging in to make appointment requests as well as apply for pet adoptions. This interaction enables him to strategically utilize the clinic's services through an online interface. Such interaction not only satisfies customers with an enhanced interface quality but also eliminates human interventions and provides smooth transactions to employees. Therefore, it can be summarized that the context diagram clearly illustrates how efficiently all users and processes are interconnected to form a comprehensive veterinary network. It also highlights the free flow of information and better branch coordination and management of their records and services that make veterinary operations highly organized and transparent

Data Flow Diagram

Figure 9 shows the Data Flow Diagram (DFD) at the Level 1 of the Veterinary Management System. A DFD at the Level 1 provides a detailed view of the internal components of a system and the interaction of the data flowing within the system. A DFD at the Level 1 extends the single process of the context diagram into multiple processes that interact with each other and provide a view of the overall processes that take place within the system. Such a diagram provides a detailed view of how the information flows and processes within the system as a whole. A DFD at the Level 1 presents the overall view of how the system handles various processes, ranging from appointment scheduling, adoption, as well as report generation. A Level 1 DFD is particularly valuable because it reveals the internal workings of the system, showing the flow of information between processes such as appointment scheduling, pet management, adoptions, medical records, and report generation. It visually represents how data moves from one process to another, how it interacts with external entities like users or administrators, and how it is stored in data repositories

for future use. This granular view ensures that all stakeholders, including developers, analysts, and administrators, can understand the system's operational logic and the dependencies between different processes.

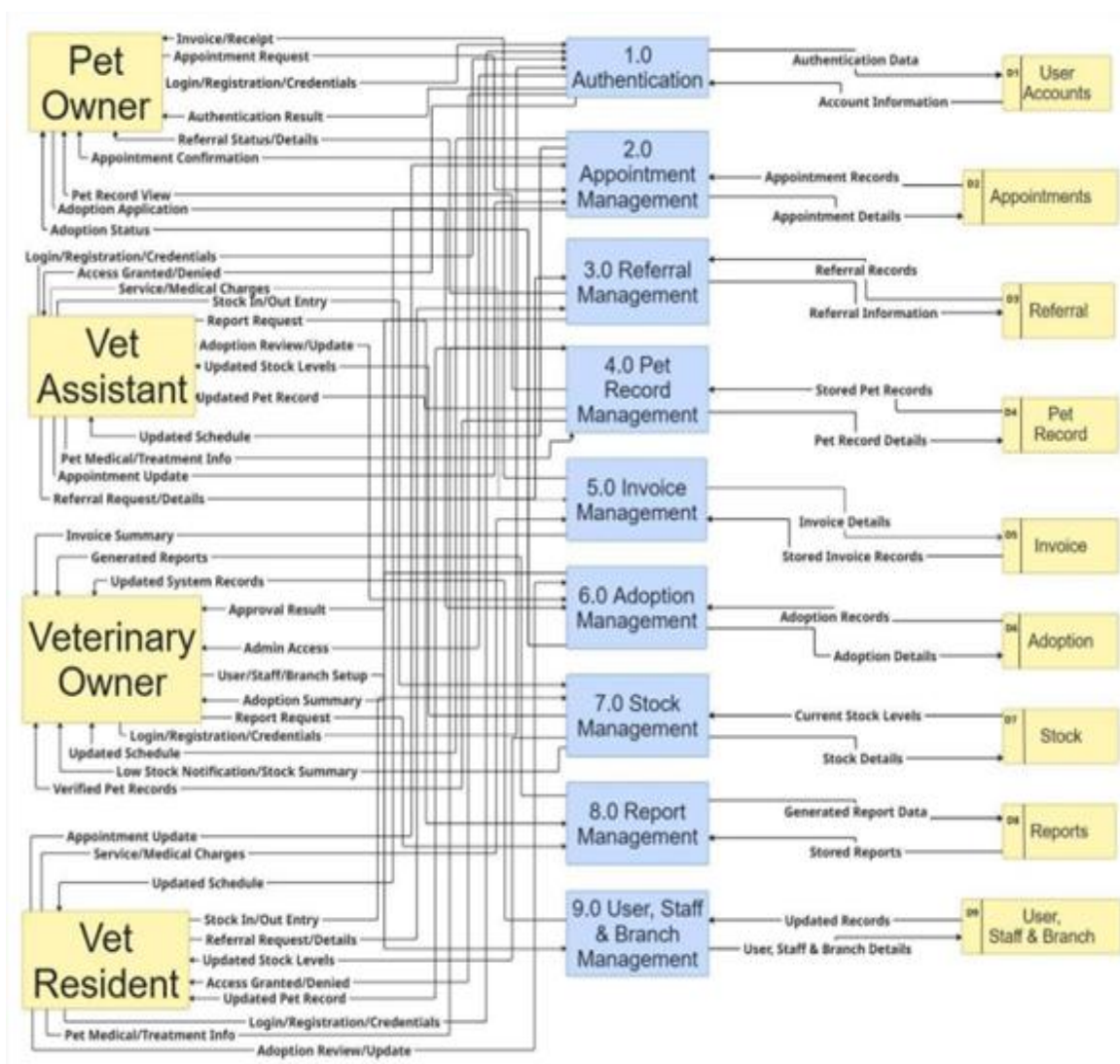


Figure 9: Data Flow Diagram Level 0

In addition to this, this diagram also highlights the logic of a well-designed Veterinary Management System through a clear depiction of how information is received, processed, stored, and transmitted between different entities of this management system. It clearly describes the roles of different entities in this management system in detail. These entities include a Veterinary Owner, Main and Sub-Branch Vets & Assistants, and Customers or Pet Owners. These entities need to

perform different roles in this management system, which can only be achieved under a well-organized process in which information moves in a smooth and efficient way throughout all processes without any interruptions in a consistent manner. All this can only be achieved through a clear depiction of a DFD Level 1 diagram in which information can travel smoothly and efficiently throughout every process in this management system.

The system is used by the Veterinary Owner to track activities, review reports, and monitor performances at both main and sub-branches. Main Branch Vets & Assistants have responsibility for managing their appointments, updating pet records, processing adoptions, creating invoices, and referring or requesting stock. The Sub Branch Vets & Assistants carry out branch-level operations like adoption handling, appointment coordination, and report submission to the main branch.

Meanwhile, the Client or Pet Owner interacts with the system for registration, appointment, adoption of pets, and follow-up or notification concerns regarding transactions. In general, the DFD Level 1 illustrates the way the data flows smoothly and in an orderly manner between the system components, fostering effective coordination and accuracy of records to be conducted at the various veterinary branches.

Figure 10 represents the Level 1 DFD for the ‘Login and Registration’ of the Veterinary Management System. Through the DFD above, the ability of the Veterinary Management System to create accounts for users such as the Customers, the Veterinary Assistants, and the Veterinary Owners can be understood. This DFD provides an opportunity to understand how the Veterinary Management System ensures security in the validation of the users’ information as well as the different classes of users and the ability of users to access the applications’ features depending on the roles of the users. From the DFD above, the ability of the Veterinary Management System to store and retrieve users’ information from the database can be understood. Additionally, the DFD shows how the system controls user access by verifying login details and user roles.

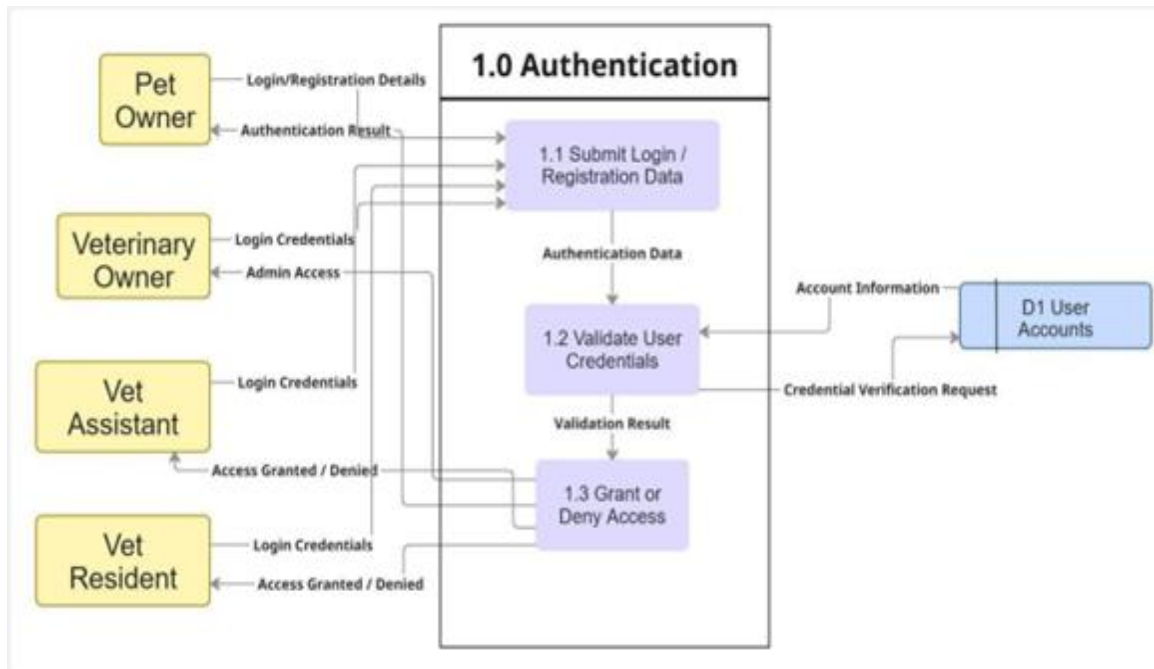


Figure 10: DFD Level 1-Login

The Authentication process (1.0) ensures effective and secure user registration and login mechanisms. It involves submitting login/registration information (1.1), verifying the information against the User Accounts data store (D1) (1.2), and either granting or denying entry (1.3). The Authentication process enables verification and granting entry for different types of users, including Pet Owners, Vet Assistants, Vet Residents, and Veterinary Owners.

Figure 11 presents the DFD Level 1 for the Appointment Management in the Veterinary Management System. This diagram highlights the processing of the request, scheduling of an appointment, as well as the rescheduling of an appointment in the system involving the pet owner, veterinary assistants, residents, and the owner of the clinic itself.

The diagram further illustrates how the appointment data flow in the system is orchestrated to achieve organized scheduling, effective coordination, and accurate tracking of branch clinic availability. It also shows how this data interacts with other system modules, such as patient records, user accounts, notifications, and ensuring seamless communication.

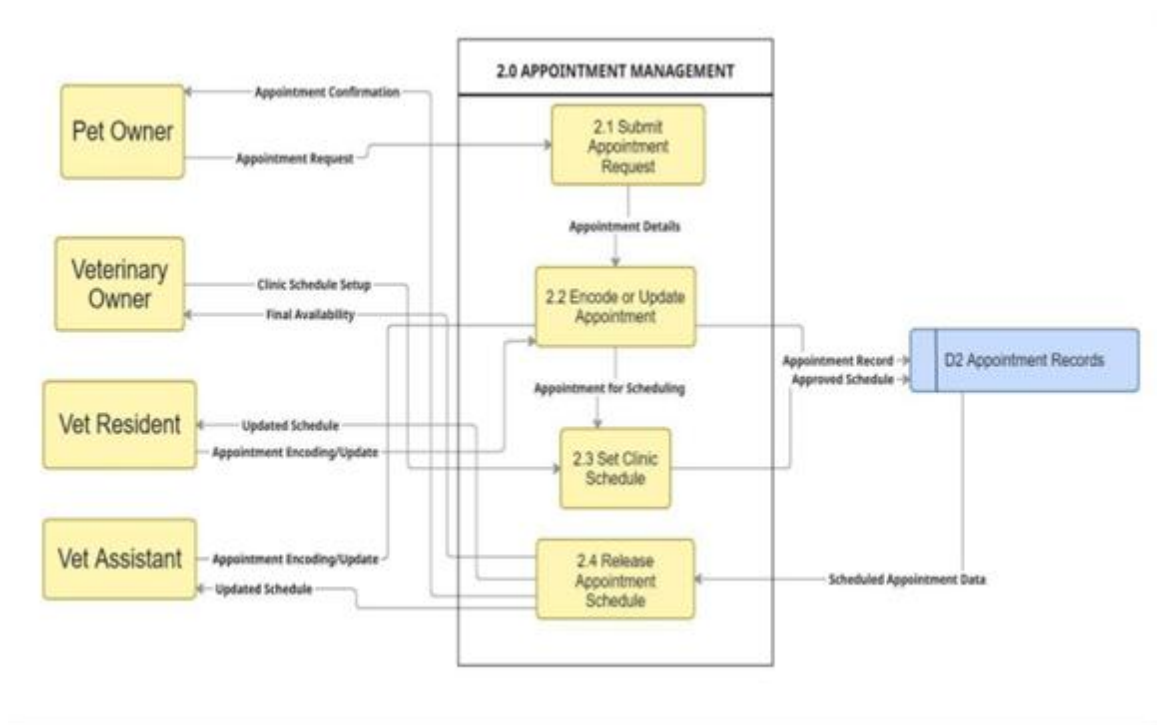


Figure 11: DFD Level 1-Appointment Management

For the Level 1 DFD for the Application Management Process, the main process is divided into two main processes or subprocesses. They include the Appointment Scheduling Process and the Appointment Approval/Rescheduling Process. For the Appointment Scheduling Process, Customers or Pet Owners are the ones that make requests to visit a veterinary clinic, providing information regarding the date they want to visit, the kind of service they need, or the status of their visit. The information is processed by the Sub and Main Branch vets and assistants, while the information is stored in the Appointments (D2) data store.

Figure 12, DFD Level 1 Referral Management, shows how communication and notifications are processed in the veterinary system as indicated in the diagram above. This diagram suggests the internal procedures involved in the processing of messages and how these are communicated within the system. It also highlights the flow of information between different modules and users, ensuring that referrals are tracked accurately and handled in a timely manner. It also shows how referral data is stored for efficient tracking and follow-up.

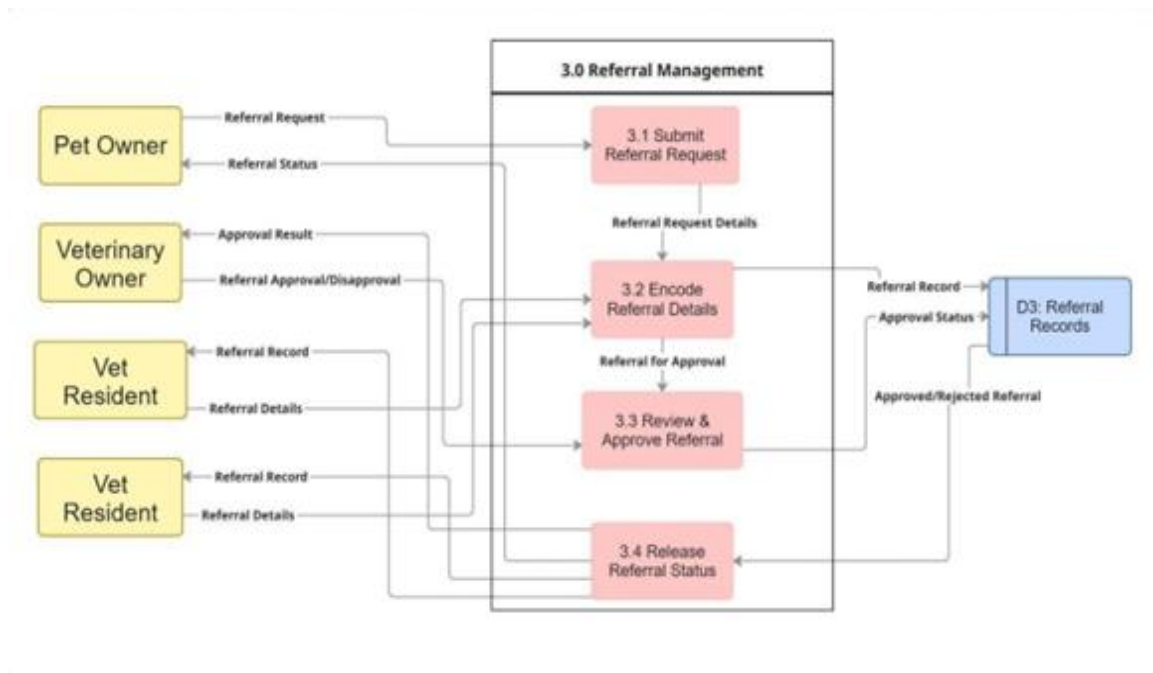


Figure 12: DFD Level 1-Referral Management

Requests for referrals and information about referrals received from users such as pet owners, staff at the branch, and the veterinary owner are managed by a referral management module. These undergo processes including encoding information about the referral, evaluating the results presented by the medicine experts, and either approving or disapproving the referrals. Referral information is stored in D3 as a new referral and the updates of the referrals are made for tracking.

Figure 13 depicts the Level 1 DFD for the Pet Record Management in the VMS. It shows the processes completed to access, update, and manage the information of pets by various entities including the Pet Owner, the Sub Branch Assistant, the Main Branch Vets, and the Veterinary Owner. It facilitates the free flow of information while keeping the records of pets consistent. It also highlights the interaction between different processes, such as data input, verification, storage, and retrieval, which collectively contribute to efficient record management.

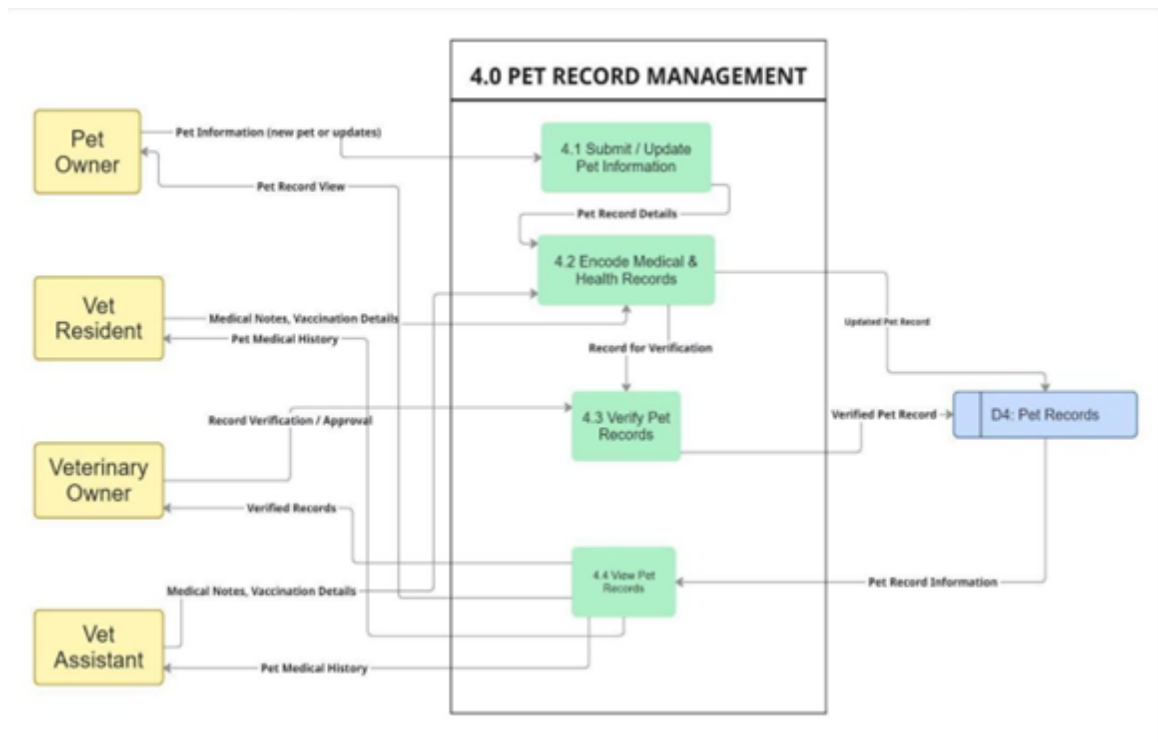


Figure 13: DFD Level 1-Pet Record Management

The level 1 DFD for Pet Record Management has three major sub-processes: Pet Record Access, Pet Record Update, and Referral Handling. In Pet Record Access, the pet owner or user is able to view or request updates to information in a pet's medical record, which first needs to be verified before updates are made and then retrieved from the Pet Records Data Store (D2).

The Pet Record Update process involves ensuring that new information such as possible diagnoses or treatments in pet records is correctly and professionally recorded and updated in the database. In Referral Handling, pet owners have the option to have their pets referred to different branches for further and more professional treatment, with updated records and information available for both referral and receiving vets. All these are crucial for ensuring that pet records are kept and managed in a professional manner to ensure easy and effective pet management systems are in place. This process also helps maintain continuity of care and supports better clinical decision-making across all veterinary branches.

. Figure 14 shows the level 1 DFD diagram associated with the Invoice Management within the Veterinary Management System

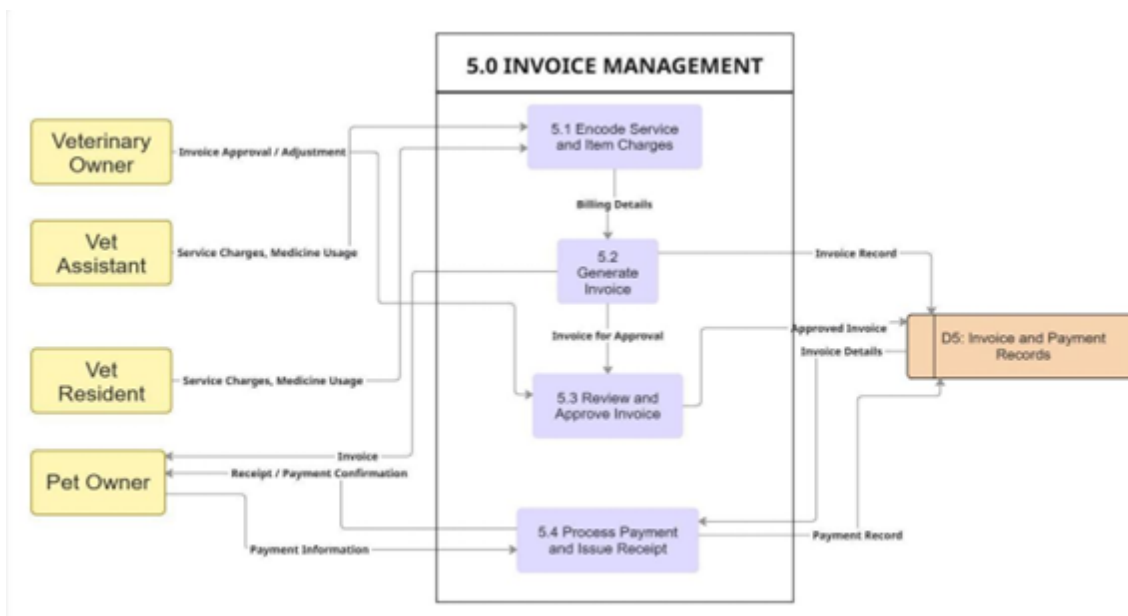


Figure 14: DFD Level 1-Invoice Management

The major subprocesses of the Invoice Management Level 1 DFD include three parts: Invoice Creation, User Login Verification, and Account Validation & Access Control. Invoice Creation captures the necessary information like the type of treatment, medication, attending veterinarian, and pet information to create itemized invoices. User Login Verification identifies that only authorized users will be granted access to view the invoice information to ensure integrity in transactions. Account Validation & Access Control identifies whether to approve or decline invoices based on the validated credentials and policies of the clinic set by the Veterinary Owner. After its approval, the invoices will be safely stored and accessible for relevant users to make sure all the billing operations are correct, compliant, and well-managed across the veterinary system.

Figure 15 shows the Level 1 DFD for adoption management in the Veterinary Management System. The figure shows how adoption applications are processed through submissions, approvals, and rejections by actors such as Customers, Branch Assistants, Vets, and Veterinary Owner. Figure 15 shows the Level 1 DFD for adoption management in the Veterinary Management

System. The figure shows how adoption applications are processed through submissions, approvals, and rejections by actors such as Customers, Branch Assistants, Vets, and Veterinary Owner.

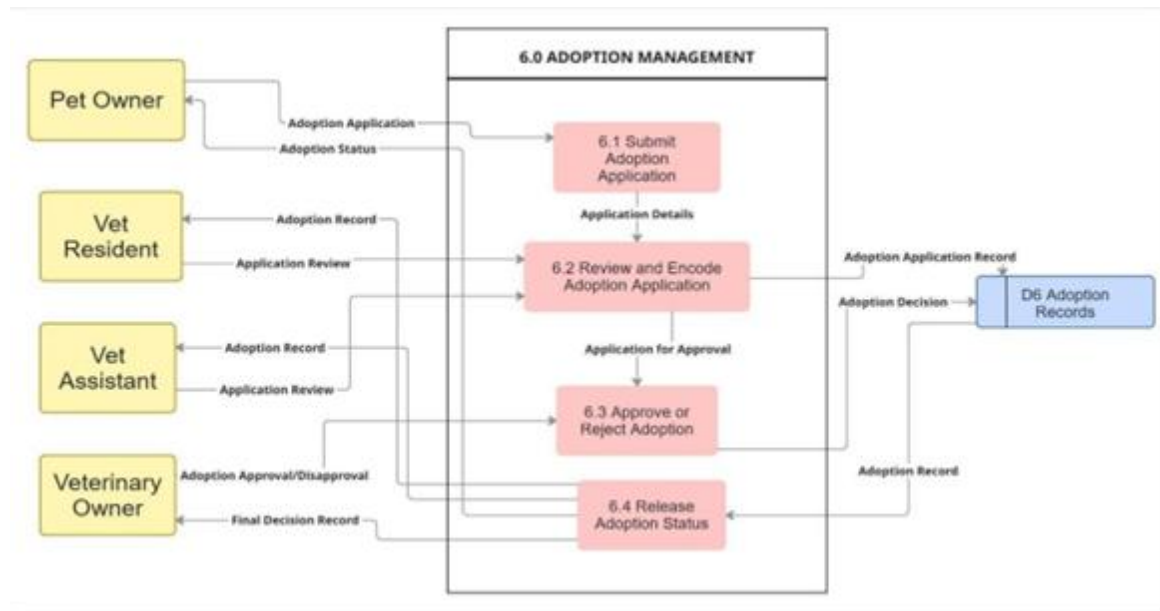


Figure 15: DFD Level 1-Adoption Management

This DFD illustrates the way the adoption process is carried out systematically through various stages, as depicted above. It begins with the Adoption Application process, where the user provides their personal information as well as their preferences. This is followed by the Application Review and Final Evaluation process, which evaluates the eligibility of the application as well as the availability of the pet. This information is shared with the involved user, as indicated by the feedback. It ends with the approval of the user, where the approved results are stored in the Adoptions Data Store.

Figure 16 illustrates the Level 1 Data Flow Diagram (DFD) for Stock Management in the Veterinary Management System. It shows how stock-related activities are monitored, validated, and distributed across all branch levels. The diagram also highlights how data flows between users and system processes to ensure accurate inventory tracking and timely restocking. Additionally, it

demonstrates how stock updates are recorded in the database in real time. This helps reduce errors and prevents stock shortages or overstocking. Overall, the DFD supports efficient and reliable stock control within the system.

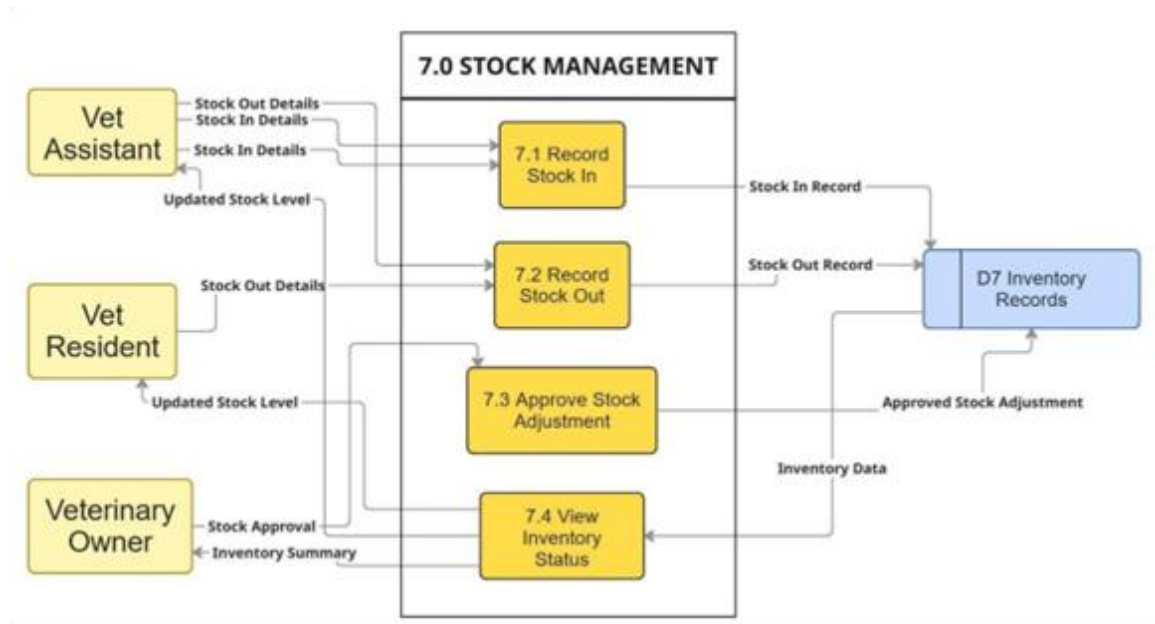


Figure 16: DFD Level 1-Stock Management

The Stock Management process includes user involvement in the form of Sub-Branch Vets, Sub-Branch Assistants, Main Branch Vets, Main Branch Assistants, and the Veterinary Owner. It commences with the generation of stock monitoring requests, which involves validating and approving the request before proceeding. Following this, the process includes the allocation of stock to ensure that inventory levels are maintained accurately and efficiently across all branches. These steps ensure that stock distribution is organized, traceable, and aligned with the operational needs of the veterinary system.

Figure 17 presents the Level 1 DFD for Report Management in the Veterinary Management System, providing a structured overview of the report handling processes within the context of analytics and decision-making. It illustrates the movement of data through multiple interconnected processes as the system collects, organizes, and analyzes information from stakeholders, including

sub-branch and main branch vets, assistants, and pet owners. The diagram also highlights how reports are generated, reviewed, and disseminated, ensuring that accurate and timely information is available to support operational decisions and improve overall system efficiency.

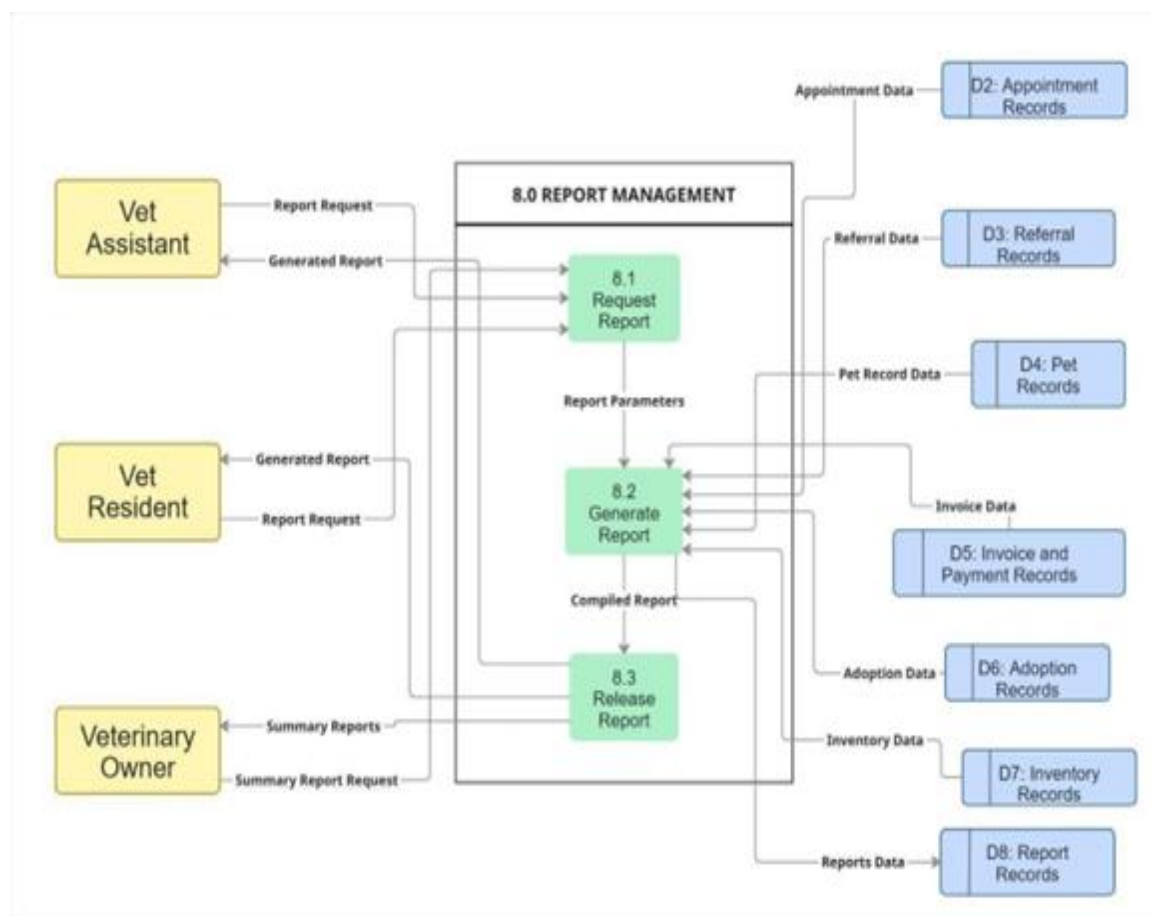


Figure 17: DFD Level 1-Report Management

The diagram highlights three subprocesses of the "7.0 Report Management" module, which include creating monitoring requests at 7.1, submitting reports at 7.2, and checking and updating distributions at 7.3. In these subprocesses, user information is actively utilized to generate accurate and comprehensive reports that are stored in the "D7: Reports" repository. These reports are then accessible to relevant parties, such as the veterinary owner or assistants, and undergo a series of reviews to ensure accuracy, which may lead to updates regarding stock levels and other operational

adjustments, ultimately supporting better decision-making and efficient management within the system.

Figure 18 illustrates the Level 1 DFD of the Veterinary Management System, focusing on the administration of User, Staff, and Branch Management. This diagram depicts the specific areas involved in administering the system, showing how tasks such as adding, updating, and monitoring users, staff, and branch details are managed. By visually representing these administrative processes, the system allows for easier oversight, efficient control of operations, and effective management of all user roles and branch activities within the veterinary system.

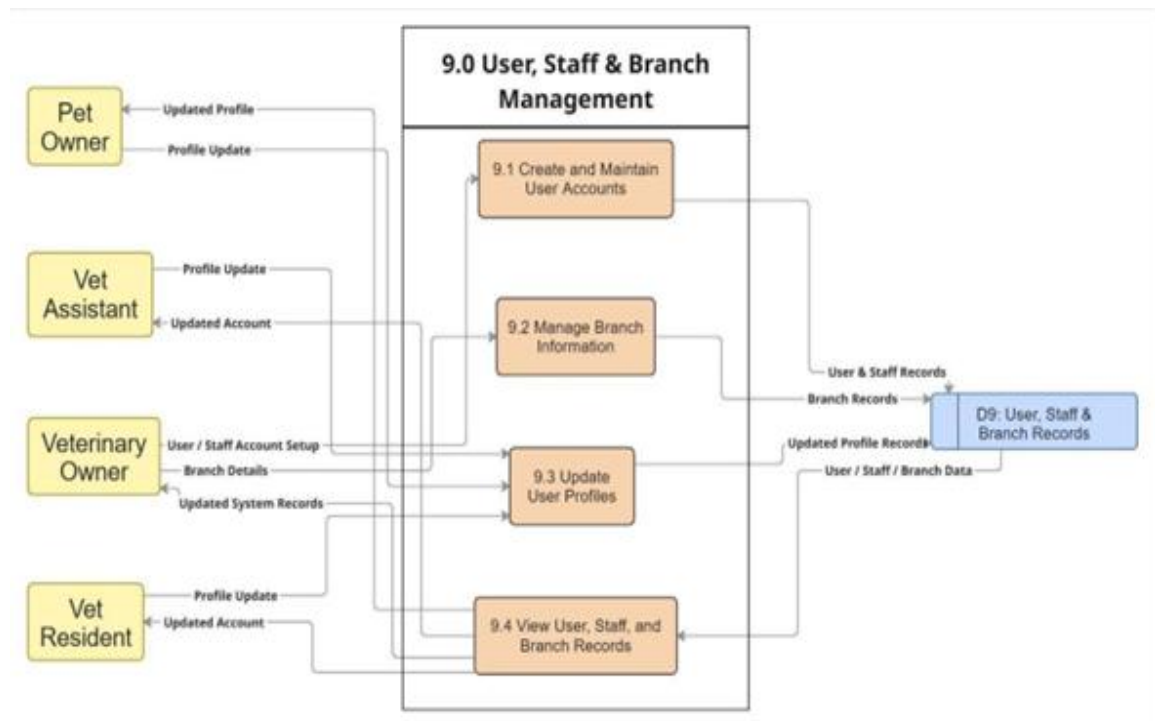


Figure 18: DFD Level 1-User, Staff, Branch Management

This indicates how the operation of administrative functions works in the context of the veterinary management system. This section explains how the user accounts are managed under the supervision of the veterinary owner. On the other hand, the operation of the staff information and branch information are updated and managed under the supervision of the vets and assistants of the main branch as well as the supervision of the assistants of the sub-branch. This information

is processed in the context of the three functions described above; the information is then updated in relation to the user accounts and staff information databases, ensuring accurate and consistent records across the system.

4.3 Development

Lastly, the deployment phase entails the actual usage of the Petkho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada. This phase ensures the Petkho Portal works properly. Based on the presented, Agile methodology has shown good results in the area of project management. Nevertheless, Agile methodology needs a cultural shift, proper information, as well as a detailed insight concerning the usage of Agile methodology[4]. This phase includes creating user accounts for the admin, the entire staff, as well as the pet owners. On the other hand, some minor tweaks ensure a stable as well as a highly responsive performance in the real world. This phase represents the actual shift of the Petkho Portal from the development phase towards actual utilization. This phase ensures the entire Petkho Portal works properly. This phase emphasizes compatibility, stability, as well as a guarantee concerning the entire Petkho Portal. It deals with any concerns arising during the actual deployment. Further, some feedback from the actual user results in possible improvements, guaranteeing a good performance.

4.3.1 Admin Modules

Figure 19 displays a screenshot of the landing page of the PetKho portal. It is also essential to note that it functions as an interface for all users of the system. The portal would introduce users to PetKho as their partner in pet healthcare and adoption services. Moreover, it would help users register or login to their accounts and make various pet adoption and healthcare services available to them. The interface would feature an appealing graphical user interface as evidenced by the image of different kinds of pets featured prominently on it. This reflects PetKho's commitment to animal welfare as expressed by their mission and vision. It presents an impression to all users that

directs them smoothly to explore all its services and facilities. The layout is designed to be intuitive, allowing users to easily navigate between sections such as appointments, pet records, and adoption listings. Clear call-to-action buttons guide users toward key functions without confusion or delay. Overall, the landing page establishes a welcoming and professional first impression that encourages user engagement.

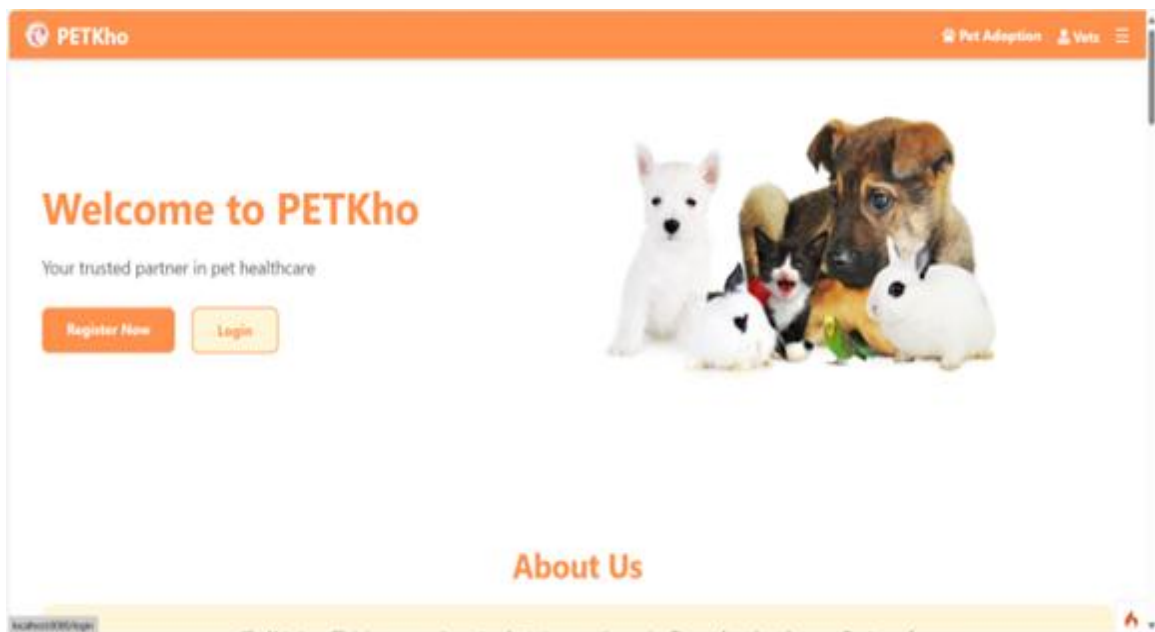


Figure 19: Landing Page

The landing page designed for PetKho is a great introduction to the features that are going to be offered on the platform, specifically on the aspect of pet healthcare and adoption. The top section features the PetKho logo, along with other links to Pet Adoption and Vets. This makes the landing page user-friendly, as users can easily get to the important features whenever they need to. The prominent section displays a "Welcome to PetKho Your trusted partner in pet healthcare" message, accompanied by a friendly image of various pets, including puppies, kittens, and rabbits. The section is filled with prominent "Register Now" and "Login" buttons to encourage user interaction on the platform. The "About Us" section begins under the "Welcome" section, where the company's mission and values are probably mentioned. The overall look is very friendly, along

with a sense of the direction that PetKho is heading as a trusted companion in pet healthcare.

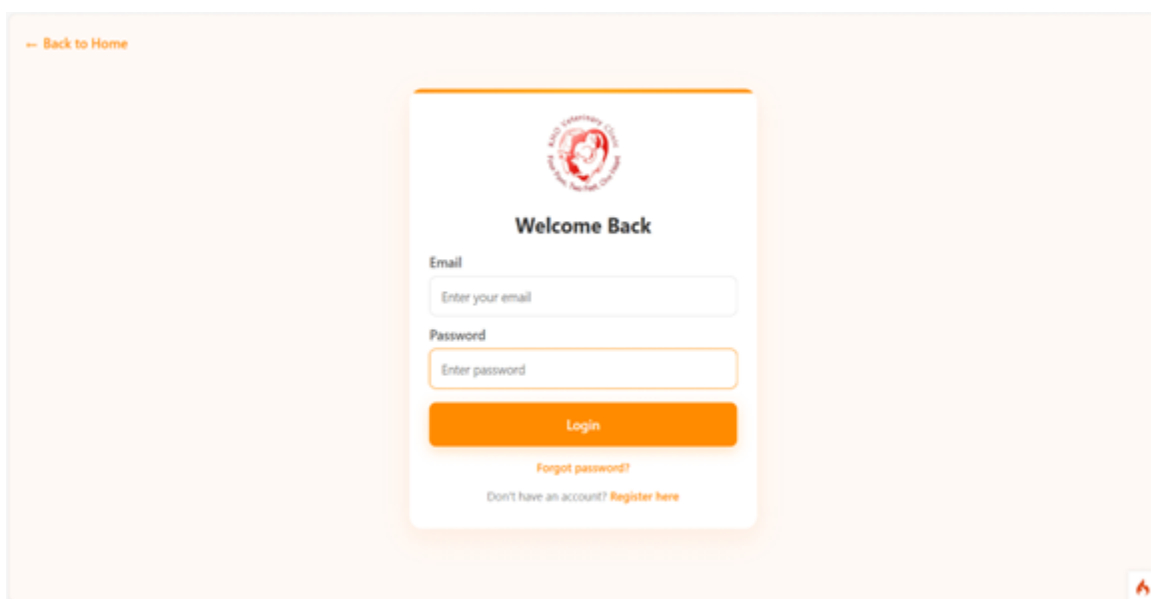


Figure 20: Login Page

Figure 20 illustrates the interface design for the login process of the Kho Veterinary Clinic Management System, specifically tailored for the administrative or veterinary owner's access. This interface is designed to be user-friendly, allowing authorized personnel to securely enter their credentials and gain access to the system's administrative functions.

Figure 20 represents the 'Login Page' of the 'Admin' or 'Veterinary Owner' version of the 'Kho Veterinary Clinics Management System.' To gain protected access to the 'Admin' section of the 'Management System' developed for the 'Veterinary Clinics,' the 'Admin' is required to enter their registered email address and password correctly. This 'Admin' login ensures that not everyone can access sensitive information in the 'Veterinary Clinics Management System' and perform various operations in the 'Management System' in general. It establishes controlled access to the system's backend, maintaining data integrity and protecting confidential clinic information from unauthorized use.

Figure 21 below presents the Admin or Veterinary Owner Dashboard in the PetPortal system. This serves as the main control panel through which an administrator can effectively run

the clinic by providing real-time information as well as quick access to the main system modules. The dashboard consolidates critical data such as appointments, inventory levels, and staff activity, allowing the administrator to make informed decisions efficiently. It also provides management tools for monitoring clinic performance, approving stock requests, and overseeing service operations across multiple branches.

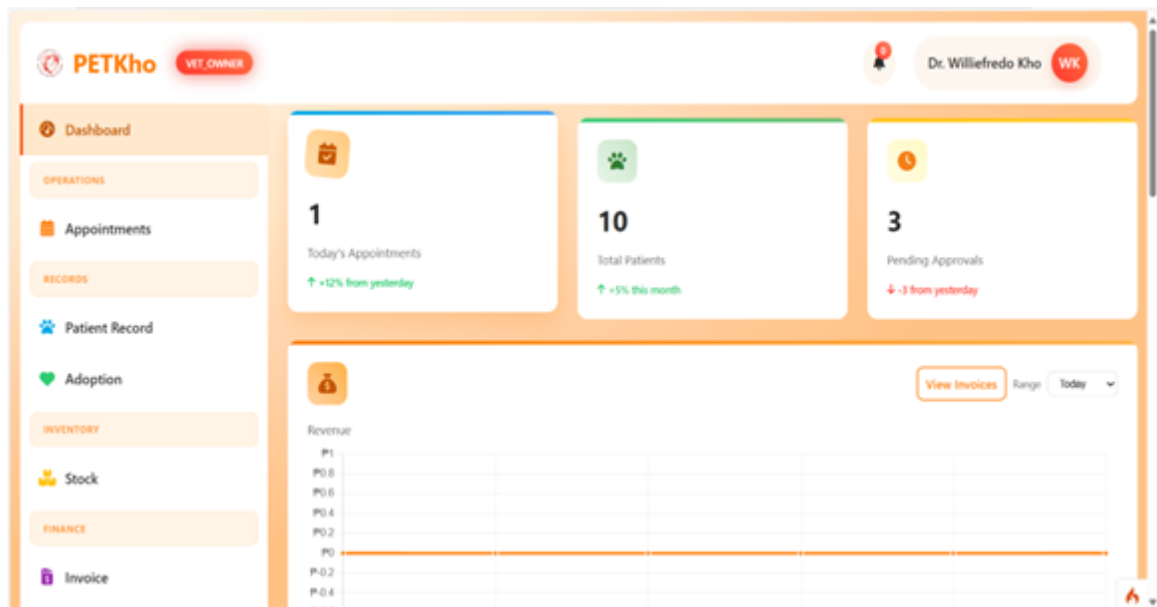


Figure 21: Admin/Vet Owner Dashboard

Figure 21 depicts the Admin/Vet Owner Dashboard of the PetKho Portal system, designed to function as a key interface for efficient management and control over critical operations from the administrator's as well as the vet owners' side. The dashboard displays a comprehensive summary of various key performance indicators, such as the total registered pets, pet owners, appointments scheduled for the day, along with the total revenue generated, depicting comparative percentage increase over the last month to analyze the growth trajectory over time. Additionally, the interface is augmented with personalized messages, like a personalized hello, to increase the overall interactivity and user-friendliness of the dashboard. The overall appearance is maintained through the presentation of various visualization tools like the depiction of revenue generated,

shown through the analytics section on the dashboard, dubbed 'Revenue Analytics.' This section displays critical information regarding income generated throughout the month, as well as patient distribution, depicting statistics on the type of animals to efficiently control resource allocation. The overall interface is augmented through side-bar navigation, ensuring ease of access to critical operations like Appointments, Pet Records, Invoices, Adoptions, Inventory, Reports, and User Management, allowing for smooth and centralized control of all veterinary management functions.

Figure 22 shows the Appointment Management module of the PetKho Portal Veterinary Management System. This module enables veterinary professionals to effectively manage all the appointments of pets under their care through the scheduling of these appointments. Each of the appointments is accompanied by relevant information like the scheduling time, the patient's name, the owner of the particular pet, the branch, the type of service to be provided, as well as the status of the particular appointment. The module enables the effective organization of the appointments, thus facilitating the elimination of scheduling problems.

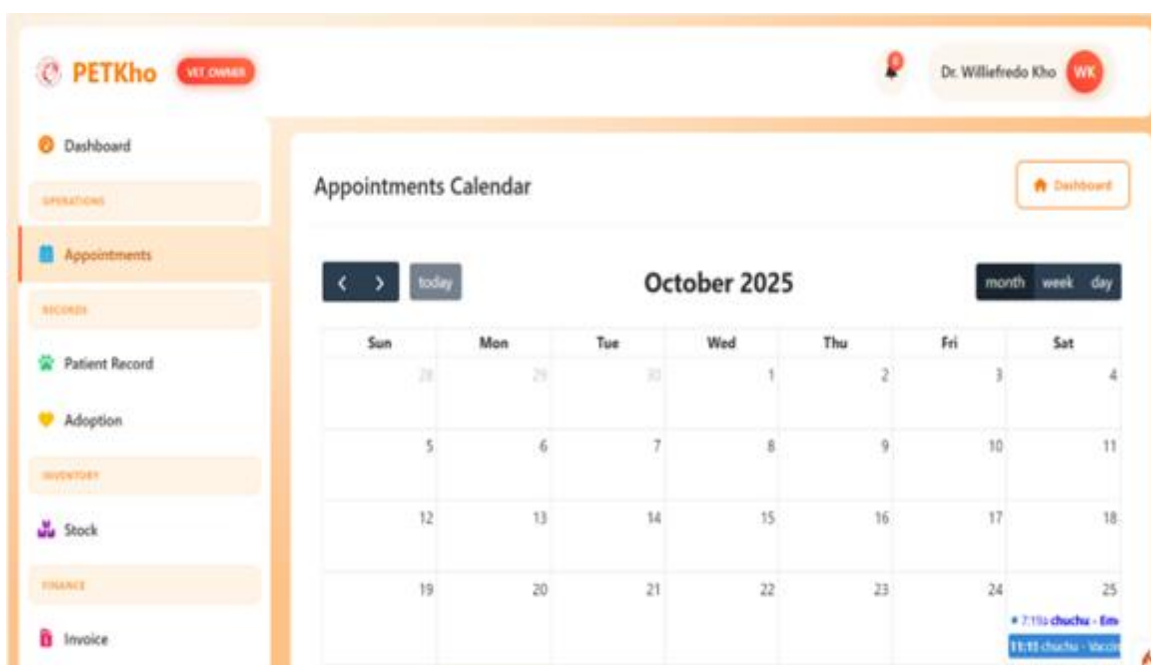


Figure 22: Appointment Management

The interface of PetKho Veterinary Clinic Management System's Appointment Management section enables users to have access to an interactive calendar that can display information for monthly, weekly, and daily appointments. Each entry contains the essential information that needs to be filled in, like the name of the pet, the details of the owner, the type of service, time, etc., aiding the efficient handling of the clinic's day-to-day activities. Such a structure can help in clarity with the potential usage of different colors on the entries based on the type of pet appointment, like vaccination, check-ups, or even pet grooming. On the left side, easy access to other modules like Patient Records, Adoption, Stock, Invoice, etc., can help in the smooth functioning of the workflow.

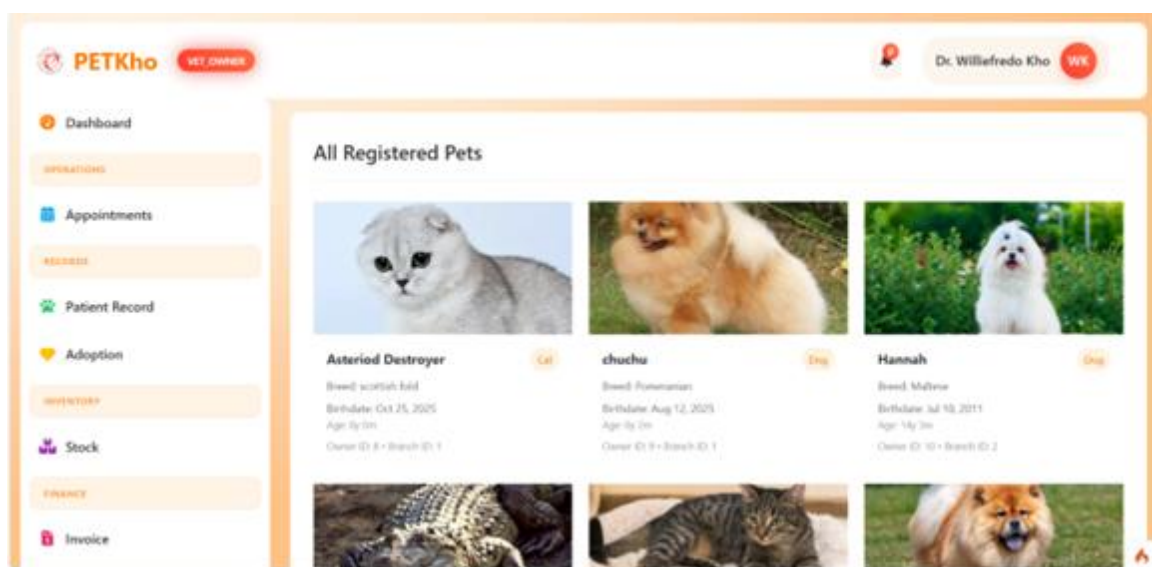


Figure 23: Pet Record Page

Figure 23 illustrates the appearance of the Pet Records Page, which enables the admin or the vet owner to see, modify, and maintain comprehensive information about each pet the user has, including its health records and profile. The tool serves as a way to hold accurate, systematized pet information that's accessible at your fingertips. Additionally, this webpage can update information quickly in case the pet has received new types of treatment or evaluation methods..

The Pet Records Page could be a place where all pet records are kept and managed.

Essentially, as well as keeping all pet information records, the Pet Records Page could provide a platform where the admin or the veterinary owner could easily view and control all records pertaining to a particular pet.

Furthermore, the page has a search feature that improves navigation, as it keeps everything in order and up to date. The page also improves coordination between individuals, such as between the staff, as it has features that allow sharing of relevant information, which may be useful in managing the veterinarian services effectively.

Figure 24 illustrates the Invoice Management module where the admin or the vet owner can create and view the invoices for the various services rendered by the vets. The module aids in the accurate accounting of the finances of the

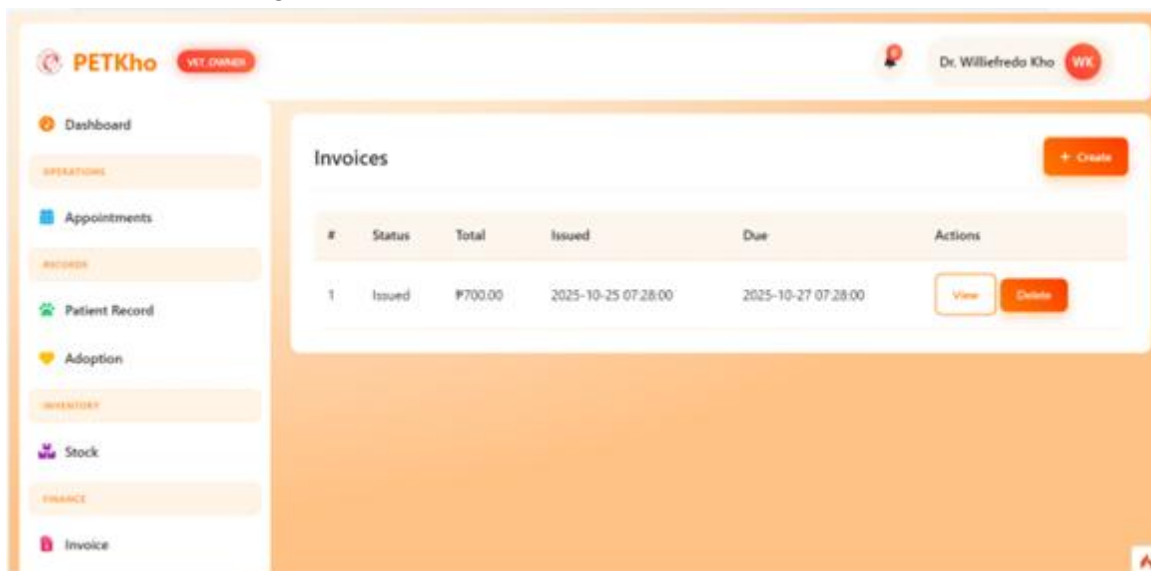


Figure 24: Invoice Page

Figure 24: The Invoice module, which plays the role of a financial management component within the PetKho Portal system, allows the admin or veterinary owner to create, review, and manage the invoices of a wide range of veterinary services performed on animals. It makes sure that financial transactions in the clinic are correctly invoiced and tracked transparently. The module provides a comprehensive record of every transaction involved, comprising a description of the service offered, information about the client, and the status of the payment. Because of such a

feature, the clinic will effectively track its revenues and reduce some possible errors in billing. This maintains sound financial records, which promote orderly accounting and business operation principles. The invoice module facilitates smooth payment processing, improves client record integrations, and service histories that minimize manual workload and discrepancies. It also enhances financial oversight through the creation of comprehensive reports aimed at aiding in the analysis of the clinic's general performance and profitability.

Figure 25 illustrates the Kho Veterinary Clinic Management Adoption Page System. It allows users to view available animals for adoption and submit adoption requests through the system.

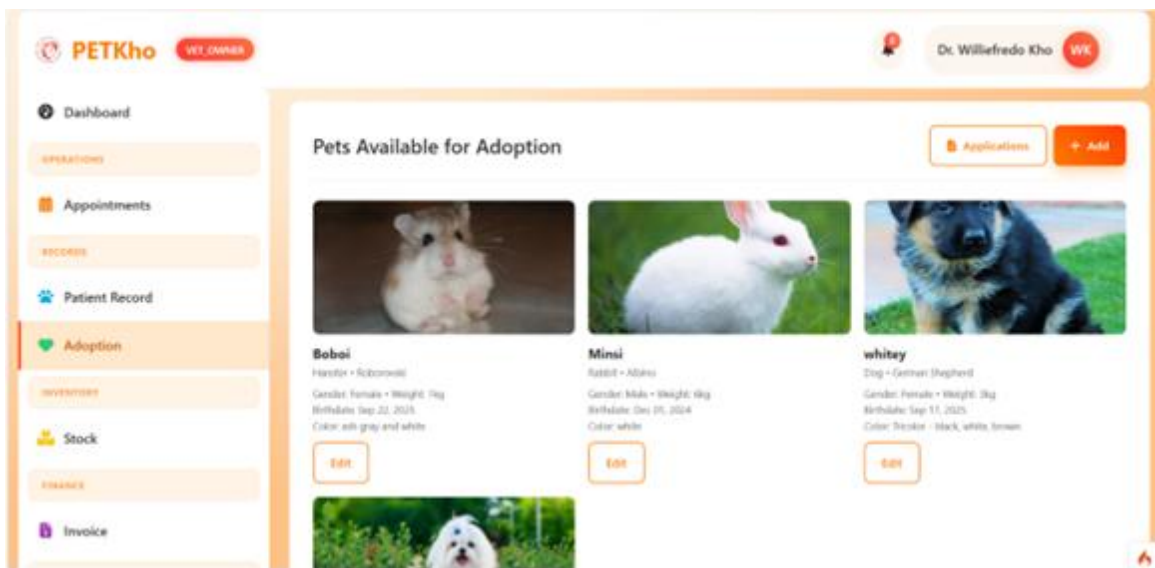


Figure 25: Adoption Page

This interface is intended to support the efficiency of veterinary staff in managing the whole process related to the adoption of pets by providing a comprehensive overview of all activities related to adoptions. It outlines categorized lists of available, pending, and adopted animals for clarity on the current status of each pet. Summary statistics across the top include the total number of pets, available pets, pending applications, urgent cases, and successful adoptions. On the page, there is also an Add Pet button for adding new animals, and filtering options to help

sort adoption records more effectively. Each animal entry indicates important information like the name of the pet, breed, age, and photo, so it can be identified easily by both staff and potential adopters.

Figure 26: Inventory or Stock Management Module that enables the administrator or the owner of the veterinary clinic to effectively track and monitor the clinic's medicine, supplies, and equipment.

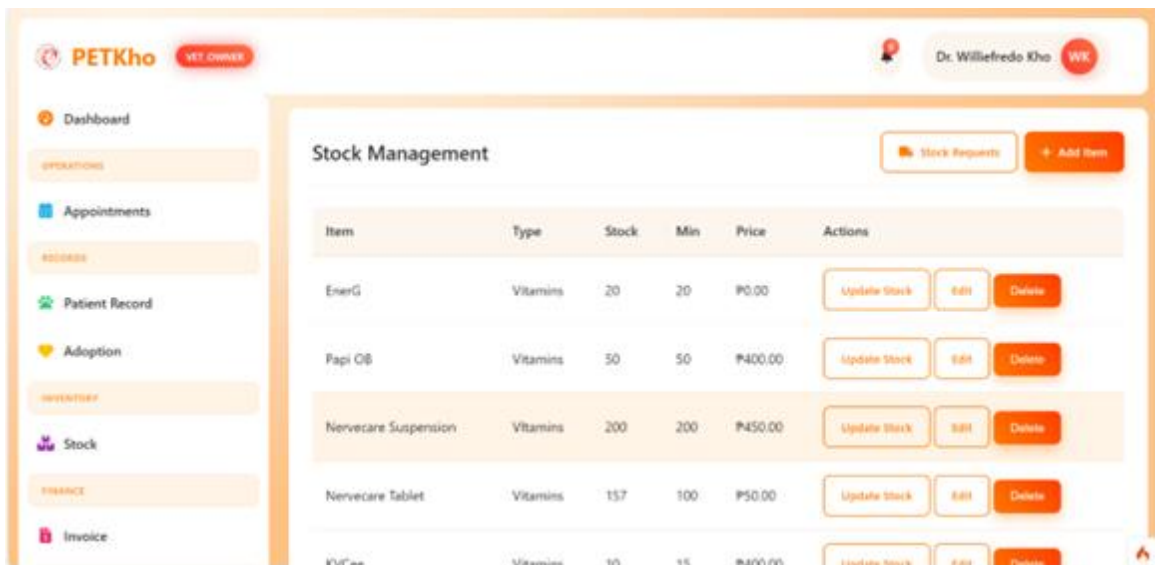
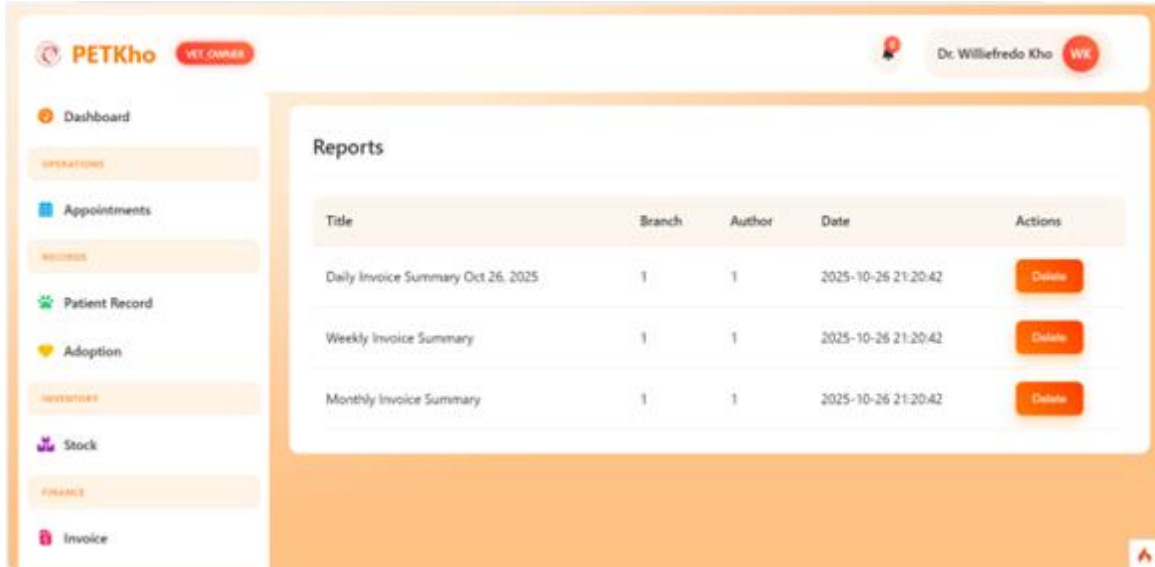


Figure 26: Stock Management

Figure 26 illustrates what is referred to as the Inventory page, which is a very essential part of the entire PetKho Portal system meant to support both the resident and assistant vets in efficiently monitoring their medical supplies in a bid to effectively manage these supplies in strict adherence to state regulations meant to promote efficient pet care services in the country.

Part of this page includes useful information on Animal Vaccines, Antirabies Medicine Supplies, Medicine Supplies, Animal Supplies, a list of "Low Stock Items," and "Recently Requested Items from Other Branches," which may or may not include details on item requests from different branches in terms of names, quantities, and approval statuses, promoting transparency and efficient inventory coordination across the clinic.

Figure 27 illustrates the report management module that enables the clinic to create, organize, and maintain reports concerning its activities or branches effectively. This contributes to



Title	Branch	Author	Date	Actions
Daily Invoice Summary Oct 26, 2025	1	1	2025-10-26 21:20:42	Delete
Weekly Invoice Summary	1	1	2025-10-26 21:20:42	Delete
Monthly Invoice Summary	1	1	2025-10-26 21:20:42	Delete

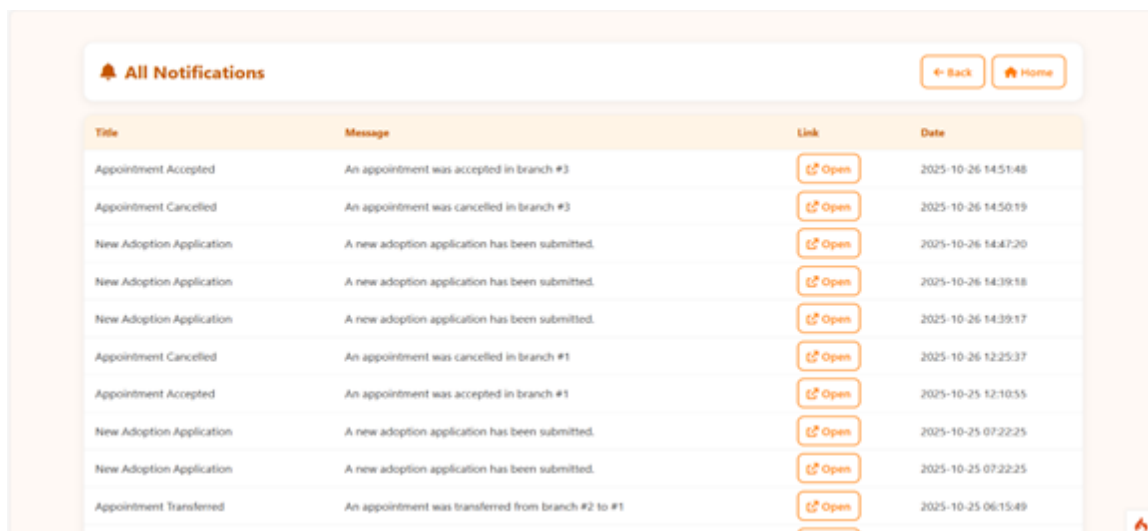
Figure 27: Report Management

the efficient documentation and accessibility of significant information to achieve efficient management of the clinic.

Figure 27 shows the Report Management page of the Kho Veterinary Clinic Management System, which facilitates the staff in generating various reports about the operations and branch performance of the clinic, managing, and monitoring them with ease. It offers a systematic table of all reports submitted, which are classified according to report title, type, branch, reporting period, status, and creator. It has actions that enable one to view, edit, or update reports. This module ensures that in the system, there is organized documentation; transparency is assured, and this facility/institution can make better decisions whenever it has up-to-date information on operations.

Figure 28: Notification Management module, this let the admin manage user accounts, roles, and permissions effectively while enforcing security. It also helps in ensuring that crucial system notifications and alerts are timely delivered to targeted users at the right time. In fact, this module further supports centralized notification control, which will basically enable administrators

to monitor, customize, and manage message distribution with the aim of enhancing communication and user awareness within the system.

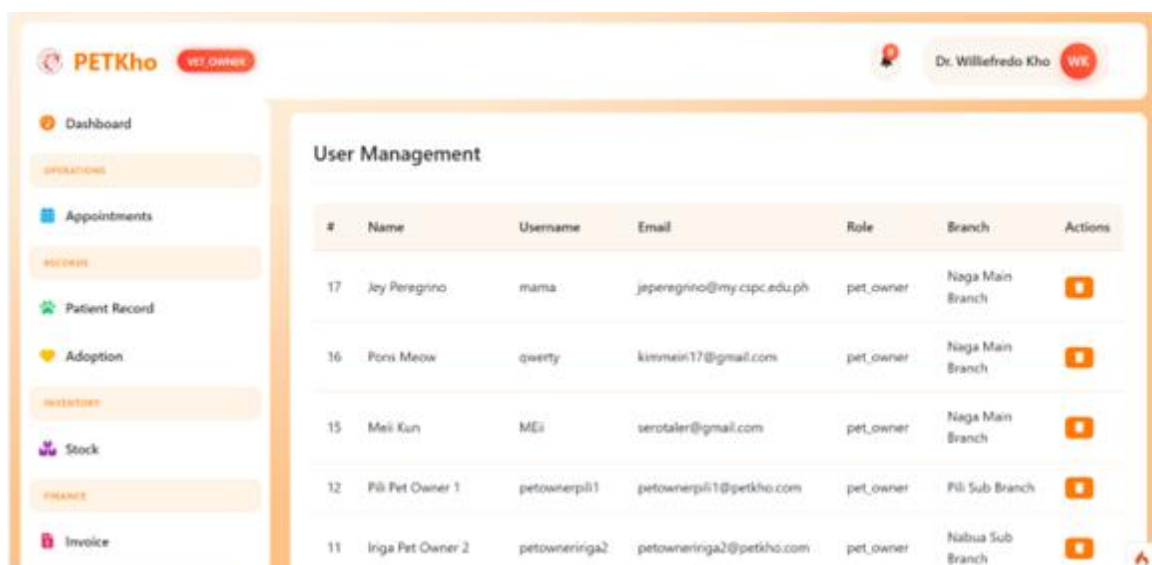


Title	Message	Link	Date
Appointment Accepted	An appointment was accepted in branch #3	Open	2025-10-26 14:51:48
Appointment Cancelled	An appointment was cancelled in branch #3	Open	2025-10-26 14:50:19
New Adoption Application	A new adoption application has been submitted.	Open	2025-10-26 14:47:20
New Adoption Application	A new adoption application has been submitted.	Open	2025-10-26 14:39:18
New Adoption Application	A new adoption application has been submitted.	Open	2025-10-26 14:39:17
Appointment Cancelled	An appointment was cancelled in branch #1	Open	2025-10-26 12:25:37
Appointment Accepted	An appointment was accepted in branch #1	Open	2025-10-25 12:10:55
New Adoption Application	A new adoption application has been submitted.	Open	2025-10-25 07:22:25
New Adoption Application	A new adoption application has been submitted.	Open	2025-10-25 07:22:25
Appointment Transferred	An appointment was transferred from branch #2 to #1	Open	2025-10-25 06:15:49

Figure 28: Notification Management

Notifications in the Kho Veterinary Clinic Management System are generated and sent by various users, including customers, main and sub-branch staff, and the veterinary owner. This process is managed through the system administration module and involves three main functions: management of notifications for employees, customer notifications, and overall communication within the system. All messages are stored in the "D8: Notifications" database, ensuring that they can be tracked, monitored, and accessed by relevant parties when needed, which helps maintain effective communication and operational coordination across the clinic.

Figure 29 displays the interface of the User Management tool within the Kho Veterinary Clinic system. This interface serves as a central hub for system administrators, allowing them to efficiently manage all user accounts registered on the platform. It provides functionalities such as adding, updating, and removing user accounts, assigning roles, and controlling access levels, ensuring that user management is organized, secure, and aligned with the operational needs of the clinic. It also allows administrators to quickly view user activity and account status, supporting



#	Name	Username	Email	Role	Branch	Actions
17	Joy Peregrino	mama	jeperegrino@my.cispc.edu.ph	pet_owner	Naga Main Branch	[Edit] [Delete]
16	Poni Meow	qwerty	kinmei17@gmail.com	pet_owner	Naga Main Branch	[Edit] [Delete]
15	Meli Kun	MEli	serotaler@gmail.com	pet_owner	Naga Main Branch	[Edit] [Delete]
12	Pili Pet Owner 1	petownerpili1	petownerpili1@petkho.com	pet_owner	Pili Sub Branch	[Edit] [Delete]
11	Iriga Pet Owner 2	petowneririga2	petowneririga2@petkho.com	pet_owner	Nabua Sub Branch	[Edit] [Delete]

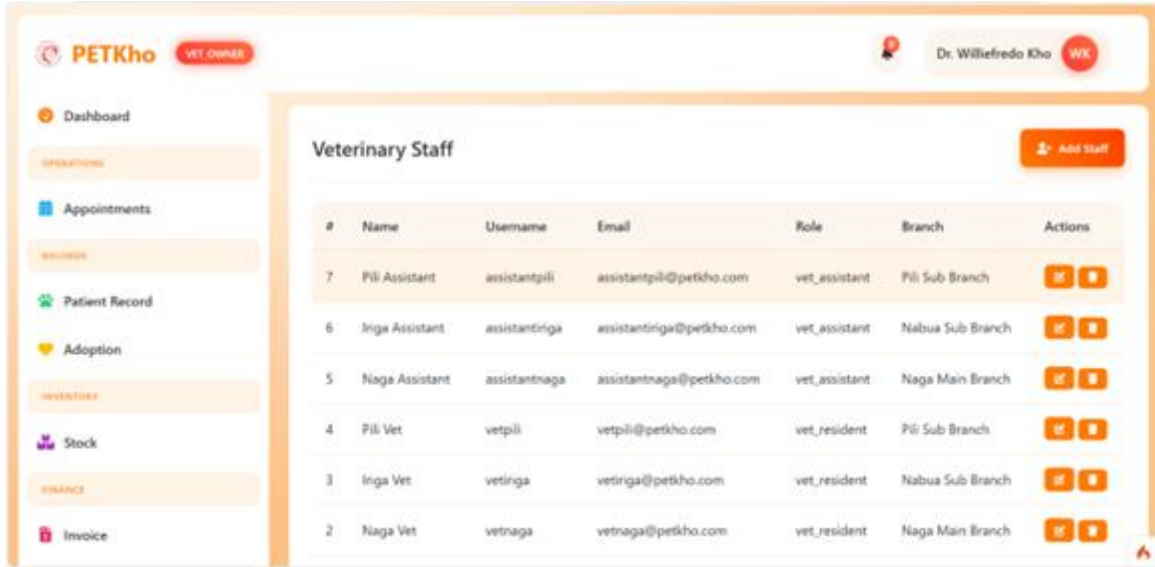
Figure 29: User Management

effective monitoring and oversight of system usage. Overall, this interface enhances administrative control by streamlining user management tasks and maintaining system security and accountability.

In figure 29, we have the User Management interface for the Kho Veterinary Clinic. It shows a general overview of user accounts with relevant metrics. From the interface, we can also search, view user roles, status, as well as user verification. It also includes the user name, user role, user status, last login, creation date, as well as actions regarding user information.

Figure 30 below depicts the Veterinary Staff Management module, where the administrator is able to The Notification Management module provides automated notifications for major events happening within the system, such as appointment reminders, stock updates, and adoption notifications. These notifications play an integral role in assisting the administrator to reduce the workload for such occurrences while keeping both employees and clients up to date with the happenings within the system. It's therefore possible for the users to customize the notifications, such as the information to be sought or the updates to be sent to the users, thereby keeping the users more related to the system for better coordination among the various branches of the veterinary management system. Additionally, this module improves communication efficiency by ensuring

timely and relevant information is delivered to the right users at the right time. Overall, the module contributes to a more organized and responsive veterinary management system.



#	Name	Username	Email	Role	Branch	Actions
7	Pili Assistant	assistantpili	assistantpili@petkho.com	vet_assistant	Pili Sub Branch	[Edit] [Delete]
6	Iriga Assistant	assistantiriga	assistantiriga@petkho.com	vet_assistant	Nabua Sub Branch	[Edit] [Delete]
5	Naga Assistant	assistantnaga	assistantnaga@petkho.com	vet_assistant	Naga Main Branch	[Edit] [Delete]
4	Pili Vet	vetpili	vetpili@petkho.com	vet_resident	Pili Sub Branch	[Edit] [Delete]
3	Iriga Vet	vetiriga	vetiriga@petkho.com	vet_resident	Nabua Sub Branch	[Edit] [Delete]
2	Naga Vet	vetnaga	vetnaga@petkho.com	vet_resident	Naga Main Branch	[Edit] [Delete]

Figure 30: Veterinary Staff Management

In Figure 30, Veterinary Staff Management is an interface in the Kho Veterinary Clinic system for managing and keeping track of all veterinary residents and assistants. For ease of management and understanding, a number of details concerning members of the veterinary staff is included. These details include identification, email, branch, and account dates, ensuring efficiency and ease in managing and keeping track of the members of staff.

Figure 31 below is an image depicting the Branch Management module as part of the Kho Veterinary Clinic application. This interface allows administrators to efficiently manage all branches of the veterinary clinics through a centralized platform. The module provides tools to monitor staff performance, track inventory levels, and assess service utilization at each branch. Additionally, it enables administrators to generate detailed reports comparing the performance of different branches over time, helping to identify areas for improvement. By centralizing branch management, the system not only enhances operational efficiency and ensures consistent service quality across all locations but also supports informed decision-making and strategic planning for

the clinic's overall growth. It further allows for quick adjustments and interventions at individual branches to address issues promptly and maintain smooth operations.

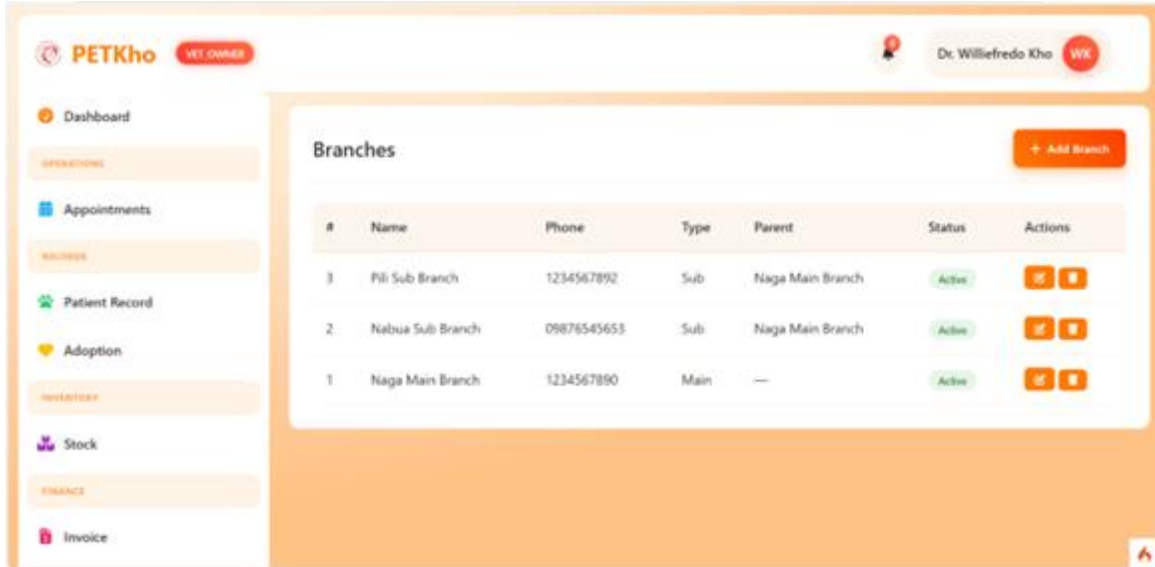


Figure 31: Branch Management

On the Branch Management page, the general public will be able to see an extensive listing of all branches that have been registered to date through this application, with relevant branch information such as the branch's name and address provided for easy identification by clients who wish to visit them for such services as creating an account with their clinic by use of the login credentials provided in this application for efficient and well-coordinated services by this application with its centralized management to help efficiently supervise operations in this clinic that have branches in diverse places as seen with the PetKho Portal application.

4.3.2 Client Module

Figure 32 shows the User Login Page of the PetKho Portal. This page serves as the secure entry point for all users, including pet owners, veterinarians, and clinic administrators. It validates user credentials to ensure that only authorized individuals gain access to the system. Additionally, it provides options for new users to register and for existing users to recover forgotten passwords, enhancing accessibility and user convenience. The straightforward design ensures that users can

navigate the login process quickly and securely, while also supporting a consistent and professional interface that aligns with the overall usability goals of the portal.

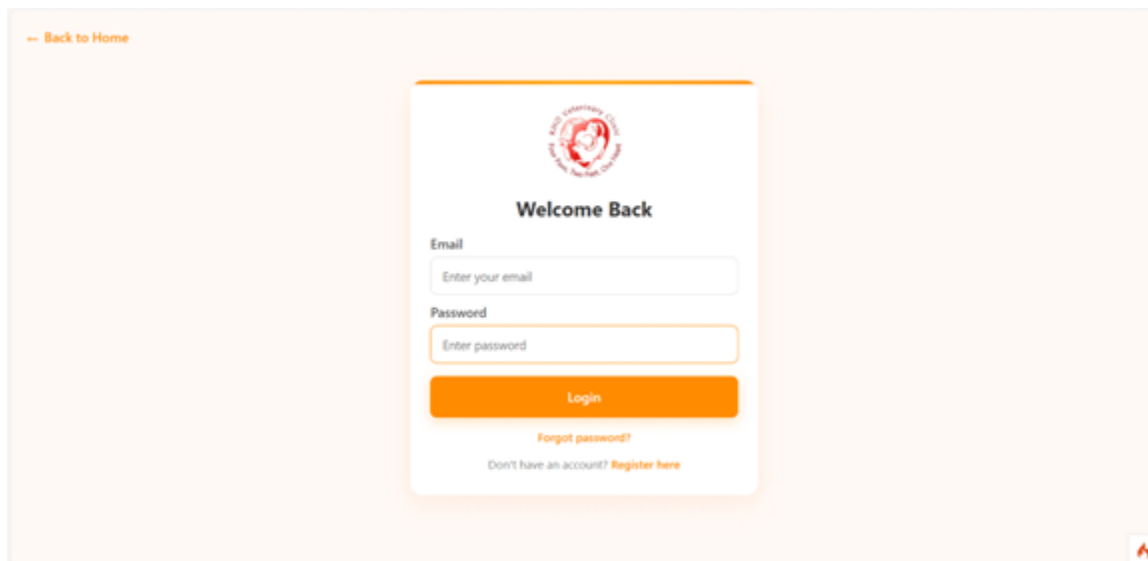


Figure 32: User Login Page

This section acts as an interface for users of the system, as it helps them to successfully and safely login to their respective veterinary management accounts. Information such as an email and a password are requested of users to enable them to gain access to their accounts. Nevertheless, users have an alternative option of creating a new account or recovering their forgotten passwords; this interface acts as an informative section to those visiting Kho Veterinary Clinic and introduces them to their respective branches and an appropriate description of their services. Moreover, this section acts as a security component that enables users to have an easy and safe handle of their personal and pet-related information. It also acts as an important component that adds value to users as it simplifies their user interface and directs them to their respective dashboard after an effective login.

Figure 33 shows the User Dashboard of the Kho Veterinary Clinic system. The user dashboard serves as the primary interface where users can view and manage all their interactions with the system. It provides quick and convenient access to key features such as appointment

scheduling, pet records, treatment history, and notifications, allowing users to stay informed and engaged. The dashboard is designed to be intuitive and visually organized, ensuring a seamless and user-friendly experience while helping users navigate the system efficiently.

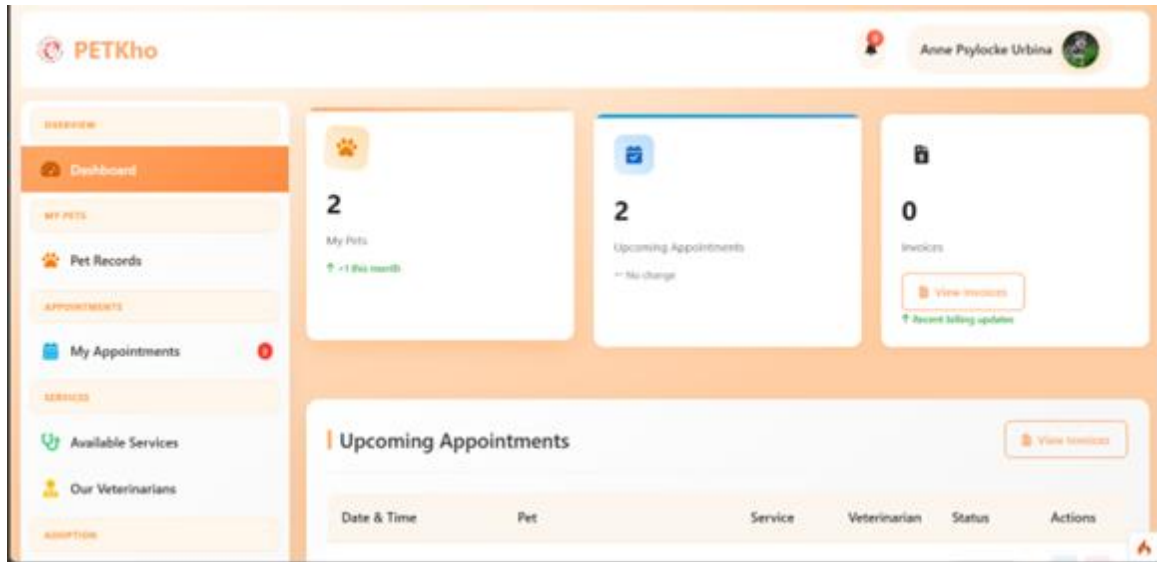


Figure 33: User Dashboard

This section of the User Dashboard allows users to view important information such as upcoming appointments they have scheduled, detailed pet information including profiles and vaccination records—whether pending or completed—and the history of past consultations. The interface is designed to help pet owners efficiently monitor their pets' health and stay informed about the various services offered by the clinic. Moreover, the User Dashboard serves as a central gateway for interacting with the PetKho Portal's features, enabling users to access additional tools and resources easily. This functionality promotes better communication between the clinic and pet owners, helping them stay engaged in their pets' care and overall health management.

Figure 34 illustrates the Appointment Management Page, which allows users to view, schedule, and manage all pet-related appointments in an organized manner. The page provides detailed information about each appointment, including the date, pet name, type of service, and current status, ensuring that users can track their appointments efficiently and with peace of mind.

By consolidating all appointment-related details in one interface, the system helps reduce scheduling conflicts, improves time management for pet owners, and enhances overall user experience when interacting with the veterinary clinic.

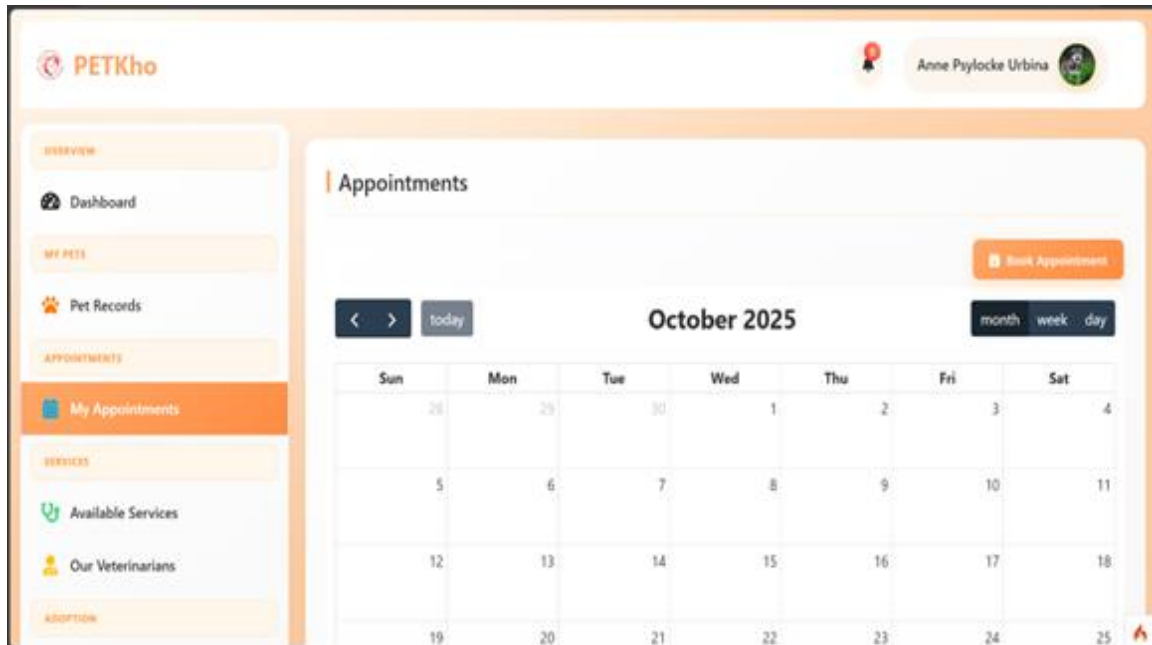


Figure 34: User Appointment Management Page

This is the Appointment Management dashboard from the Customer/Pet Owner perspective, which supports the client-side of the 2.1 Appointment Scheduling process. It allows direct access to the customer's pet appointments, sourced from the D2: Appointments store. It succinctly highlights all relevant information, such as the name of the pet, Shih; the servicing branch, Pili Sub Branch; the service, Vaccination; and the status, completed. Most importantly, the big + Schedule Appointment button allows the pet owner to create new bookings, while the 'Actions' column provides the ability to manage or query about existing appointments, thus placing power over services with the schedule in the hands of the pet owner. Besides offering a historical record of services provided to the customer, this system offers real-time interaction with the scheduling of the clinic-a very important component of effective client service.

Figure 35 User Pet Record Page allows clients to see and manage detailed information

about their registered pets. Full access is given to health data such as breed, gender, weight, and medical notes for each of the pets, which helps the clients be well informed about the general status and case history of their pets.

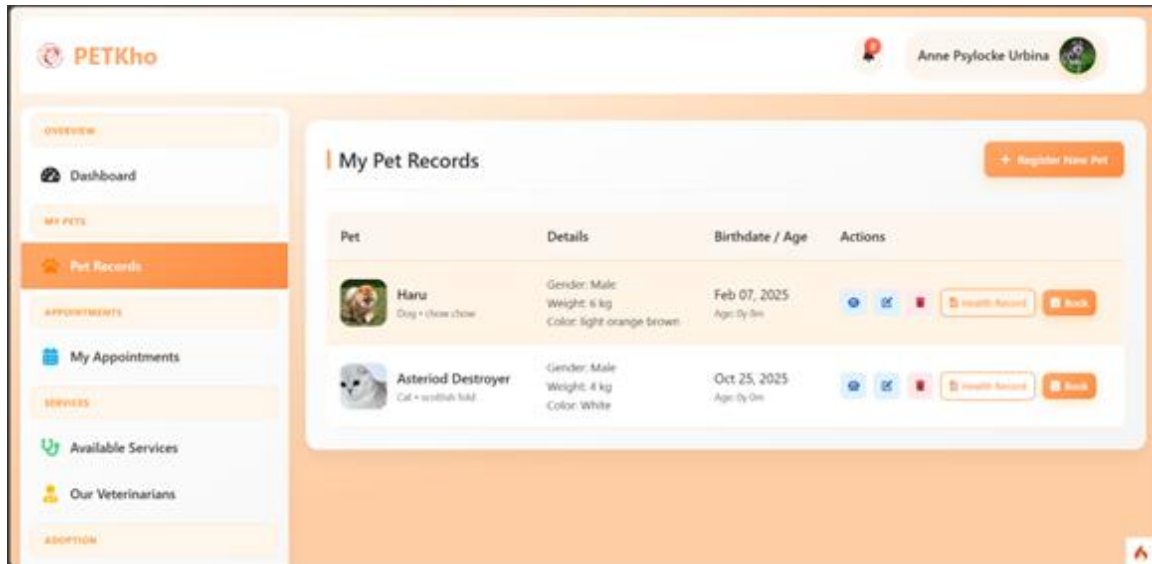


Figure 35: User Pet Record Page

This is the User's Pet Records dashboard as viewed by the Customer/ Pet Owner. The customer can ensure he remains fully informed of the status and history of his pet. While the main action, here, is to view, yet the + Add New Pet button helps the owner to create 3.1 Add / Update Pet Records by providing new information about the patient, directly into the system. which enables them to perform 3.2 View / Download Pet Records from the D3: Pet Records store. The interface provides a card-based detailed view of their registered pets along with key attributes: species, breed, physical stats and vital Medical Notes.

Figure 36 shows the User Adoption Page, which allows pet owners to browse through the various pets available for adoption and monitor the status of their requests. The page offers a systematic overview of adoption statuses, making it easy for users to navigate, identify available pets, submit applications, and track approvals or updates. This interface ensures that the adoption process is transparent, user-friendly, and efficiently managed, helping pet owners engage

effectively with the clinic's adoption services.

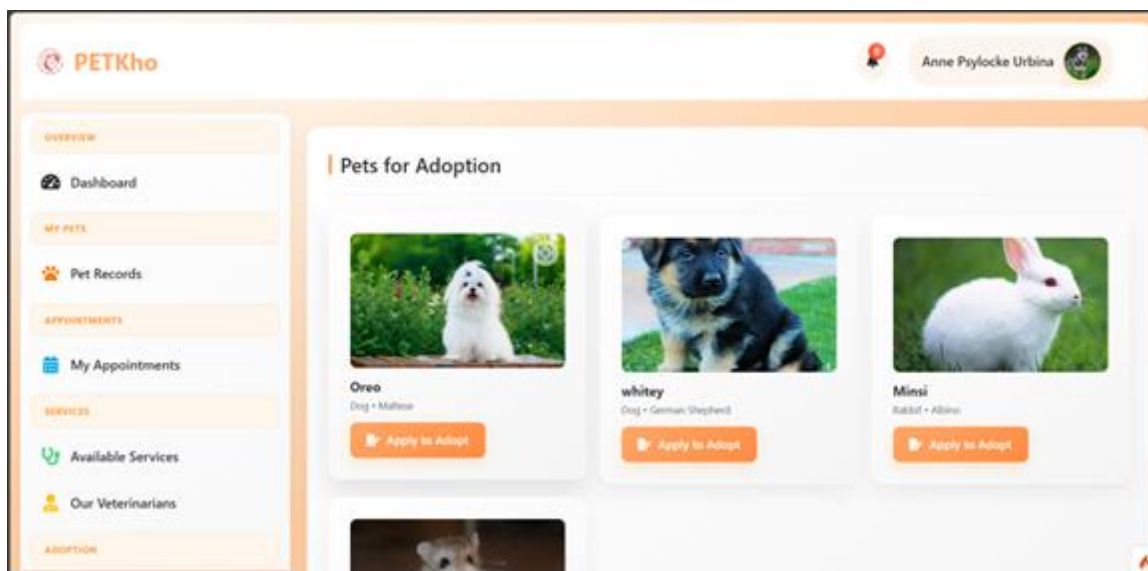
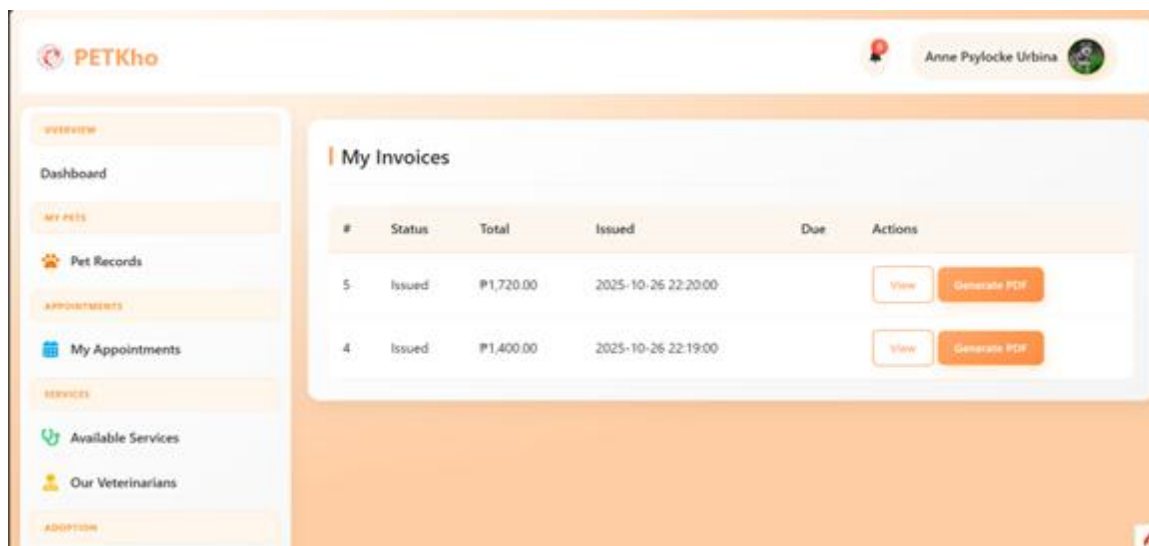


Figure 36: User Pet Adoption Page

The "User - Pet Adoptions" dashboard, which is being viewed by the customer/pet owner, Kyle, is the customer-facing side of the 5.0Pet Adoption application. The application is fully connected to the back-end information store, DesignD5: Adoptions. The main "Browse Pets" tab enables the customer to actively view all the available pets while also being provided with an overview of the application as a whole through vital stats like the '5 Total Pets' and the '3 Urgent pets' statistic. The visual style of the application, including the 'Available' and 'Urgent' badges on the pets like 'kore' and 'Buddy,' is necessary to direct the clients' attention to the 'urgent' animals to ensure their timely adoptions.

Secondly, the "My Applications" tab is a crucial element that plays a key role in the self-service aspect of the customer. This enables the client to easily monitor the progress of any applications submitted without having to directly contact the clinic. This suggests that the customer is empowered in the entire pet adoption process, starting with the expression of interest attraction until the staff present the client with the official results through 5.3 Approve / Reject Adoption Request.

Figure 37 shows the User Invoice Page, where customers can view and manage their billing records.



#	Status	Total	Issued	Due	Actions
5	Issued	₱1,720.00	2025-10-26 22:20:00		View Generate PDF
4	Issued	₱1,400.00	2025-10-26 22:19:00		View Generate PDF

Figure 37: User Invoice Page

The "User Invoices Page" dashboard provided here represents the customer/pet owner's (Kyle) view for his/her own financial records, which facilitates and directly involves the 4.0 "Billing Management" process and data extracted from the "D4: Invoices." Obviously, under the "Billing & Payments," all the required billing information for the follow-up appointments has been listed in detail, including "Invoice," "Pet (Shih)," "Service Date & Time," "Items," and "Total Amount." The very important "Print" option under "Actions," however, allows the customer to obtain his/her invoice, which represents the final stage in the customer-side data flow for "4.3 Send Invoice to Customer," ensuring full customer involvement and understanding in his/her financial transactions with the clinics.

Figure 38 illustrates the User Notification Page, which provides users with access to important updates related to their pets, appointments, and other clinic activities. Through this page, users can stay informed about appointment confirmations, changes in schedules, reminders for vaccinations or treatments, and other relevant notifications from the clinic. By consolidating all updates in one interface, the system ensures that clients remain connected, engaged, and well-informed about their

pets' care, promoting timely actions and improving overall communication between the clinic and its users.

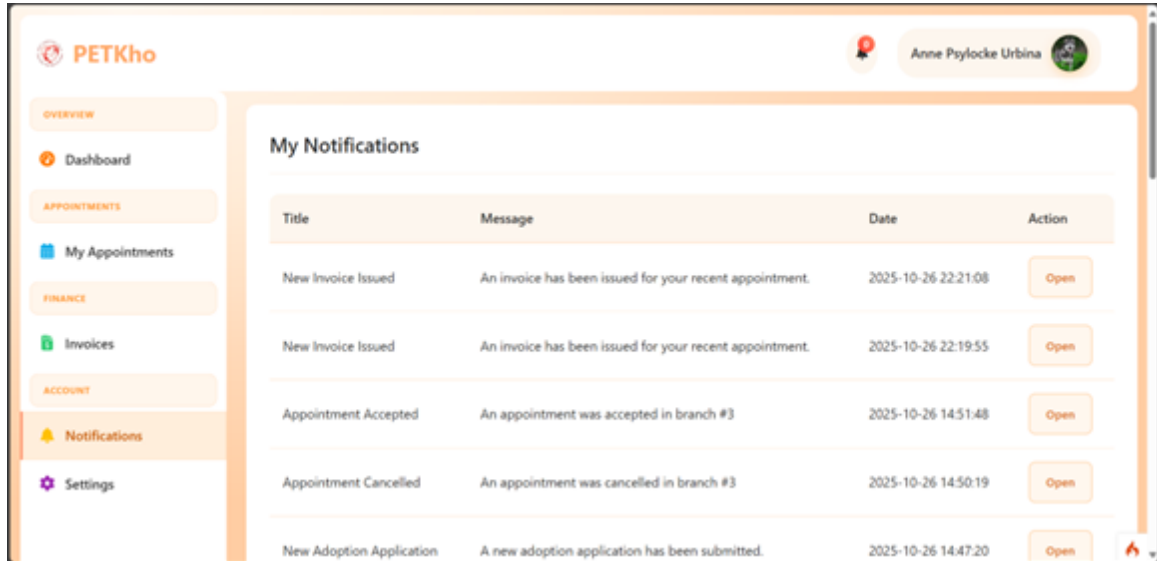


Figure 38: User Notification Page

This User Notifications Page Dashboard serves as the client-side interface for the Customer or Pet Owner to receive communication and updates in a clear and user-friendly manner. It directly supports the 9.2 Receive Notifications process and interacts with the D8: Notifications data store within the system, ensuring that all relevant messages are captured and delivered efficiently. The interface presents a collection of user-specific, event-based notifications, keeping the customer informed about activities related to their pets as well as appointments they have scheduled within the system. Notifications include key system events such as "Pet Updated," "Follow-up Created after visiting the Vet," "Appointment Confirmed," and other critical updates. Users can click on individual notifications to view more details or take immediate actions, such as rescheduling appointments or updating pet information. Additionally, the dashboard provides options to filter and sort notifications, allowing users to quickly access the most relevant updates and maintain an organized view of all communications, thereby improving engagement and ensuring timely responses to important events. This system also helps build trust with pet owners by keeping them consistently informed and connected with their pets' care and clinic services.

Network Model

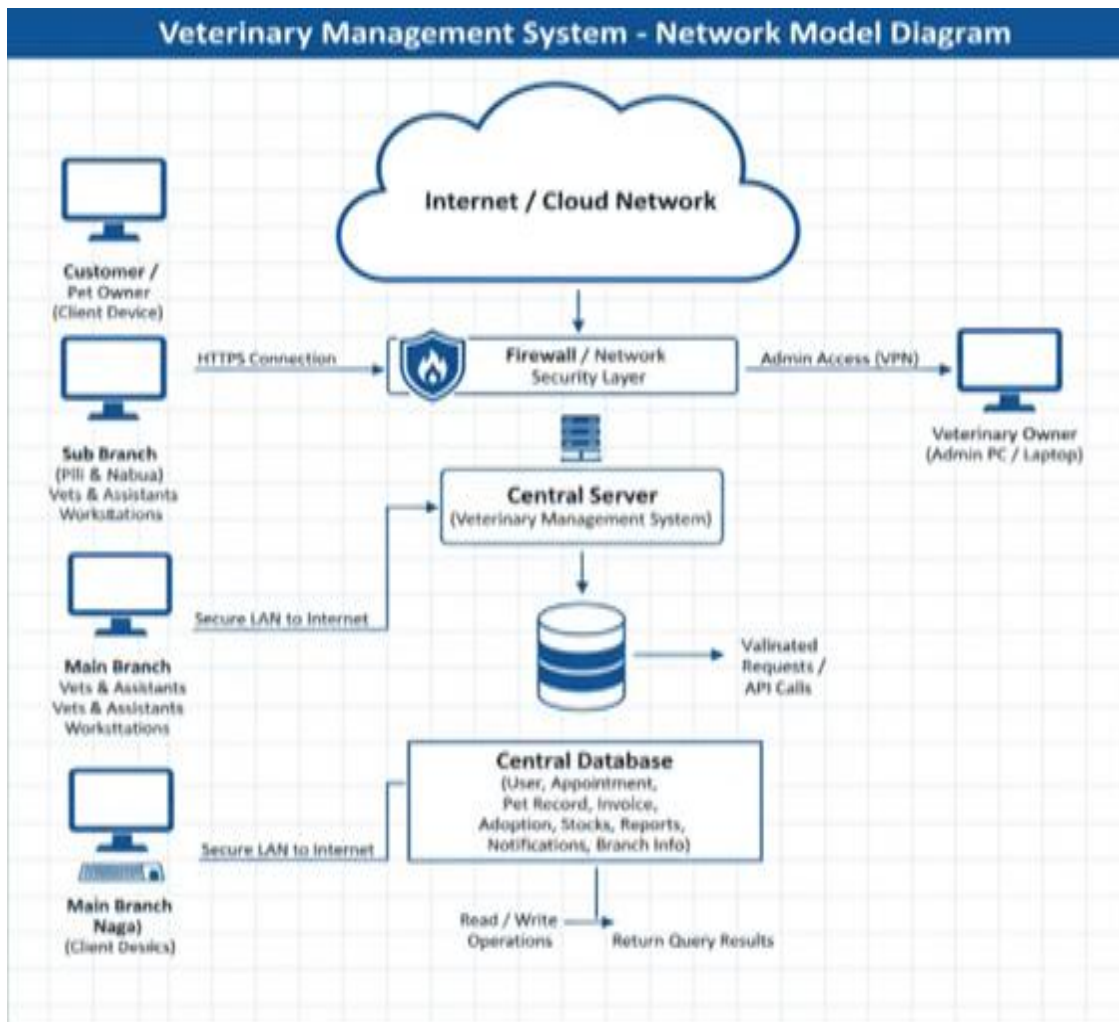


Figure 39: Network Model

The network model describes how the Veterinary Management System keeps all users and clinic branches connected safely. Through HTTPS, customers can securely access the system to book appointments or check pet records for requested services. Their actions go directly to the central server, while the vets and assistants in both main and sub-branches remain connected through a secure internet connection. All requests pass through a firewall, an abstract gate that ensures only protected and valid data enters the system.

This central server, after passing the firewall, interacts with the central database for information storage or retrieval. The veterinary owner can also access the system from any location through a VPN, thus being able to monitor the activities and manage all other branches securely. Generally, the network model represents a secure, well-organized, and integrated structure that supports the smoothness of the clinic's operation.

4.4 Testing

The testing phase is an important phase in the construction of the system since it enables the verification of whether the features of the system are functioning effectively as needed by the project. From this process, the researchers were also able to learn the functionality of the system when used in real situations. This phase includes the testing phase by the researchers to ascertain that the system meets the intended needs as desired by the project while working effectively. By the adoption of Agile Methodology, the researchers were referring to an approach that supports the continuous iteration of the system testing and development phases throughout the life cycle of the project as indicated in [7].

This table shows the results of performance testing conducted for the Pet Owners' side of the PetKho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada. It assesses the speed and stability of the system, whether it is responsive, as pet owners navigate the features and perform various actions. The purpose of this testing is so that all functions can load smoothly, respond quickly, and work efficiently without delays or any other performance issues for the end user. It tests the load or how well the system is performing during peak usage or when several pet owners access the portal simultaneously. Moreover, it looks into the system's capacity to maintain consistent performances while handling such requests as scheduling appointments, accessing pet records, and receiving notifications. The results of this testing help identify any bottlenecks or areas where optimization may be required. Performance testing also

ensures that the system can scale effectively as the number of users grows over time. Any issues discovered during testing can then be addressed to further enhance system reliability and user satisfaction.

Table 14: Pet Owners Performance Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Speed	4.00	Strongly Agree
TC-2	Usability	4.00	Strongly Agree
TC-3	Effectiveness	4.00	Strongly Agree
TC-4	Reliability	4.00	Strongly Agree
TC-5	Satisfaction	4.00	Strongly Agree
TC-6	Responsiveness	4.00	Strongly Agree
TC-7	Stability	4.00	Strongly Agree
TC-8	Efficiency	4.00	Strongly Agree
TC-9	Performance	4.00	Strongly Agree
TC-10	Usefulness	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 14 presents results from the Performance Testing conducted on the Pet Owners' side of the PetKho Portal. In all the dimensions, including speed, ease of use, effectiveness, and reliability, the ratings of the system were rated highest. This implies that pet owners found the portal fast, smooth, easy to use, and stable; reliable features provide an experience found the system helpful, responsive, ready to use for their daily needs.

This table summarizes the performance testing on Vet Owners/Resident Doctors' side of PetKho Portal and details how brisk and nimble the system is at managing appointments, pet records, and client information. The testing ensures that the system works smoothly, handles multiple actions, and supports daily clinic work efficiently without any jitters or stutters for a steady

experience provided to the veterinarians. Any issues discovered during testing can then be addressed to further enhance system reliability and user satisfaction.

Table 15: Vet Owners/Resident Doctor Performance Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Learnability	4.00	Strongly Agree
TC-2	Usability	4.00	Strongly Agree
TC-3	Effectiveness	4.00	Strongly Agree
TC-4	Reliability	4.00	Strongly Agree
TC-5	Satisfaction	4.00	Strongly Agree
TC-6	Navigation	4.00	Strongly Agree
TC-7	Guidance	4.00	Strongly Agree
TC-8	Efficiency	4.00	Strongly Agree
TC-9	Acceptance	4.00	Strongly Agree
TC-10	Usefulness	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 15 presents an illustration of the result of Performance Testing of the Veterinary Owners/Resident Doctors of PetKho Portal under the Performance Testing. The system was assessed based on several factors such as how easy it is to learn, how easy it is to use, how good it is to work with, and how reliable it is. All test cases scored a rating of 4.00 as it means “Strongly Agree.” This illustrates that all veterinarians had an easy time learning and using the system as it presents easy instructions and features that provide easy and effective operations. Moreover, it illustrates that the system meets the purpose and satisfaction of the users very effectively.

This above table shows the results obtained in the Performance Testing, which targeted the IT Experts portion of the PetKho Portal. Performance Testing is a test that measures the efficiency of a given system. In this case, the Performance Testing measured how effectively the PetKho

Portal performs, including how it can carry multiple tasks at a given time and how effectively the PetKho Portal can operate.

Table 16: IT Experts Performance Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Performance	4.00	Strongly Agree
TC-2	Performance	4.00	Strongly Agree
TC-3	Reliability	4.00	Strongly Agree
TC-4	Reliability	4.00	Strongly Agree
TC-5	Efficiency	4.00	Strongly Agree
TC-6	Performance	4.00	Strongly Agree
TC-7	Stability	4.00	Strongly Agree
TC-8	Efficiency	4.00	Strongly Agree
TC-9	Robustness	4.00	Strongly Agree
TC-10	Reliability	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 16 shows the ratings for performance, reliability, efficiency, stability, and robustness. All test cases scored 4.00, or “Strongly Agree,” showing that IT experts found the system fast, reliable, and efficient. The portal handles multiple tasks smoothly without errors or slowdowns. Overall, the results confirm that the PetKho Portal is stable, dependable, and ready for everyday use.

This table shows the results of the User Acceptance Testing (UAT) for the Pet Owners’ side of the PetKho Portal. The testing checks if the system meets pet owners’ needs, like scheduling appointments, viewing pet records, and getting reminders. It ensures all features work well, are easy to use, and provide a smooth experience. The goal is to confirm the system is ready for real-world use and satisfies its users.

Table 17: Pet Owners User Acceptance Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Responsiveness	4.00	Strongly Agree
TC-2	Speed	4.00	Strongly Agree
TC-3	Stability	4.00	Strongly Agree
TC-4	Reliability	4.00	Strongly Agree
TC-5	Efficiency	4.00	Strongly Agree
TC-6	Performance	4.00	Strongly Agree
TC-7	Navigation	4.00	Strongly Agree
TC-8	Multi-tasking	4.00	Strongly Agree
TC-9	Learnability	4.00	Strongly Agree
TC-10	Satisfaction	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 17 shows the results of the UAT conducted for the Pet Owners' side of the PetKho Portal. The present system was assessed for the following key sub-characteristics: responsiveness, speed, stability, reliability, efficiency, performance, navigation, multi-tasking, learnability, and overall satisfaction. All test cases received the highest rating of 4.00, which means "Strongly Agree," that the pet owners strongly agreed that in all concerns, the system performs very well.

From the results obtained, it is safe to conclude that it is easy to schedule an appointment and to view pet records and reminders. Moreover, it is evident that the pet owners are comfortable working with the portal as it proves to be useful and reliable. It should also be noted that pet owners are highly satisfied with the portal and that it is easy to learn and utilize. Therefore, based on the findings of UAT, it is evident that the PetKho portal serves all the purposes of pet owners and is ready to serve them efficiently and effectively. It is also evident that it is reliable and easy to utilize as it proves to be useful to pet owners.

This table depicts the results of the User Acceptance Testing done for the Vet Owners/Resident Doctors side of the PetKho Portal. This testing is targeted at ensuring the system meets the needs and expectations of veterinarians in performing daily tasks in the clinic, such as appointment management, updating of pet records, and maintenance of client information. It helps to ensure all features work as intended, are easy to use, and support efficient workflow. The objective of this UAT is confirmation that the system is ready to go live and meets its end-user requirements prior to going full-scale live in the clinic. The testing also generates feedback from the veterinarians relative to the usability, layout, and overall functionality of the system. It also brings into view possible enhancements that may further improve system performance and user experience

The feedback obtained from the User Acceptance Testing (UAT) is carefully analyzed and systematically incorporated into the development of the PetKho Portal to refine its features, improve functionality, and address any usability concerns that may arise. This iterative approach allows the development team to closely align the system with the real-world workflows and operational needs of veterinary staff, ensuring that tasks such as appointment scheduling, pet record management, and report generation are both intuitive and efficient. Any issues identified during testing, including navigation difficulties, incomplete or missing information, or suggestions for new features, are carefully documented, prioritized, and addressed before the final deployment of the system. By actively involving veterinarians, clinic staff, and other end-users in this testing phase, the PetKho Portal not only meets its technical and functional requirements but also receives practical validation from those who will be using it on a daily basis. This collaborative process increases user confidence, encourages adoption, and ensures that the system is not only technically robust but also user-friendly and responsive to real operational needs. Additionally, the insights gained from UAT help anticipate potential challenges and streamline workflows, further enhancing the efficiency and reliability of the system. Overall, this careful integration of user feedback

guarantees that the PetKho Portal is practical, dependable, and easy to use, providing a solid foundation for long-term adoption and success in supporting veterinary clinic operations.

Table 18: Vet Owners/Resident Doctor User Acceptance Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Responsiveness	4.00	Strongly Agree
TC-2	Speed	4.00	Strongly Agree
TC-3	Stability	4.00	Strongly Agree
TC-4	Reliability	4.00	Strongly Agree
TC-5	Efficiency	4.00	Strongly Agree
TC-6	Performance	4.00	Strongly Agree
TC-7	Navigation	4.00	Strongly Agree
TC-8	Multi-tasking	4.00	Strongly Agree
TC-9	Stability	4.00	Strongly Agree
TC-10	Satisfaction	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 18 outlines the results of User Acceptance Testing for Vet Owners/Resident Doctors. Here, all tests generated similar results with an average score of 4.00 that translates to “Strongly Agree,” an outcome that confirms that the system is “easy to use,” “reliable,” and “efficient,” able to “accomplish tasks,” ready for everyday use in the “clinic,” among others.

The table shows the Use Case Testing of Pet Owners. This checks whether the system is assisting in making proper appointments and handling reminders. It tests whether the system is performing well and is ready for use. In addition, the test checks how easily users can interact with the system. It measures the system's efficiency and correctness in handling simultaneous requests. It helps in ensuring that the portal is smooth and user-friendly for all pet owners. This confirms that the system meets the practical needs of its users.

Table 19: Pet Owners Use Case Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Speed	4.00	Strongly Agree
TC-2	Usability	4.00	Strongly Agree
TC-3	Effectiveness	4.00	Strongly Agree
TC-4	Reliability	4.00	Strongly Agree
TC-5	Satisfaction	4.00	Strongly Agree
TC-6	Responsiveness	4.00	Strongly Agree
TC-7	Stability	4.00	Strongly Agree
TC-8	Efficiency	4.00	Strongly Agree
TC-9	Performance	4.00	Strongly Agree
TC-10	Usefulness	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 19 presents the results of the Use Case Testing on the Pet Owners' side of the PetKho Portal. The tests scored 4.00, "Strongly Agree," which means the system is easy to use, reliable, and efficient. It works as it should: smooth appointment scheduling, reviewing records of pets, and getting reminders-the basics, but helpful in daily life.

This table illustrates the results of testing the Use Case for the Vet Owners and Resident Doctors, focusing on the system's ability to efficiently manage appointments, pet records, and client information. The testing process ensures that all functionalities operate correctly and reliably, confirming that the system is capable of handling daily veterinary tasks without interruptions. Additionally, checks are performed to verify that each feature supports a smooth workflow, meets the practical needs of the veterinarians, and aligns with real-world operational requirements. By evaluating the system in this manner, the testing validates that the PetKho Portal is not only reliable

and user-friendly but also fully capable of supporting daily veterinary operations, improving efficiency, and enhancing the overall experience for both clinic staff and clients.

Table 20: Vet Owners/Resident Doctors Use Case Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Correctness	4.00	Strongly Agree
TC-2	Functionality	4.00	Strongly Agree
TC-3	Effectiveness	4.00	Strongly Agree
TC-4	Accuracy	4.00	Strongly Agree
TC-5	Realism	4.00	Strongly Agree
TC-6	Guidance	4.00	Strongly Agree
TC-7	Completeness	4.00	Strongly Agree
TC-8	Reliability	4.00	Strongly Agree
TC-9	Robustness	4.00	Strongly Agree
TC-10	Effectiveness	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 20 presents the Use Case Testing result of the PetKho Portal under the Vet Owners/Resident Doctors entity of interest. Herein, system performance was assessed based on sub-characteristics such as correctness, functionality, effectiveness, accuracy, realism, guidance, completeness, reliability, and robustness. All test cases scored 4.00 or “Strongly Agree” means that all the veterinarians strongly agreed to all of these as their assessments of system performance.

Based on these results, it can be concluded that the portal indeed enables a veterinarian to perform day-to-day tasks in a clinical environment in an efficient and accurate manner by providing clear instructions on every required function and process, implementing all processes in a realistic way following day-to-day clinical environments, and making sure that all components of the portal

perform successfully. As a matter of fact, these results conclude that the PetKho Portal is indeed highly reliable and ready to be utilized day to day in a clinical environment.

This table indicates the test results from the Security Testing for the "IT Experts" side of the PetKho Portal. This kind of testing checks the system for the adequacy of its security in handling its user data from unauthorized sources.

Table 21: IT Experts Security Testing

TEST CASE ID	SUB-CHARACTERISTICS	RESULT	INTERPRETATION
TC-1	Security	4.00	Strongly Agree
TC-2	Security	4.00	Strongly Agree
TC-3	Security	4.00	Strongly Agree
TC-4	Security	4.00	Strongly Agree
TC-5	Security	4.00	Strongly Agree
TC-6	Security	4.00	Strongly Agree
TC-7	Security	4.00	Strongly Agree
TC-8	Security	4.00	Strongly Agree
TC-9	Security	4.00	Strongly Agree
TC-10	Security	4.00	Strongly Agree
RESULTS		4.00	Strongly Agree

Table 21 shows the result obtained for the Security Testing for the IT Experts for the PetKho Portal. All test cases achieved the highest rating, which was 4.00 and stood for "Strongly Agree," implying that IT Experts considered the system very secure. The test examines whether the system secures data or not, whether it prevents unauthorized persons from accessing data or not, and so on. Moreover, the test recognized that the system's security features, such as login protection, data or unauthorized access protection, and so on, are working appropriately or not. The PetKho Portal results reveal that the system is reliable and secure and will operate safely in the clinic.

4.4.1 Overall Result

The table represents the test case scenarios for the Pet Owners' end of the Pet Kho Portal System, such as Performance Testing, User Acceptance Testing, and Use Case Testing, to assess the functionality of the system, its usability, and its effectiveness as a whole for everyday activities like booking an appointment or reminding the pet owners of their schedules.

Table 22: Test Case Result for Pet Owners

TYPES OF TESTING	OVERALL RESULT	LEVEL OF FUNCTIONALITY
Performance Testing	4.00	Strongly Agree
User Acceptance Testing	4.00	Strongly Agree
Use Case Testing	4.00	Strongly Agree
Total	4.00	Strongly Agree

With all test types receiving the highest rating of 4.00, "Strongly Agree," it indicates that pet owners find the PetKho Portal easy to use, fast, and reliable. This high rating demonstrates that the system performs exceptionally well in key functions such as booking appointments, accessing and reviewing pet records, and receiving timely reminders. The results confirm that the portal is fully ready for daily use, providing practical support to pet owners and enhancing their experience with the clinic's services while ensuring smooth, efficient, and dependable operation.

This table depicts the results of test cases performed on the Vet Owners and Resident Doctors side of the PetKho Portal, encompassing Performance Testing, User Acceptance Testing, and Use Case Testing. The evaluation examines how effectively the system supports veterinarians in carrying out their daily responsibilities within the clinic, including managing appointments, updating and accessing pet records, and maintaining accurate client information. By assessing these areas, the tests provide insights into the system's reliability, usability, and efficiency, ensuring that it meets the practical needs of veterinary staff while streamlining workflows and enhancing overall clinic operations.

Table 23: Test Case Result for Vet Owners/Resident Doctors

TYPES OF TESTING	OVERALL RESULT	LEVEL OF FUNCTIONALITY
Performance Testing	4.00	Strongly Agree
User Acceptance Testing	4.00	Strongly Agree
Use Case Testing	4.00	Strongly Agree
Total	4.00	Strongly Agree

The results showed that all the tests got the rating of 4.00, which is the strongest agreement and signifies that the veterinarians found the system very reliable and efficient. The system performs tasks very well and produces precise results, with clear navigation that facilitates the workflow. These results prove that the developed system is very reliable and can offer significant support by facilitating the daily operations of veterinarians with ease. Overall, the PetKho Portal is a viable platform that can support daily operations with ease.

This table presents the comprehensive results of the Test Cases conducted by IT Experts on the PetKho Portal, covering both Performance and Security Testing to ensure the system operates reliably under real-world conditions. Performance Testing evaluates the system's ability to handle multiple simultaneous tasks, maintain stability during extended or intensive use, and efficiently manage large volumes of data, which is crucial for supporting multiple users such as pet owners, veterinarians, and clinic staff interacting with appointments, pet records, and reports at the same time. Security Testing assesses the system's ability to protect sensitive information, including client details and pet medical records, through robust authentication, access controls, and encryption measures, ensuring data integrity and confidentiality are maintained at all times. By combining these assessments, potential bottlenecks, vulnerabilities, and areas for optimization are identified before full deployment, allowing the development team to enhance stability, efficiency, and security. The results confirm that the PetKho Portal is capable of supporting high workloads

while safeguarding critical information, providing a dependable, secure, and high-performing platform for all users, ensuring smooth veterinary operations and fostering confidence in the system's long-term reliability.

Table 24: Test Case Result for IT Experts

TYPES OF TESTING	OVERALL RESULT	LEVEL OF FUNCTIONALITY
Performance Testing	4.00	Excellent / Strongly Agree
Security Testing	4.00	Excellent / Strongly Agree
Total	4.00	Excellent / Strongly Agree

Both Performance and Security Testing had the highest rating, 4.00, “Strongly Agree,” that IT experts found the system very reliable, technically strong, secure, and capable of processing several actions with consistency in response time without losing performance or allowing security breaches to sensitive information like records of pets, owners' data, and appointments. These scores demonstrate that the system will not have errors, delays, or allow security breaches under real operational conditions. In summary, the PetKho Portal has been validated as a reliable, high-performance, secure platform that allows daily clinical operations to be carried out smoothly, safely, and reliably for IT experts, veterinarians, and owners. It also proves that the system is not only technically strong but also ready to be applied completely in the real clinical environment.

4.5 Deployment

The deployment phase marked the transition of the Petkho Portal from development to real-world use at Kho Veterinary Clinic Rinconada. After successful testing and validation, the system was installed and configured within the clinic's operational environment. This deployment covers server setup, database, and user account creation, and makes sure that the system is accessible and stable. This involves orientation and training of the clinic staff to make them familiar with the features and workflows of the platform. Initial feedback was gathered at deployment for any minor

tuning required to result in smooth adoption. At this phase, it will be ensured that the system is functional and accessible for day-to-day running of the clinic smoothly and performs the job of enhancing service delivery and making administrative works easier. Furthermore, during and after deployment, agile project management best fitted practices have also been discussed[5].

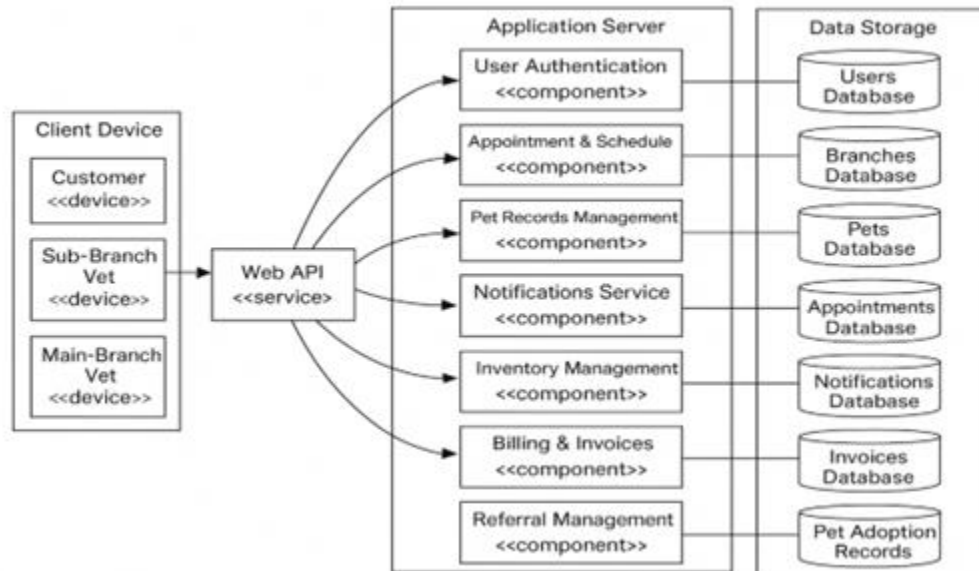


Figure 40: UML Deployment Diagram

The UML Deployment Diagram illustrates the overall system architecture of the PetKho Portal, showing how software components are distributed across different hardware and system nodes. It shows the interaction between the client devices, the web application server, as well as the data storage layer. For instance, the client devices, which represent the customers, the sub-branch veterinarians, as well as the main branch veterinarians, use a Web API service, which acts as a communication bridge so as to allow secure communication between the user interface as well as the back-end processes.

The Application Server is also home to various important software components such as User Authentication, Appointments/Schedules Management, Pet Records Management, Notifications Service, Inventory Management, Billing/Invoices, and Referral Management,

respectively. These components are instrumental in the effective running of the major functionalities of the system through the various user components. The Data Storage section includes various databases such as the Users Database, Branches Database, Pets Database, Appointments Database, Notifications Database, Invoices Database, and the Pet Adoption Records Database, respectively. From the above discussion, one is able to appreciate the various aspects depicted by the Pet Kho Portal System through the system design depicted in the above image.

Notes

- [1] Samer Al-Saqqa, Mohammed Sawalha, and Hesham AbdelNabi.. 2020. Agile software development: Methodologies and trends. *International Journal of Interactive Mobile Technologies*. 14, 11 (2020). <https://doi.org/10.3991/ijim.v14i11.13269>.
- [2] José Manuel Álvarez and Elena Roibás-Millán. 2021. Agile methodologies applied to Integrated Concurrent Engineering for spacecraft design. *Research in Engineering Design*. 32, 4 (2021). <https://doi.org/10.1007/s00163-021-00371-y>.
- [3] Sajjad Mahmood Cheema, Muhammad Aleem, Muhammad Arslan. 2023. A natural language interface for automatic generation of data flow diagram using web extraction techniques. *Journal of King Saud University - Computer and Information Sciences*. 35, 2 (2023). <https://doi.org/10.1016/j.jksuci.2023.01.006>.
- [4] Emmanuel Chibuike Daraojimba, Oluwaseun Stephen Adebayo, Abdulrahman Oladipo Aremu. 2024. COMPREHENSIVE REVIEW OF AGILE METHODOLOGIES IN PROJECT MANAGEMENT. *Computer Science & IT Research Journal*. 5, 1 (2024). <https://doi.org/10.51594/csitrj.v5i1.717>.
- [5] Abdulrahman Hassan, Ahmed Ali, and Mohammed Alshammari. 2021. Agile methodology in an saas deployment project: A pre- And post-saas deployment study. *International Journal of Information Technology Project Management*. 12, 2 (2021). <https://doi.org/10.4018/IJITPM.2021040101>.
- [6] Carlos Alberto Torres Romero, María Soledad Ramírez Montoya, and José Ramón Valdés Lozano. 2021. Software Architecture for Planning Educational Scenarios by Applying an Agile Methodology. *International Journal of Emerging Technologies in Learning*. 16, 8 (2021). <https://doi.org/10.3991/ijet.v16i08.20603>.
- [7] Thomas Hamilton 2022. Agile Methodology: What is Agile Model in Software Testing? *Guru99*. (2022).

CHAPTER 5

SUMMARY, FINDINGS, CONCLUSION AND RECOMMENDATIONS

The study provides a wide overview with a discussion on its scope, a highlight on the major findings, and a conclusion on important conclusions and recommendations. The study reviews insights based on how its research aims were attained. It focuses on valuable conclusions and improvements in its practical applications.

5.1 Summary

This study aims to create the PetKho Portal: A Smart Veterinary Care and Monitoring System for the Kho Veterinary Clinic Rinconada, aiming to innovate the current operation of the clinic to enhance the delivery of service to veterinarians and animal owners alike. A traditional approach was used in the clinic's operation prior to the introduction of information technology approaches, characterized by the use of traditional methods such as pen-and-paper records and an unstructured approach to communication.

In order to address these challenges, the researchers created a system that is based on a web application that helps to automate many functions of a clinic, such as appointment schedules, records related to pets, inventory control, billing, referrals, and notifications. Overall, the system is intended to create a hub that helps to improve accuracy, efficiency, and communication between those who run a clinic and those that seek such services.

The objectives were to identify the existing challenges in operation, design a workable and accessible system, perform black-box testing through performance, security, use case, and finally, usability testing, and assess its overall effectiveness.

The system involves two types of roles: Admin, who controls and updates information regarding pets, appointments, stock, invoices, and referrals; and Clients, who can easily create appointments, check the history of their pets, get updates, and send messages to the clinic. Also,

operating based on these kinds of roles allows the facilitation of less manual error and the maintaining of neat records of data.

The Agile Methodology was used in carrying out the research, which allowed continuous feedback and improvement after each development cycle. Black-box testing was performed to evaluate how the system responds to various inputs by using performance, security, workflow accuracy, and user satisfaction as parameters. The results revealed that the system responded quickly, maintained data integrity, executed workflows appropriately, and users offered favorable feedback about the system in all test categories.

The study concluded, in summary, that the portal was found to be a very reliable, secure, and efficient tool for addressing long-standing operational problems by way of automation, improved data accuracy, and better communication-a needed shift of the clinic toward digital transformation.

5.2 Findings

1. The study showed that Kho Veterinary Clinic Rinconada experienced several areas of inefficiency with its manual approach, which relates to the first objective of highlighting difficulties in clinic administration:

1.1 The current way of managing patient records involved a manually conducted system. There were challenges associated with retrieving past information about the patients. Personnel within the facility explained that retrieving past records took a lot of time. This manual system further increased the chance that records might be misplaced or incomplete, which again contributed to the overall precision and consistency of animal care.

1.2 The appointment system presently employed involves phone-based booking, which often resulted in double bookings and missed appointments. Notably,

numerous cases of appointment conflicts were observed, causing frustrations among staff and pet owners.

1.3 The means of communication with pet owners was mainly in person or through phone calls, which created delays in informing them of appointments or treatments scheduled. Some of the challenges cited by many of the pet owners were last-minute reminders of appointments or scheduled treatments.

2. Identifying the component system as an objective to develop for the alleviation of identified challenges, the PetKho Portal was able to be developed and implemented by taking into consideration a central portal where:

- 2.1 Authentication and authorization ensured that the accesses were secure for clinic staff and reduced any unauthorized use.

- 2.2 Appointments and pet records managed a centralized patient information system wherein the veterinarians and staff could get the records in no time and manage the appointment schedules quite efficiently.

- 2.3 Referral management facilitated the process of referring to and from veterinarians, lessening paper work and manual follow-up.

- 2.4 Inventory management tracked medical supplies and warned the staff when levels became low to prevent shortages.

- 2.5 Automated billing and invoicing successfully generated invoices and maintained financial records to minimize manual errors.

- 2.6 Notification and communication features sent reminders and updates to pet owners, ensuring timely follow-ups and increasing client satisfaction.

3. The results obtained, which were used to test the management system, are as follows:

- 3.1 The performance test showed that the overall System was efficient and did not hang much, even when performing many operations at the same time. All the

respondents rated the speed and response of the system as 4.00, interpreted as Strongly Agree, which meant that it was stable and reliable in performance.

3.2 Security testing was ensured so that user data and clinic records were safeguarded from unauthorized access. The mean score given by the respondents accounted for 4.00, which is interpreted as Strongly Agree, so this confirmed that sensitive information was duly safeguarded.

3.3 Use case testing showed that all functions, from appointment booking and records retrieval, to referral processing and billing, are working as expected. This means that all tested scenarios were successfully accomplished. A mean score of 4.00 interpreted as Strongly Agree was gotten, indicating that the system is fully functional and operating correctly.

3.4 User acceptance testing showed that staff and selected pet owners found the system user-friendly. In its entirety, the automated notification and smooth workflow highly improved the respondents' experience, as evidenced by the mean score of 4.00, interpreted as Strongly Agree.

4. The Petkho portal was successful in being deployed in the clinic to reduce errors and lost documents. The management of appointments was now well organized and increased in efficiency. The level of customer satisfaction was also increased through improvements in customer communication. The overall efficiency of the clinic was increased to allow employees to focus more on the patients.

5.3 Conclusions

On the basis of the findings, the following conclusions were made:

1. To identify challenges in existing processes in managing a veterinary clinic, the study concluded that in order to manage Kho Veterinary Clinic in Rinconada more efficiently, there were major inefficiencies in its existing manual processes. Manual processes were a

major draw back in managing customer records; it was a time-consuming process and often resulted in a lot of mistakes. Moreover, managing appointments through phone calls and using a logbook often resulted in duplicate appointments and missed appointments. Additionally, there were a lack of communication channels with Pet Owners.

2. To develop a component for the animal veterinary care and monitoring system, the study established that the proposed system through the PetKho Portal was able to address the challenges identified by successfully incorporating the identified features that promoted easier workflow management through the inclusion of authentication/authorization functionality, appointment management functionality for patients and records management, as well as invoicing and notifications.
3. To test how well it manages the system, it was evaluated that the PetKho Portal indeed adheres to and meets all the necessary standard criteria in terms of its functioning, efficiency, security, and usability. However, in terms of its efficiency, based on its performance testing, it was clear that it works well with its multiple operations in a single instance. Moreover, based on user acceptance, there was strong agreement on how well it works and how satisfied users are with its ease of use. Use case and user acceptance were affirmed based on strong satisfaction. Security was assured based on strong user data.
4. To deploy the developed system to Kho Veterinary Clinic Rinconada, The study concluded that the PetKho Portal works effectively in the real clinical environment. Its implementation reduced manual errors and lost records, improved appointment management, and overall communication with the pets' owners. Generally, operational efficiency improved and allowed the staff to focus more on patient care, which showed the effectiveness of the system in supporting day-to-day veterinary operations.

5.4 Recommendations

The following recommendations are proposed for future researchers who aim at

developing/improving upon various aspects of veterinary care management systems. These recommendations for future studies are based upon the outcomes of the current studies and aim at further improving efficiency, usability, and communication in a veterinary clinic.

1. Research operational processes and user needs at the veterinary clinic. In a future study, it is recommended to carry out an in-depth analysis of workflows in clinics and user needs, including more veterinary clinics, in order to have more specific operational issues. The design will then be robust enough to consider different environments.
2. Improve the functionality and integration of features within the system. It is recommended that researchers undertake more investigations into features that may enhance the usability of the program, duly considering mobile application integration and even the utilization of features like artificial intelligence to enforce appointment and vaccine schedules.
3. Improve the usability of the system and the user experience. It would be recommended that in the future, researchers assess the usability of the program over a period of time. It would benefit in illustrating the long-term effect of the program on clinic efficiency and client experience.
4. Assess system performance, reliability, and effectiveness in the long run. Researchers could investigate more advanced functions in role management and data analytics to further reduce manual errors in minimizing optimizations of clinic data organization, ensuring security and usability for multiple user types.
5. The researcher further recommends strengthening the security and enhancing the transaction processes. Future researchers are also encouraged to create a comprehensive billing and payment management module, including payment tracking and flexible payment options such as down payments for a complete billing system that would mean greater convenience and transparency for the client.

BIBLIOGRAPHY

Book Sources

Holger Görges. 2023. *Solutions and prospects for digital documentation of treatment data on-farm*. Smart Agricultural Technology. Elsevier B.V.

Olufunke Afuolabi. 2023. *Exploring the impact of emotional intelligence on veterinary nursing: Enhancing patient care and well-being*.

Gabrielle R. Cabrera. *A centralized mobile app for dog rabies vaccination reporting by private and city veterinarians in Davao City*.

Barak N. Hadar. *Emerging Technology and Health Informatics in Veterinary Medicine: Technology-Enhanced Weight-Loss Program, Disease Patterns and Predictions from Insurance Data, and Perceptions of Telemedicine in the COVID-19 era*.

Dedi Iskandar. *Development of a technology-based animal electronic medical record information system for animal health services*.

Chao-Yu Kong. *Enhancing AI Accessibility in Veterinary Medicine: Linking Classifiers and Electronic Health Records*.

Davide Vecchio. *Development of a data-driven method for assessing health and welfare in the most common livestock species in Switzerland: The Smart Animal Health project*.

Christian Nägele. 2024. *I E CER VETNET Proceedings 2024 Nicosia*. VII, (2024).
<https://doi.org/10.5281/zenodo.13624225>

Shubham D. Pasalkar. *Agrisense Integrated Health Surveillance for Cattle*.

Sherwin S. Chua, John A. Alis, and Mark Joseph Oria. 2024. *WEB-Based Veterinary Clinic with e-Mail and SMS Notification*, 2024 9th International Conference on Business and Industrial Research (ICBIR), Bangkok, Thailand, pp. 1160-1165. doi: 10.1109/ICBIR61386.2024.10875747

Putri Kartika Ayuningtyas. 2023. *Performance and Functional Testing With The Black Box Testing Method*. International Journal of Progressive Sciences and Technologies. 39, 2 (2023).
<https://doi.org/10.52155/ijpsat.v39.2.5471>

Finland Oy Justus Jaskari, W. *Ensuring Customer Value through Comprehensive Reporting*.

Stanimir Stoyanov. 2025. *Survey on Hardware Components Providing Information Security*. AIP Conference Proceedings (Mar. 2025).

Ana Maria Nogueira da Cruz. 2020. *An open web-based platform for enhancing the visibility of Brazilian research*. Proceedings - 2020 International Conference on Computational Science and Computational Intelligence, CSCI 2020 (2020)

Journal Articles

Rosmani, Arifah Fasha and Mokhtar, Muhammad Harith. 2023. *An Online Scheduling Platform for Veterinary Appointments*. Jurnal Intelek 18, 2 (Aug. 2023), 16–26.
<https://doi.org/10.24191/ji.v18i2.22092>

Benis, Arriel; Haghi, Mostafa; Deserno, Thomas M.; and Tamburis, Oscar. 2023, May. *One Digital Health Intervention for Monitoring Human and Animal Welfare in Smart Cities: Viewpoint and Use Case*. JMIR Medical Informatics 11 (May 2023), e43871.
<https://doi.org/10.2196/43871>

Katrin Beyer. 2023. *Unlocking the potential of ICT innovation in veterinary healthcare: The pathway to improve practices and business model*. Procedia Computer Science (2023), 4775–4784.

Péter B. Bök and Dario Micucci. 2024. *The future of human and animal digital health platforms*. Journal of Reliable Intelligent Environments (September 2024). <https://doi.org/10.1007/s40860-024-00232-0>

Mohammad Dayoub. 2023. *Applications of smart agriculture to improve animal production: Opportunities, challenges, solutions, and benefits*. Preprints (2023).
<https://doi.org/10.20944/preprints202310.1229.v1>

Nurul Badriah Jaffar. 2021. *The development of veterinary clinic management system using structured approach*. Applied Information Technology and Computer Science 2, 2 (2021), 1555–1567. <https://doi.org/10.30880/aitcs.2021.02.02.100>

Marinette Babila and Hui Xiao. 2020. *The principal as a curriculum-instructional leader: A strategy for curriculum implementation in Cameroon secondary schools*. International Journal of Education and Research 8, 4 (2020).

Sasmita Swain. 2024. *Smart livestock management: Integrating IoT for cattle health diagnosis and disease prediction through machine learning*. Indonesian Journal of Electrical Engineering and Computer Science 34, 2 (May 2024), 1192–1203.
<https://doi.org/10.11591/ijeecs.v34.i2.pp1192-1203>

Zahra Aghababaeyan. 2023. *Black-box testing of deep neural networks through test case diversity*. IEEE Transactions on Software Engineering 49, 5 (May 2023), 3182–3204.
<https://doi.org/10.1109/TSE.2023.3243522>

Anna Cecilia Ragnarsdóttir. 2023. *Bridging gaps in vocational education and training systems in Norway*. Journal of Vocational Education and Training (2023).

<https://doi.org/10.1080/13636820.2023.2255992>

Rosmani Arifah Fasha and Mokhtar Muhammad Harith. 2023. *An online scheduling platform for veterinary appointments*. Jurnal Intelek 18, 2 (Aug. 2023).

<https://doi.org/10.24191/ji.v18i2.22092>

Katrin Beyer. 2023. *Customer experiences drive the need of innovation in veterinary practices*. Procedia Computer Science (2023), 3094–3103.

Anders Herlin. 2021. *Animal welfare implications of digital tools for monitoring and management of cattle and sheep on pasture*. Animals. MDPI AG.

Nina Janke. 2021. *Pet owners' and veterinarians' perceptions of information exchange and clinical decision-making in companion animal practice*. PLoS ONE. 16, 2 February (Feb. 2021).

<https://doi.org/10.1371/journal.pone.0245632>

Maria K. Koytcheva. 2021. *A Systematic Review of Environmental Sustainability in Veterinary Practice*. Topics in Companion Animal Medicine. 44, (Aug. 2021).

<https://doi.org/10.1016/j.tcam.2021.100550>

Diego R. Marquez. 2023. *Assessing US small animal veterinary clinic adaptations and their impacts on workforce COVID-19 preparedness and response*. Health Security. 21, 6 (2023).

<https://doi.org/10.1089/hs.2023.0041>

Amanuel Wondie Mekonen. 2024. *Assessment of veterinary pharmaceutical warehouse management practices and its associated challenges in four selected zones and Bahir Dar city of*

Amhara regional state, Ethiopia. *Frontiers in Veterinary Science*. 11, (2024).

<https://doi.org/10.3389/fvets.2024.1336660>

Xin Zhao. 2025. *Remote Vital Sensing in Clinical Veterinary Medicine: A Comprehensive Review of Recent Advances, Accomplishments, Challenges, and Future Perspectives*. *Animals*. MDPI.

Xiaojun Han. 2021. *A Study of Performance Testing in Configurable Software Systems*. *Journal of Software Engineering and Applications*. 14, 09 (2021).

<https://doi.org/10.4236/jsea.2021.149028>

Dawid W. Sroczynski. 2025. *From disorganized data to emergent dynamic models: Questionnaires to partial differential equations*. *PNAS Nexus*. 4, 2 (Feb. 2025).

<https://doi.org/10.1093/pnasnexus/pgaf018>

Khairul Fahmi Syah and Sukirman Sukirman. 2023. *Sistem Peminjaman Buku Perpustakaan dengan Identifikasi Kode Quick Response (QR)*. *Edumatic: Jurnal Pendidikan Informatika*. 7, 1 (2023). <https://doi.org/10.29408/edumatic.v7i1.13372>

Samer Al-Saqqa, Mohammed Sawalha, and Hesham AbdelNabi. 2020. *Agile software development: Methodologies and trends*. *International Journal of Interactive Mobile Technologies*. 14, 11 (2020). <https://doi.org/10.3991/ijim.v14i11.13269>

José Manuel Álvarez and Elena Roibás-Millán. 2021. *Agile methodologies applied to Integrated Concurrent Engineering for spacecraft design*. *Research in Engineering Design*. 32, 4 (2021). <https://doi.org/10.1007/s00163-021-00371-y>

Sajjad Mahmood Cheema, Muhammad Aleem, Muhammad Arslan. 2023. *A natural language interface for automatic generation of data flow diagram using web extraction techniques*. *Journal*

of King Saud University - Computer and Information Sciences. 35, 2 (2023).

<https://doi.org/10.1016/j.jksuci.2023.01.006>

Emmanuel Chibuike Daraojimba, Oluwaseun Stephen Adebayo, Abdulrahman Oladipo Aremu. 2024. *Comprehensive review of agile methodologies in project management*. Computer Science & IT Research Journal. 5, 1 (2024). <https://doi.org/10.51594/csitrj.v5i1.717>

Abdulrahman Hassan, Ahmed Ali, and Mohammed Alshammari. 2021. *Agile methodology in an SaaS deployment project: A pre- and post-SaaS deployment study*. International Journal of Information Technology Project Management. 12, 2 (2021). <https://doi.org/10.4018/IJITPM.2021040101>

Carlos Alberto Torres Romero, María Soledad Ramírez Montoya, and José Ramón Valdés Lozano. 2021. *Software Architecture for Planning Educational Scenarios by Applying an Agile Methodology*. International Journal of Emerging Technologies in Learning. 16, 8 (2021). <https://doi.org/10.3991/ijet.v16i08.20603>

Online References

Iskandar, D. n.d. *Development of a Technology-Based Animal Electronic Medical Record Information System for Animal Health Services*. Retrieved February 18, 2025 from <https://www.researchgate.net/publication/development-of-a-technology-based-animal-electronic-medical-record-information-system>

Patil, H. D. n.d. *Smart Surveillance and Animal Care System Using IoT and Deep Learning*. Retrieved January 25, 2025 from <https://www.ijert.org/smart-surveillance-and-animal-care-system-using-iot-and-deep-learning>

Research Exhibition in Mathematics and Computer Sciences (REMACS 5.0). n.d. *Conference Proceedings*. Retrieved December 10, 2024 from <https://remacs.org/conference-proceedings>

What Is Black Box Testing? Techniques, Types & Example. n.d. *Software Testing Reference Article*. Retrieved November 5, 2024 from <https://www.guru99.com/black-box-testing.html>

What Is User Acceptance Testing. n.d. *Software Testing Reference Article*. Retrieved October 22, 2024 from <https://www.guru99.com/user-acceptance-testing.html>

Hamilton, T. n.d. *Agile Methodology: What Is Agile Model in Software Testing?* Retrieved September 14, 2024 from <https://www.guru99.com/agile-testing.html>

Healthray. n.d. *8 Essential Features of Clinical Management Systems*. Retrieved August 30, 2024 from <https://healthray.com/blog/clinic-management-systems/8-essential-features-of-clinical-management-systems/>

AgencyAnalytics. n.d. *Page Load Time KPI: Definition and Importance*. Retrieved July 18, 2024 from <https://agencyanalytics.com/kpi-definitions/page-load-time>

Research Exhibition in Mathematics and Computer Sciences (REMACS 5.0). n.d. *Conference Proceedings*. Retrieved June 9, 2024 from <https://remacs.org/conference-proceedings>

APPENDICES

APPENDIX A

EVALUATION TOOL

Section 1: General Information

1. **Can you tell us about your veterinary clinic?** (History, size, number of staff, specialization, etc.)
2. **What are your main services?** (Consultations, vaccinations, surgery, grooming, boarding, etc.)
3. **What tools or methods do you currently use to manage your clinic's daily operations?** (Paper records, spreadsheets, clinic management software, or a combination)
4. **How many clients do you serve on average per day/week/month?**
5. **What are your biggest operational challenges in running the clinic?**

Section 2: Appointment & Patient Management

6. **How do you schedule and manage appointments?** (Manual logbook, Excel, software, etc.)
7. **What challenges do you face with appointment scheduling?** (Missed appointments, overbooking, etc.)
8. **How do you maintain and update patient medical records?** (Paper-based, electronic, mixed)
9. **How do you track pet vaccinations and medical history?**
10. **Would you be interested in an automated system for managing appointments and medical records?**

Section 3: Billing & Payment Process

11. **How do you currently process payments for your services?** (Manual receipts, printed invoices, software, etc.)
12. **What payment methods do you accept?** (Cash, credit/debit cards, mobile payments, online banking, etc.)
13. **Do you encounter any issues with billing and payments?** (Late payments, tracking finances, system inefficiencies, etc.)
14. **How do you handle refunds, discounts, or special payment arrangements for clients?**
15. **Would you find an automated billing and payment tracking system helpful for your clinic?**

Section 4: Inventory & Supplies Management

16. **How do you track and manage your clinic's medical supplies and medications?**
17. **What are the common issues you face with inventory management?** (Stock shortages, overstocking, expired products, etc.)

18. How often do you restock supplies, and what is your process for ordering new items?
19. Do you currently use any software or tools for inventory management?
20. Would an automated inventory tracking system improve your workflow?

Section 5: Communication & Customer Engagement

21. How do you communicate with pet owners for follow-ups, reminders, and updates?
22. What challenges do you face in staying connected with clients? (Missed follow-ups, lack of reminders, etc.)
23. How do you handle emergency cases or urgent pet care requests?
24. Would you be interested in an automated system for sending appointment reminders via SMS or email?
25. How do you currently manage customer feedback and inquiries?

Section 6: Pain Points & Future Improvements

26. What are the biggest challenges your clinic faces in daily operations?
27. If you could improve or automate one aspect of your clinic, what would it be?
28. Are there any additional features or tools you wish you had to make your workflow more efficient?
29. Would you be open to using a digital system designed specifically for veterinary clinics?
30. Do you have any concerns or expectations regarding implementing an online system?

PET OWNERS

PERFORMANCE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system loads quickly after login.				
2. The buttons, labels, and features were clear.				
3. I finished my tasks without confusion.				
4. The system matched what I expected it to do.				
5. I felt comfortable using the system for clinic tasks.				
6. Menus and pages respond immediately when clicked.				
7. The system completed all actions without failing or restarting.				
8. Tasks (e.g., saving data, generating receipts) are processed without noticeable delays.				
9. I experienced no lag while using key features of the system.				
10. Overall, I accept this system as useful and helpful.				

USER ACCEPTANCE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system responded quickly when I clicked buttons or menus.				
2. Pages loaded without a long waiting time.				
3. The system ran smoothly while I was using it.				
4. I did not experience freezing or sudden slowdowns.				
5. The system processed my actions right away.				
6. The speed of the system felt acceptable for daily use.				
7. I can easily locate the features or tools I need.				
8. The system worked well even when doing several tasks.				
9. I was able to use the system without needing technical help.				
10. I was satisfied with the speed and smoothness of the system.				

USE CASE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system followed the correct steps for each task.				
2. The features worked as they were supposed to.				
3. I was able to complete common clinic tasks easily				
4. The system showed the correct output after each action.				
5. The workflows matched real-life clinic processes.				
6. The system guided me properly when doing tasks.				
7. No steps felt missing or incomplete.				
8. The system displayed correct information for the selected pet.				
9. The system behaved as expected in different scenarios.				
10. Overall, the system supported my tasks smoothly and correctly.				

VET OWNER/RESIDENT DOCTORS

PERFORMANCE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system was easy to understand on my first use.				
2. The buttons, labels, and features were clear.				
3. I finished my tasks without confusion.				
4. The system matched what I expected it to do.				
5. I felt comfortable using the system for clinic tasks.				
6. The design made it simple to navigate.				
7. The instructions and steps were easy to follow.				
8. The system helped me do tasks faster.				
9. I would be willing to use this system regularly.				
10. Overall, I accept this system as useful and helpful.				

USER ACCEPTANCE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system responded quickly when I clicked buttons or menus.				
2. Pages loaded without long waiting time.				
3. The system ran smoothly while I was using it.				
4. I did not experience freezing or sudden slowdowns.				
5. The system processed my actions right away.				
6. The speed of the system felt acceptable for daily use.				
7. I could move from one page to another without delays.				
8. The system worked well even when doing several tasks.				
9. The overall performance felt stable.				
10. I was satisfied with the speed and smoothness of the system.				

USE CASE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system followed the correct steps for each task.				
2. The features worked as they were supposed to.				
3. I was able to complete common clinic tasks easily				
4. The system showed the correct output after each action.				
5. The workflows matched real-life clinic processes.				
6. The system guided me properly when doing tasks.				
7. No steps felt missing or incomplete.				
8. Most features worked without errors.				
9. The system behaved as expected in different scenarios.				
10. Overall, the system supported my tasks smoothly and correctly.				

IT EXPERTS

PERFORMANCE TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The system responded well during high user activity.				
2. Page loading remained acceptable under heavy load..				
3. The system stayed stable when multiple users were active.				
4. There were no crashes during stress testing.				
5. The system handled multiple processes at the same time.				
6. The response time stayed consistent under load.				
7. The system did not show performance drops during long use.				
8. Memory and processing usage stayed within acceptable limits.				
9. The system recovered smoothly after stress conditions..				
10. Overall, the system performed reliably during load tests.				

SECURITY TESTING				
Questions	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
1. The login process prevented unauthorized access.				
2. User accounts were properly protected.				
3. Sensitive data was not exposed during use..				
4. Access control worked correctly for different user roles.				
5. The system handled incorrect or suspicious inputs safely.				
6. Session handling was secure and stable.				
7. Data storage followed proper security practices.				
8. No major vulnerabilities were found during testing.				
9. The system protected data during transmission.				
10. Overall, the system met the needed security requirements				

APPENDIX B

DOCUMENTATION



APPENDIX C

SAMPLE INPUT/OUTPUT/REPORTS

Login Page

[← Back to Home](#)



Welcome Back

Email

Password

Login

[Forgot password?](#)

Don't have an account? [Register here](#)

Appointments

  **Anne Psylocke Urbina** 

OVERVIEW

- Dashboard
- MY PETS
- Pet Records
- APPOINTMENTS
- My Appointments**
- SERVICES
- Available Services
- Our Veterinarians
- ADOPTION

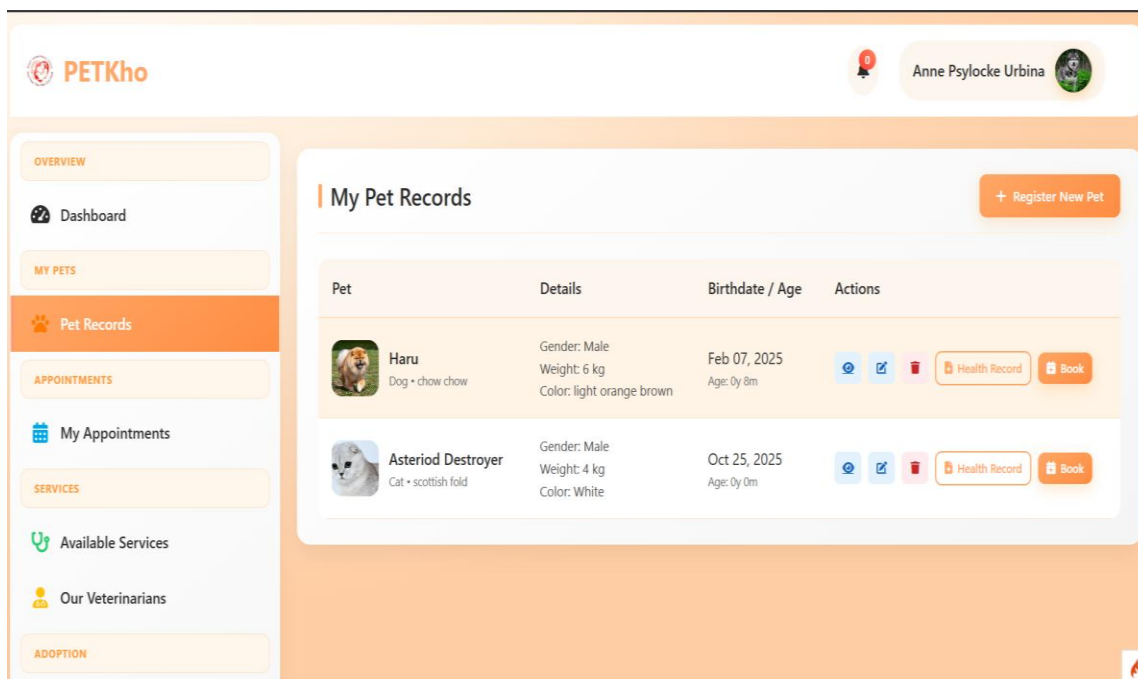
Appointments

[Book Appointment](#)

< > today **October 2025** month week day

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25

Pet Records

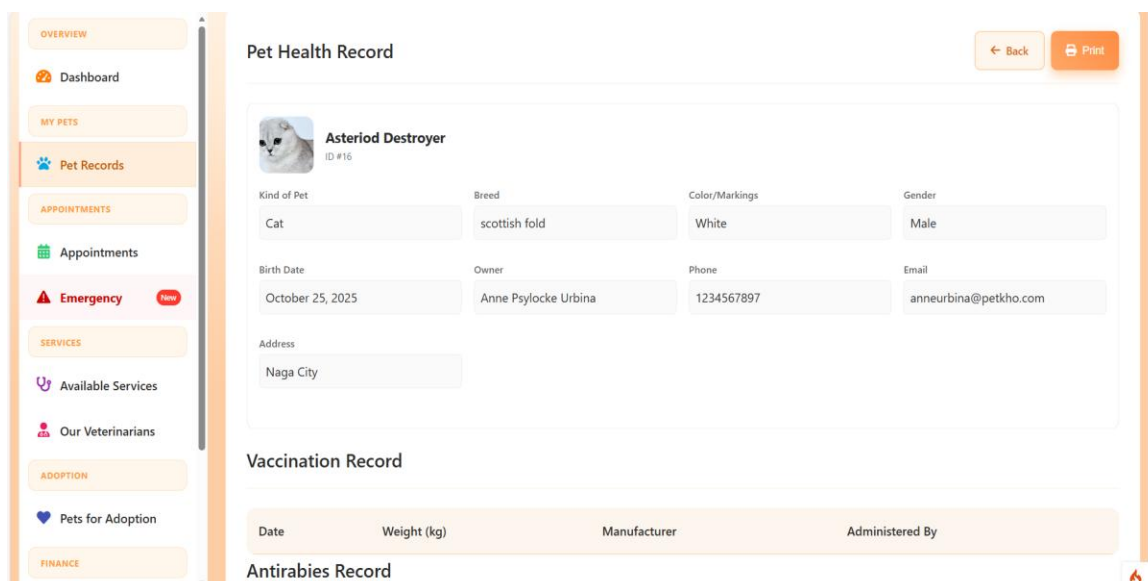


PETKho Anne Psylocke Urbina

My Pet Records + Register New Pet

Pet	Details	Birthdate / Age	Actions
 Haru Dog • chow chow	Gender: Male Weight: 6 kg Color: light orange brown	Feb 07, 2025 Age: 0y 8m	    
 Asteriod Destroyer Cat • scottish fold	Gender: Male Weight: 4 kg Color: White	Oct 25, 2025 Age: 0y 0m	    

Pet Health Record



Pet Health Record ← Back Print

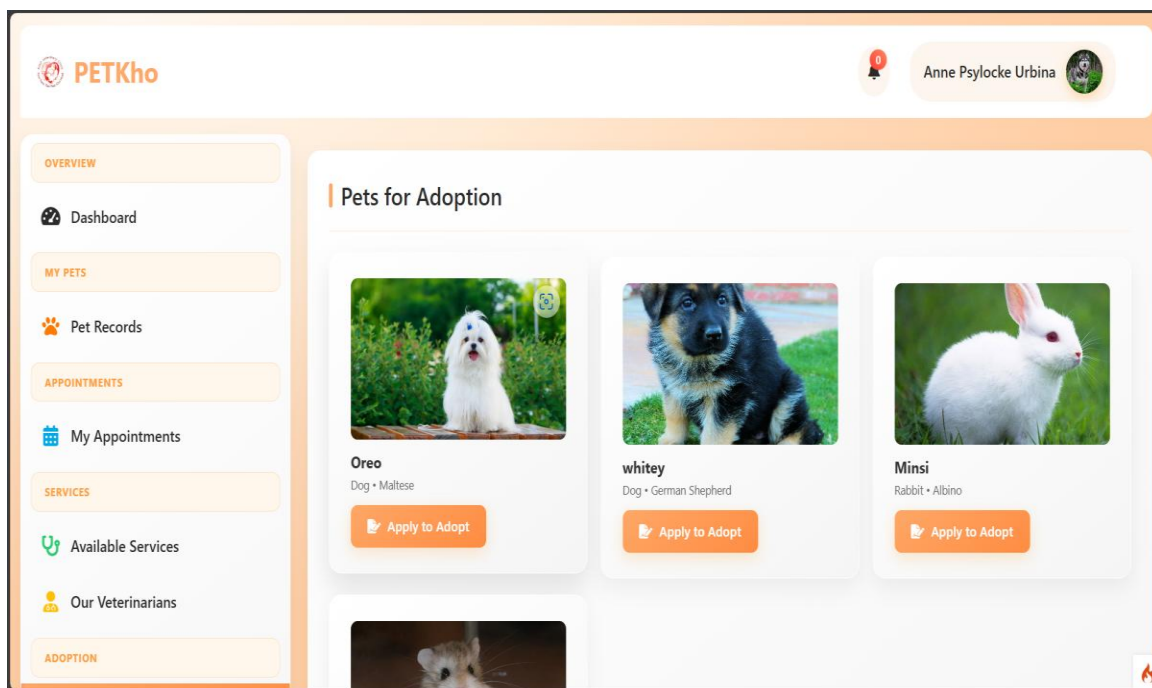
 **Asteriod Destroyer**
ID #16

Kind of Pet	Breed	Color/Markings	Gender
Cat	scottish fold	White	Male
Birth Date	Owner	Phone	Email
October 25, 2025	Anne Psylocke Urbina	1234567897	anneurbina@petkho.com
Address			
Naga City			

Vaccination Record

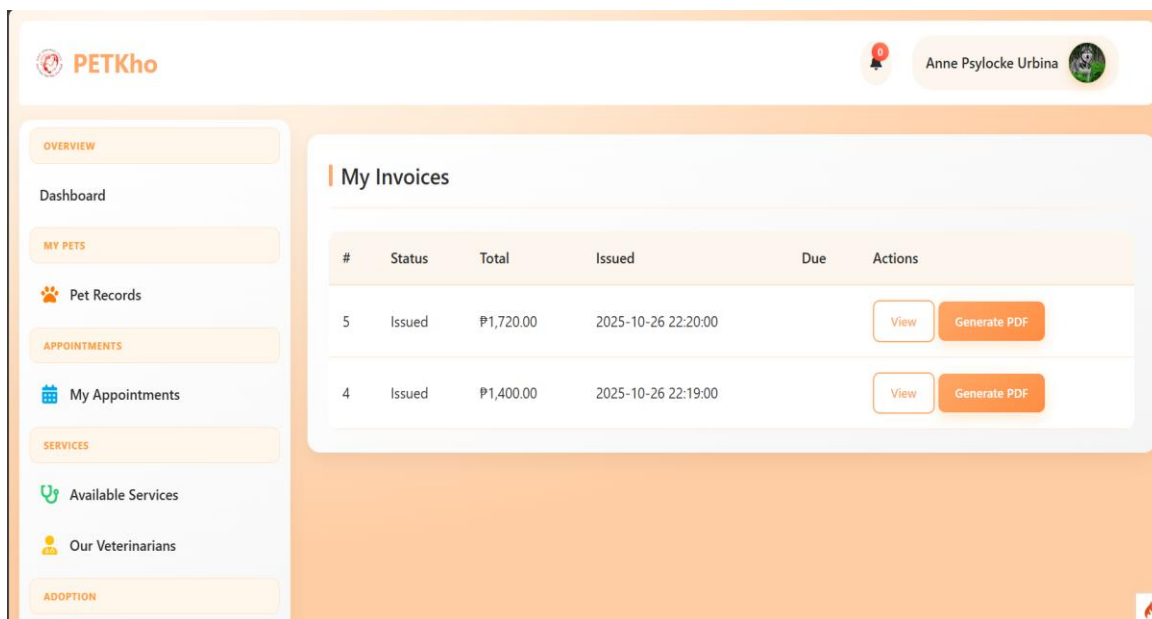
Date	Weight (kg)	Manufacturer	Administered By
Antirabies Record			

Adoptions



The screenshot shows the PETKho Adoptions page. The left sidebar contains navigation links: OVERVIEW, Dashboard, MY PETS, Pet Records, APPOINTMENTS, My Appointments, SERVICES, Available Services, Our Veterinarians, and ADOPTION. The main content area is titled "Pets for Adoption" and displays three pet cards: Oreo (Dog • Maltese), whitey (Dog • German Shepherd), and Minsi (Rabbit • Albino). Each card has an "Apply to Adopt" button. The user profile at the top right shows "Anne Psylocke Urbina".

Invoices



The screenshot shows the PETKho My Invoices page. The left sidebar is identical to the Adoptions page. The main content area is titled "My Invoices" and displays a table with two rows of invoice data. Each row has "View" and "Generate PDF" buttons.

#	Status	Total	Issued	Due	Actions
5	Issued	₱1,720.00	2025-10-26 22:20:00		<button>View</button> <button>Generate PDF</button>
4	Issued	₱1,400.00	2025-10-26 22:19:00		<button>View</button> <button>Generate PDF</button>

Notifications

PETKho Anne Psylocke Urbina

My Notifications

Title	Message	Date	Action
New Invoice Issued	An invoice has been issued for your recent appointment.	2025-10-26 22:21:08	Open
New Invoice Issued	An invoice has been issued for your recent appointment.	2025-10-26 22:19:55	Open
Appointment Accepted	An appointment was accepted in branch #3	2025-10-26 14:51:48	Open
Appointment Cancelled	An appointment was cancelled in branch #3	2025-10-26 14:50:19	Open
New Adoption Application	A new adoption application has been submitted.	2025-10-26 14:47:20	Open

Stocks

PETKho VET_OWNER Dr. Williefredo Kho WK

Stock Management Filter Types Stock Requests Add Item

Item	Type	Stock	Min	Price	Actions
Sin1	vaccines	200	200	P500.00	Update Stock Edit Delete
Amoxicilin Suspension	antibiotics	200	200	P250.00	Update Stock Edit Delete
Antirabies	vaccines	200	200	P300.00	Update Stock Edit Delete
Papi Scour	antibiotics	200	200	P450.00	Update Stock Edit Delete

Reports

Dashboard

MANAGEMENT

Appointments

Branch Patients

Invoices

INVENTORY

Stock Management

ADOPTION

Adoptions

REPORTS

Reports

ACCOUNT

Notifications

Reports

Generate Report + New

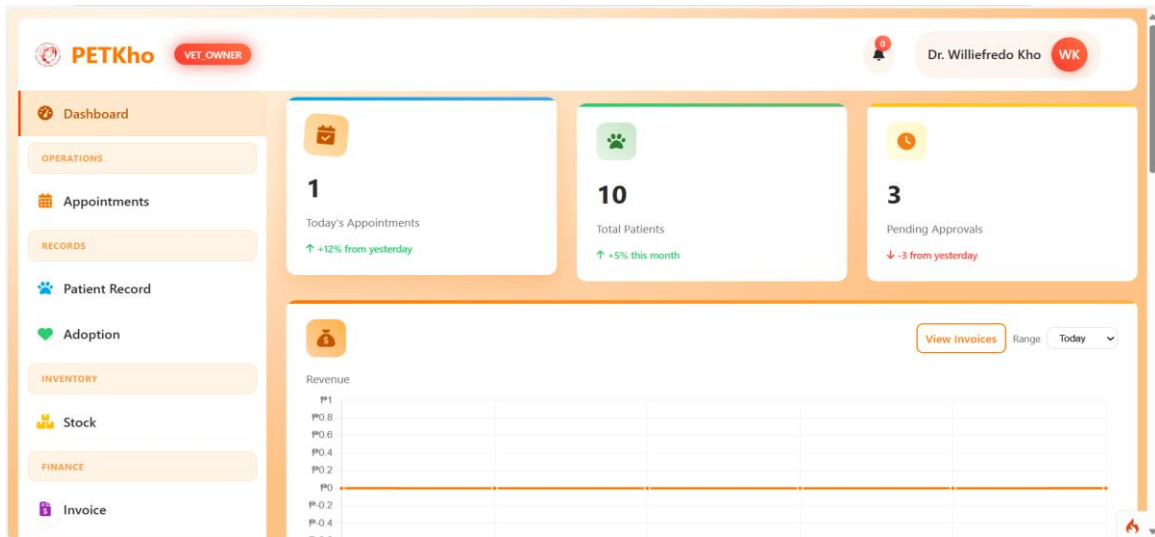
Title	Date	Actions
Sales Report - Monthly (December 2025)	2025-12-10 01:26:26	Edit Delete
Weekly Invoice Summary	2025-12-07 16:24:34	Edit Delete
Monthly Invoice Summary	2025-12-07 16:24:34	Edit Delete
Weekly Invoice Summary	2025-12-07 16:22:12	Edit Delete
Monthly Invoice Summary	2025-12-07 16:22:12	Edit Delete
Diagnosis	2025-10-27 04:04:41	Edit Delete

APPENDIX D

USER'S GUIDE

ADMIN AND STAFF SIDE

1. Admin Dashboard



The Admin Dashboard which serves as the main control panel for managing clinic operations. This page gives a quick and clear overview of the clinic's daily status using visual cards, charts, and a navigation menu, helping the vet owner make fast and informed decisions.

A. Sidebar Menu – Located on the left side, it allows easy navigation to key modules such as Dashboard, Appointments, Patient Records, Adoption, Stock, and Invoices.

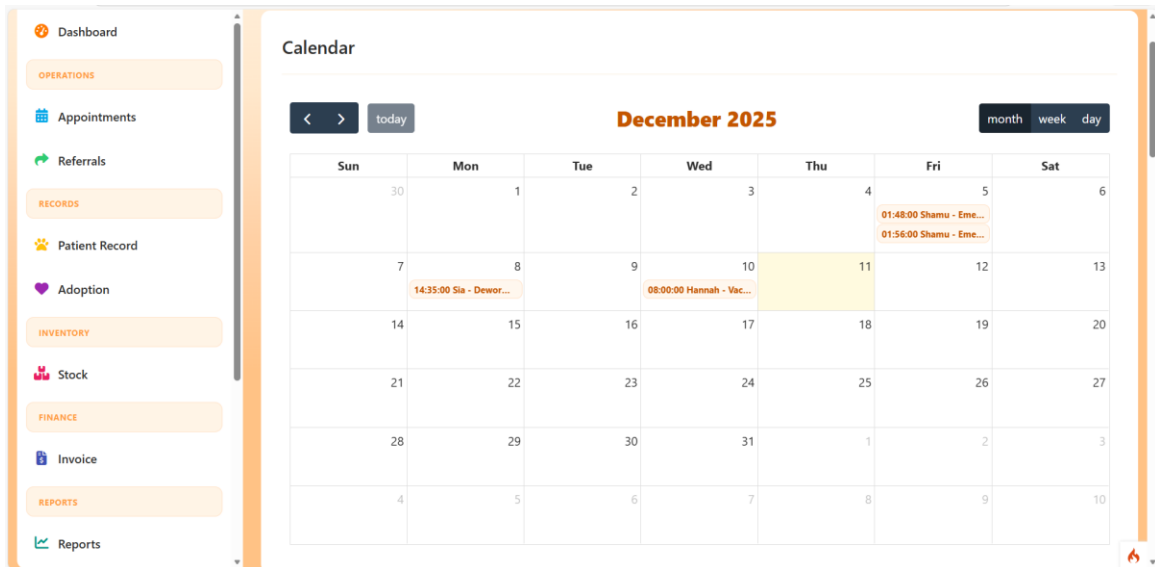
B. Summary Cards – Display important daily metrics such as Today's Appointments, Total Patients, and Pending Approvals, with indicators showing increases or decreases compared to previous days.

C. Revenue Chart – Shows a visual summary of clinic income, helping the admin track financial performance over a selected date range.

D. View Invoices Button – Provides quick access to invoice records for easier financial monitoring and review.

E. User Profile and Notifications – Found at the top-right corner, showing the logged-in vet owner's name and allowing access to alerts and account options.

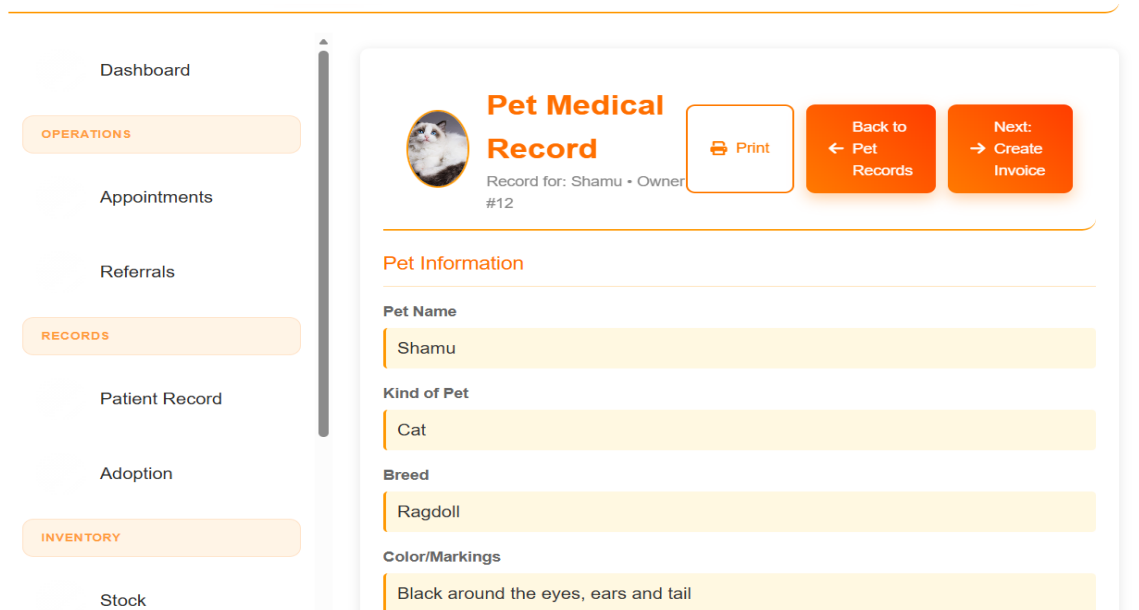
2.Appointments



The Appointment page gives the admin a clear view of all appointment requests that require verification before being finalized. It organizes each request by date, service type, pet owner, and branch, making it easy to monitor and process scheduled visits. Each record shows the pet's name, the service requested, the owner, and the appointment's current status.

- A. Approve Button (Green Check) – Approves the appointment and confirms it in the system.
- B. Reschedule Button (Red Calendar) – Approves the appointment but its confirms in the system with the rescheduled date.
- C. View Button (Blue Icon) – Opens the detailed view of the appointment, showing full information such as notes, purpose, and additional details.
- D. Calendar grid- Shows all dates within the selected month or view mode. Days display scheduled appointments in colored labels.when pet schedule is clicked it will redirected to pet medical Record

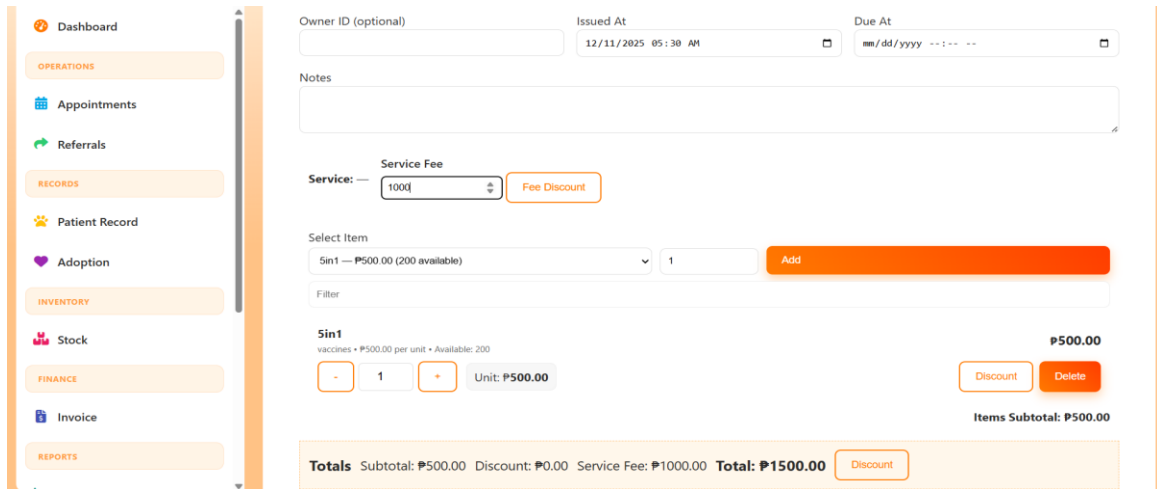
3. Pet Medical Record



The Pet Medical Record page gives the admin or veterinarian a full view of a pet's medical history and essential information. It provides details about the pet, the owner, medical notes, treatments, and other important records. This page helps staff review, update, and print records for documentation and billing purposes.

- A. Print Button – Generates a printable version of the pet's medical record for filing or giving to the owner.
- B. Back to Pet Records – Returns the user to the main list of all pet records.
- C. Next: Create Invoice – Redirects to the invoice creation page, allowing billing to be completed after reviewing the record.
- C. Add Vaccination- Adding the Vaccination information of the pet
- D. Add Antirabies- Adding the Antirabies information of the pet
- E. Add Deworming or Heartworming-Adding the Deworming
- F. Add Fleas and Tick Prevention-Adding the Fleas and Tick Prevention

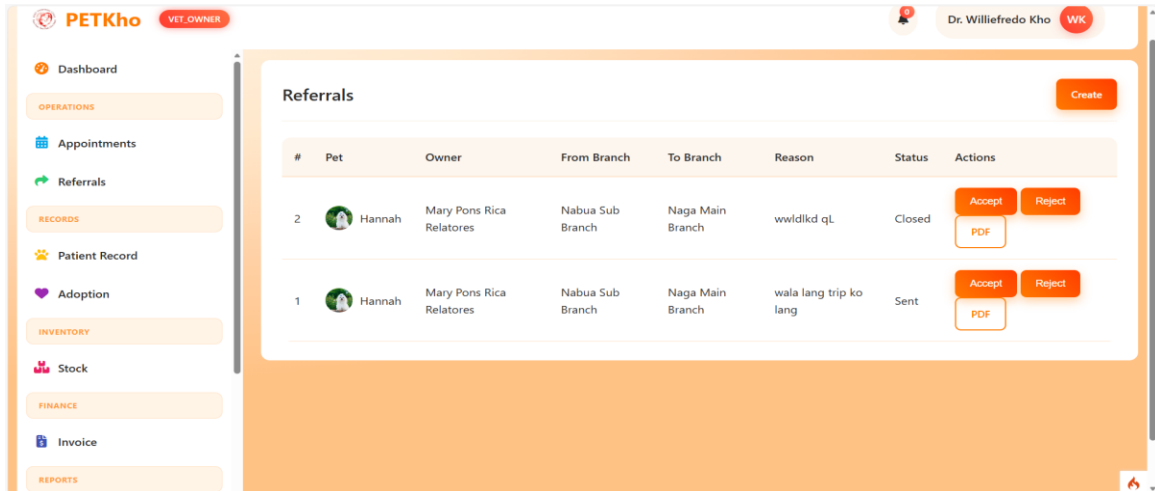
4. Invoice



The Invoice Creation page allows staff to generate complete and accurate invoices for pet owners by selecting services, adding items, entering fees, and applying discounts when applicable.

- A. Service Fee – Allows the staff to input the base service fee for the consultation, procedure, or treatment being billed.
- B. Fee Discount Button – Opens a dialog where a discount can be applied to the service fee. This may be used for promotional rates, courtesy adjustments, or special cases.
- C. Select Item – A dropdown menu displaying available inventory items such as vaccines, medications, and supplies. Each item shows its name, price, and available stock.
- D. Quantity Selector – Allows staff to enter or adjust the quantity of the selected item to be added to the invoice.
- E. Add Button – Adds the selected item and quantity to the invoice list.
- F. Item Controls (– / +) – Enables staff to increase or decrease the quantity of an already-added item.
- G. Item Discount Button – Allows staff to apply a discount to a specific item. This is separate from the service fee discount.
- H. Delete Button – Removes the selected item from the invoice.

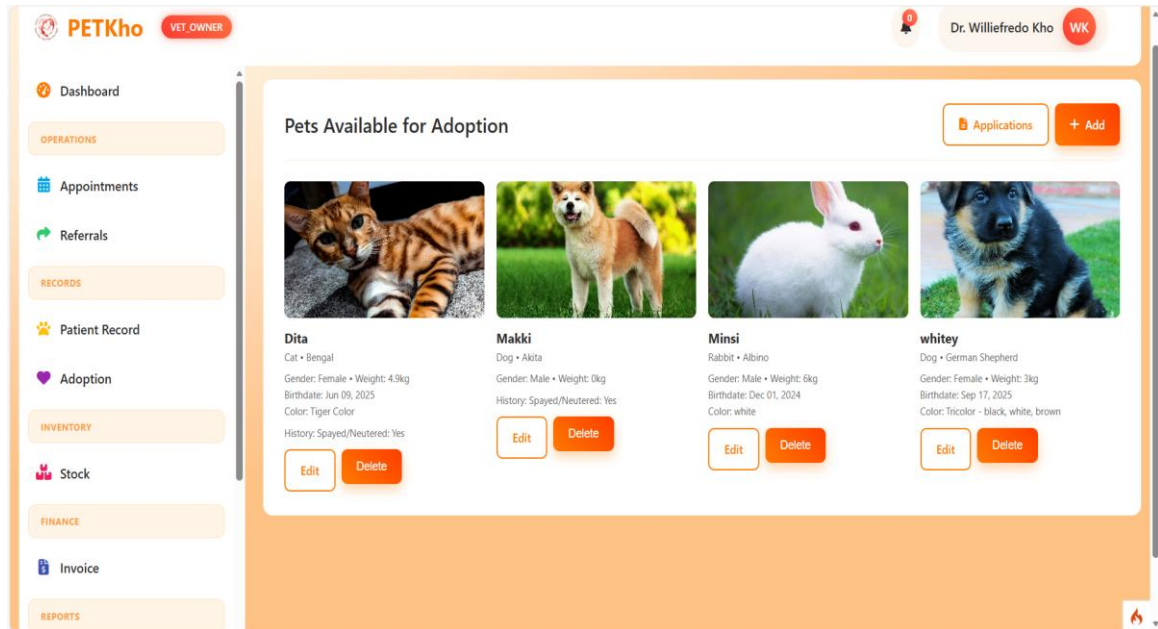
5. Referrals



The Referrals page provides the admin or veterinarian with a full overview of all referral requests made for pets between clinic branches. It helps track where the pet is coming from, where it is being referred to, and the reason for the referral. The page makes it easy to review, approve, reject, or print referral documents.

- A. Create Button – Allows the admin/vet to create a new referral request.
- B. Accept Button – Approves the referral and updates the status to indicate it has been accepted.
- C. Reject Button – Declines the referral request.
- D. PDF Button – Generates a downloadable PDF document of the referral for printing or filing.

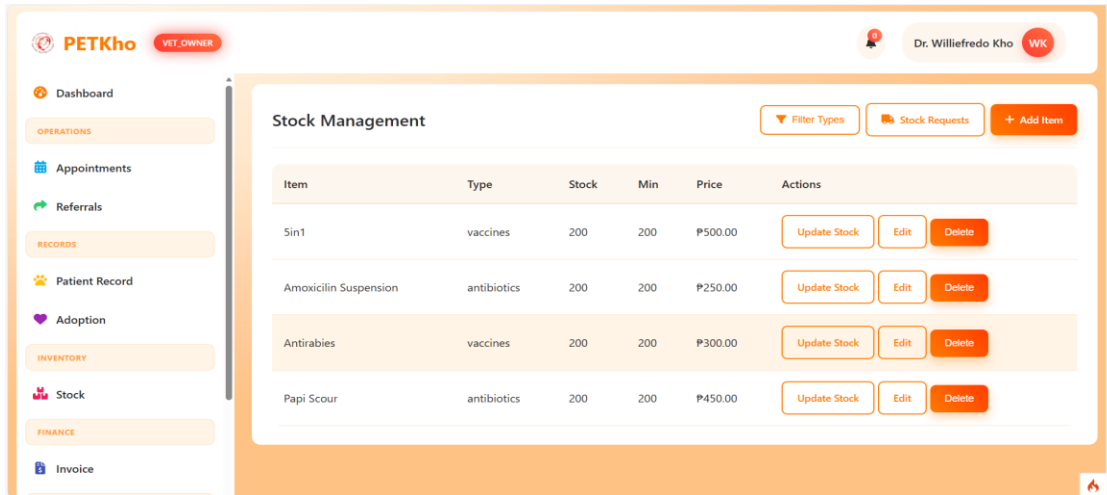
6. Adoptions



The Adoption page gives the admin full visibility over all pet adoption applications submitted by potential adopters. It helps track adoption requests, review adopter information, monitor the status of each application, and process approvals efficiently. Each record shows the pet available for adoption, the applicant's details, the reason for adoption, and the current status of the request.

- A. Applications Button – Opens the list of adoption applications submitted by potential adopters.
- B. Add Button – Allows the admin to add a new pet into the adoption list. Opens a form for uploading a photo and entering pet details such as breed, color, history, and birthday.
- C. Edit Button – Opens the pet's profile for updating information (image, medical history, age, etc.).
- D. Delete Button – Permanently removes the pet from the adoption list.

7. Stock Management



The Stock Management page allows the admin to oversee all clinic supplies such as medicines, vaccines, and other inventory items. It provides tools for adding new items, updating current stock levels, organizing by item type, and viewing stock requests from staff. Each record displays essential information including item name, type, stock quantity, minimum stock requirement, and price.

- A. Filter Types – Filters items based on their category (e.g., vaccines, antibiotics, supplies) to make searching and organizing easier.
- B. Stock Requests – Displays pending stock requests submitted by staff or branches needing additional inventory.
- C. Add Item – Opens a form to register a new stock item, including name, category, minimum stock, and price.
- D. Update Stock Button – Allows the admin to adjust item quantity (add or reduce stock), important for daily stock monitoring.
- E. Edit Button – Opens the item form for modifying details such as name, type, minimum stock, or price.
- F. Delete Button – Removes the item permanently from the inventory list.

8. Veterinary and Staff

#	Name	Username	Email	Role	Branch	Actions
7	Pili Assistant	assistantpili	assistantpili@petkho.com	vet_assistant	Pili Sub Branch	[Edit] [Delete]
6	Iriga Assistant	assistantiriga	assistantiriga@petkho.com	vet_assistant	Nabua Sub Branch	[Edit] [Delete]
5	Naga Assistant	assistantnaga	assistantnaga@petkho.com	vet_assistant	Naga Main Branch	[Edit] [Delete]
4	Pili Vet	vetpili	vetpili@petkho.com	vet_resident	Pili Sub Branch	[Edit] [Delete]
3	Adelma Morales	vetnabua	vetnabua@petkho.com	vet_resident	Nabua Sub Branch	[Edit] [Delete]
2	MAMA Vet	vetnaga	vetnaga@petkho.com	vet_resident	Naga Main Branch	[Edit] [Delete]
1	Williefredo Kho	vet	vetowner@petkho.com	vet_owner	Naga Main Branch	[Edit] [Delete]

The Veterinary Staff page allows the admin to manage all employees across different branches, including veterinarians, assistants, and residents. It provides a clear list of staff members along with their role, contact details, and assigned branch. This page helps ensure accurate staff tracking and easy updates for personnel management.

- A. Add Staff – This feature allows the admin to open a staff registration form to add new personnel. The form collects essential details such as full name, username, email, role (vet_owner, vet_resident, vet_assistant), and assigned branch. Adding staff through this form helps keep the system organized and ensures each employee is properly registered with the correct role and branch.
- B. Edit Icon (Orange) – Clicking the orange edit icon opens the selected staff member's profile, allowing the admin to update details like name, email, role, or branch assignment. This ensures that staff information remains accurate and up-to-date as roles or branch assignments change.
- C. Delete Icon (Red) – The red delete icon lets the admin permanently remove a staff member from the system. This is used when an employee resigns or is no longer part of the organization, keeping the staff database clean and current.

9. Branch Management

PETKho VET_OWNER

Dr. Williefredo Kho WK

Branches + Add Branch

#	Name	Phone	Type	Parent	Status	Actions
17	Iriga Branch	1234567891	Main	—	Active	
3	Pili Sub Branch	1234567892	Sub	Naga Main Branch	Active	
2	Nabua Sub Branch	09876545653	Sub	Naga Main Branch	Active	
1	Naga Main Branch	1234567890	Main	—	Active	

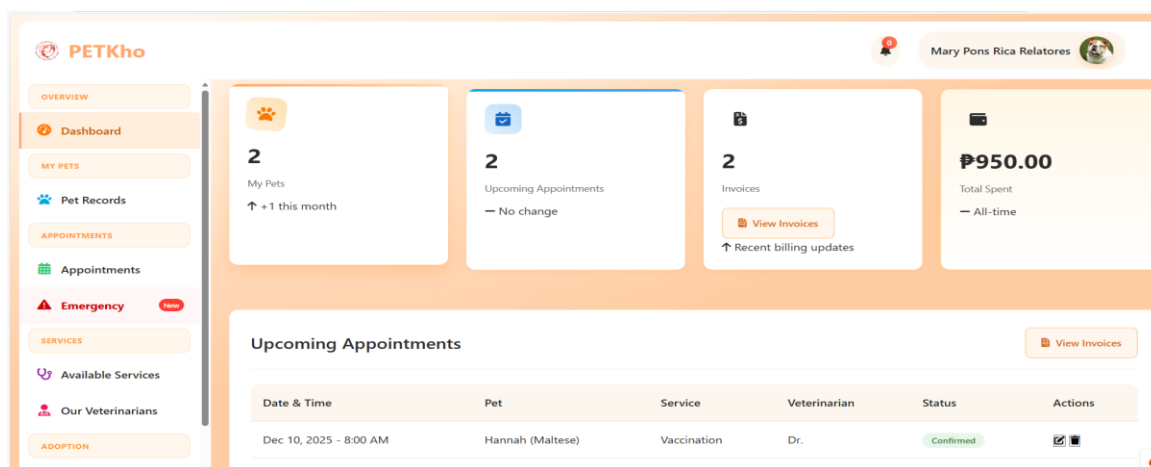
The Branches page provides administrators with a structured overview of all veterinary clinic branches registered in the system. It allows efficient management of branch information, including creation, modification, and status monitoring.

- A. Add Branch – This feature allows the administrator to open a branch creation form to register a new clinic branch in the system. The form collects important information such as the branch name, contact number, branch type (*Main* or *Sub*), and the parent branch if it is a sub-branch. Adding branches through this form helps organize the clinic’s network and ensures that each branch is correctly recorded in the system for operational and reporting purposes.
- B. Edit Icon (Orange) – Clicking the orange edit icon opens the selected branch’s profile, allowing the admin to update details like branch name, phone number, type, or parent branch assignment. This ensures that branch information remains accurate, reflecting any changes such as reclassification or restructuring within the organization.

- C. Delete Icon (Red) – The red delete icon allows the admin to permanently remove a branch from the system. This is typically used when a branch is closed or no longer part of the organization, helping maintain a clean and up-to-date branch database.

USER SIDE

1 User Dashboard

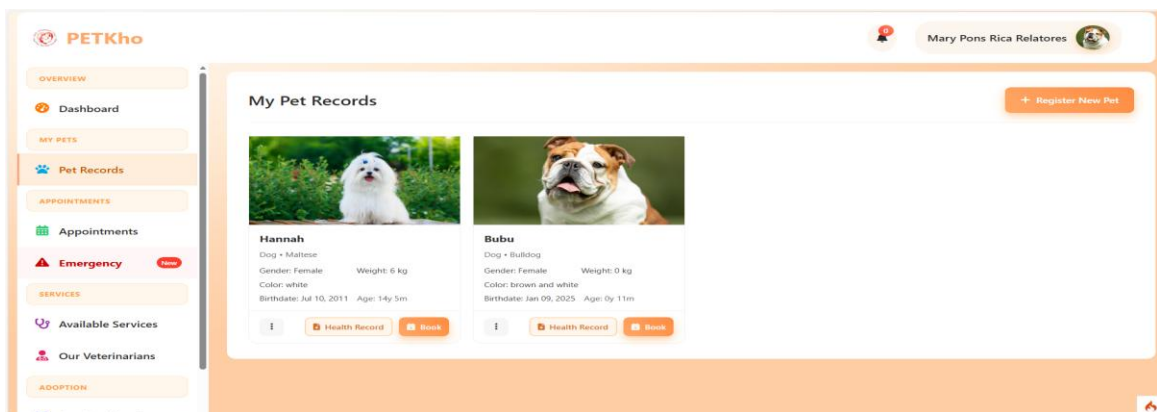


The Client Dashboard is the main hub for pet owners in the system, designed to give a complete view of their pets' care and activities. It makes it easy for clients to stay informed, organized, and proactive about their pets' health. Key features include:

- Quick overview:** Shows essential details at a glance, including the number of pets, upcoming appointments, recent invoices, and total clinic spending. Helps clients track interactions without navigating multiple pages.
- Appointment details:** Provides detailed information about upcoming visits and medical schedules, ensuring clients stay updated on their pets' care.
- Organization and convenience:** Gathers all essential data in one place, helping clients manage their pets' care smoothly and stay organized.
- Notifications and reminders:** Alerts clients about completed services, new medical notes, or schedule changes, keeping them informed in real time.

- E. Enhanced communication: Strengthens the connection between the clinic and pet owners, giving clients confidence that they are aware of their pets' health and any necessary actions.
- F. Proactive pet care: Combines convenience, clarity, and real-time updates to support responsible and proactive management of pets' health.

2. Pet Health Record

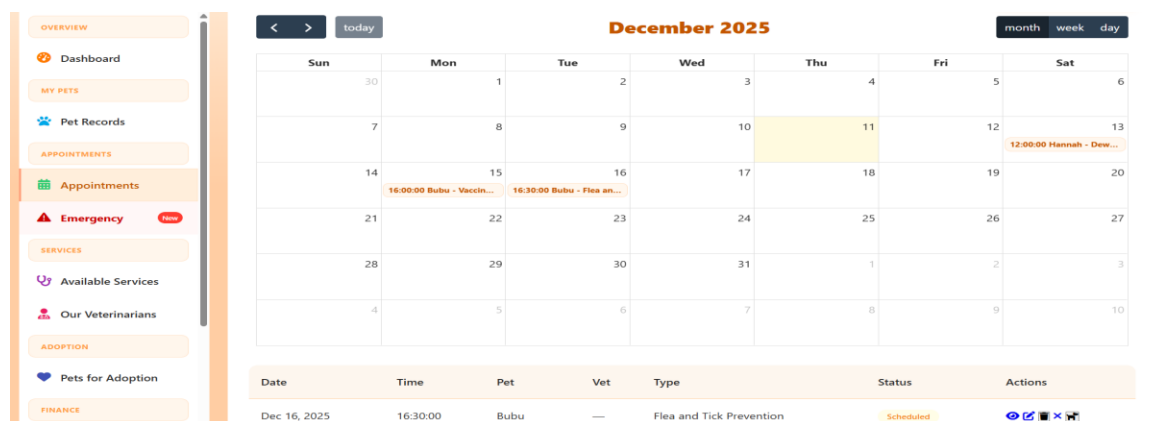


The Pet Records page allows clients to view and manage all pets registered under their account. Each pet is displayed in an organized card format showing essential information such as breed, gender, weight, age, and other identifying details. From this page, users can access medical records or schedule appointments for any of their pets. The clean layout ensures easy navigation and quick access to each pet's profile.

- A. Register New Pet – Opens the pet registration form where clients can add a new pet to their account. Information typically includes name, species, breed, gender, color, weight, and birthdate.
- B. Menu Icon (Three Dots) – Provides additional options for managing the pet's record, such as editing details or removing the pet from the system (depending on system rules).

- C. Health Record Button – Redirects the user to the pet’s complete medical history. This includes vaccination logs, checkup records, prescriptions, and past veterinary services.
- D. Book Button – Allows the user to schedule a new appointment for the selected pet. The user will be directed to the appointment form where they can choose a service, date, and preferred veterinarian.

2. Appointments

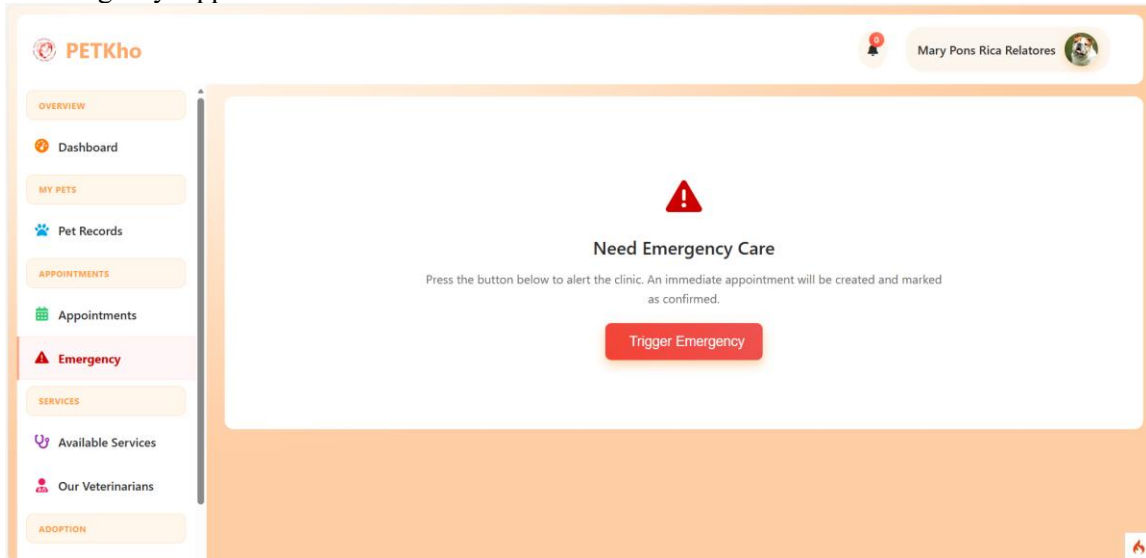


The Appointments page provides staff with a complete calendar view of all scheduled, ongoing, and completed appointments. This page supports efficient monitoring, scheduling, and management of veterinary visits.

- A. Navigation Buttons (Back / Forward / Today) – Controls that allow staff to move between months, return to the current date, or browse past and future schedules.
- B. Calendar View (Month / Week / Day) – Switches the display format. Staff may choose between:
- Month View – Full calendar overview.
 - Week View – Detailed weekly breakdown.
 - Day View – Focused schedule for a single day.

-
- C. Monthly Calendar – Shows all booked appointments for the selected month. Each appointment is displayed as a labeled event containing:
- Time
 - Pet name
 - Service type - Staff can click on an event to view more details or perform actions.
- D. Highlighted Dates – Dates with scheduled appointments appear with visible labels. Empty dates remain blank.
- E. Appointments Table (List View) – Located below the calendar, this table provides a structured list of all appointments with complete details:
- Date – The scheduled day of the appointment.
 - Time – The exact appointment time.
 - Pet – Name of the pet scheduled for the service.
 - Vet – Assigned veterinarian (if applicable).
 - Type – Type of service booked (e.g., Vaccination, Deworming, Check-up).
 - Status – Appointment status such as Scheduled, Completed, Cancelled, or Pending.
- F. Action Buttons – Icons that allow staff to manage each appointment directly from the list:
- View (Eye icon) – Opens the appointment details.
 - Edit (Pencil icon) – Modify appointment information.
 - Cancel (X icon) – Cancel the appointment.
 - Mark as Completed (Check icon) – Set the appointment status to Completed.
 - Invoice (Document icon) – Generate an invoice for completed or billable appointments.

3. Emergency Appointment

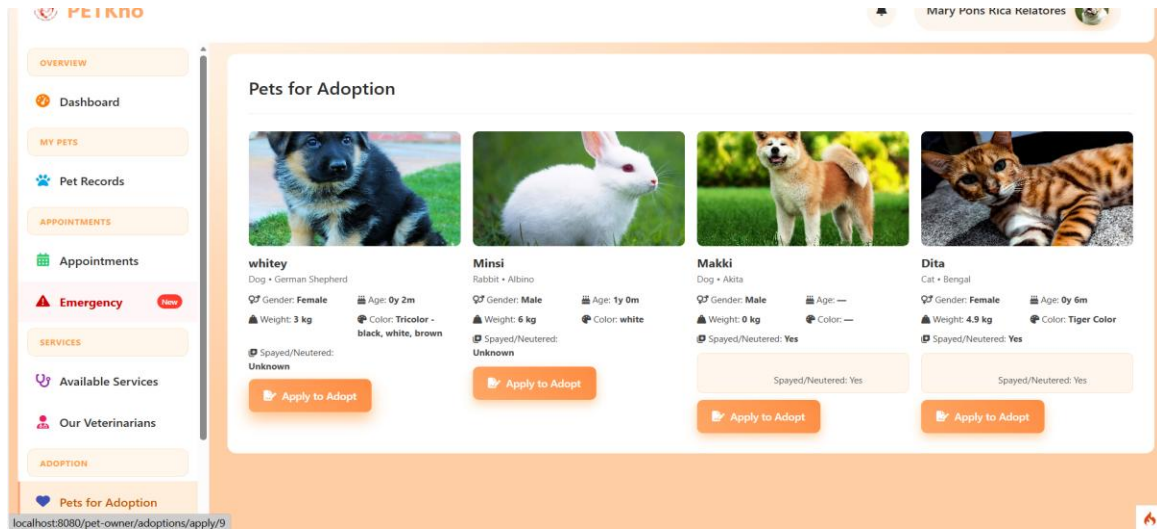


The Emergency page provides clients with a rapid and reliable way to request immediate veterinary assistance during critical situations. Instead of navigating the usual appointment process, users can instantly alert the clinic with just a single action, allowing the system to automatically generate an emergency appointment that is marked as confirmed. This ensures that the veterinary team is notified without delay, enabling them to prepare the necessary resources and respond as quickly as possible. Designed with urgency and accessibility in mind, the page offers a straightforward interface that clients can use even under stressful circumstances, giving them a direct and uninterrupted channel to signal emergencies. By streamlining communication and minimizing steps, the Emergency page helps safeguard pets by ensuring they receive timely care when every second matters.

- A. **Trigger Emergency Button** – This feature allows users to send an immediate emergency alert directly to the veterinary clinic. Once the button is pressed, the system instantly notifies the clinic staff that urgent assistance is needed, providing details such as the user's information, location (if enabled), and the nature of the emergency. This ensures that the clinic can respond quickly to critical situations,

helping protect the health and safety of the pet. The emergency button is designed to be easily accessible and highly responsive, giving pet owners peace of mind knowing that help is just a click away whenever a sudden medical issue arises.

4. Adoptions

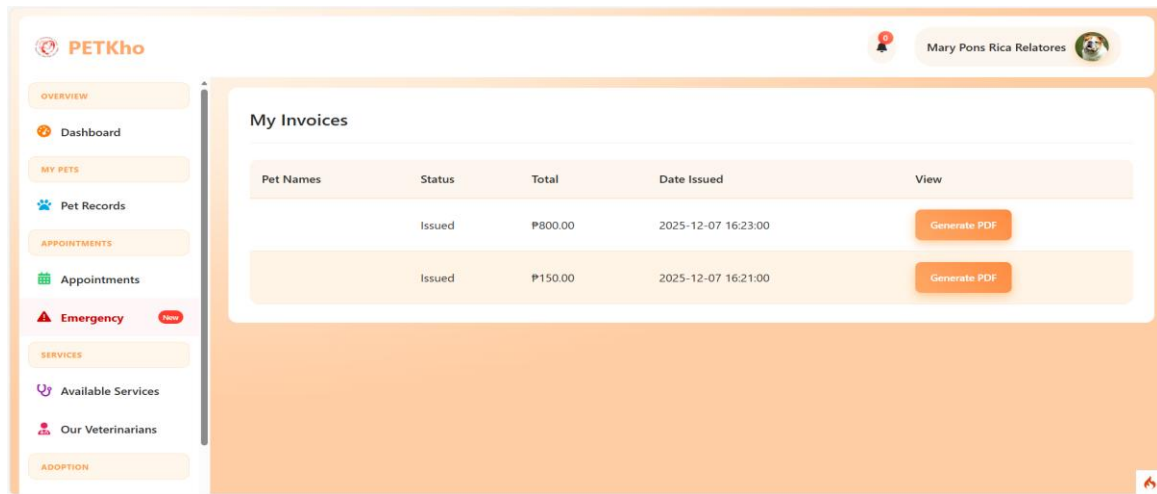


The Pets for Adoption page provides pet owners and visitors with a complete list of animals currently available for adoption in the clinic. Each pet profile displays essential information such as species, breed, gender, age, weight, color, and spay/neuter status. This page allows users to review details of animals they may be interested in and submit an adoption request directly.

- A. **Apply to Adopt Button** – This feature allows users to submit a formal adoption request for the selected pet. When the button is clicked, the user is redirected to a dedicated adoption application form where they are required to provide important personal information, household details, and any other relevant data needed to assess their suitability as a pet owner. This process ensures that the adoption request is complete and organized, giving the shelter or clinic all the necessary information to review and approve applications efficiently. By guiding users through a clear and structured application process, the button helps streamline pet adoptions, making it easier for

prospective adopters to connect with pets while ensuring the welfare and safety of the animals.

5. Invoice



The My Invoices page provides pet owners with a complete list of all invoices issued for their pets' services, treatments, or appointments. It allows users to easily review charges, verify payment details, and download official copies of their invoices.

- A. **Generate PDF Button** – This button enables the user to create a digital PDF version of the invoice, which can be downloaded or viewed directly within the system. By clicking this button, the system compiles all relevant billing details, including services rendered, itemized charges, taxes, and total amounts due, into a neatly formatted PDF document. The generated PDF can then be saved locally for personal recordkeeping, printed for physical documentation, or submitted for reimbursement and administrative purposes. This feature ensures that users have easy access to an accurate and professional copy of the invoice, improving transparency, record management, and overall convenience for both clients and clinic staff.

APPENDIX E

COMMUNICATION LETTERS

DR. WILLIEFREDO V. KHO
Properiority/ Kho Veterinary Clinic
Kho Veterinary Clinic Rinconada
Iriga City, Camarines Sur

Subject: Request for Interview

Dear Dr. Kho,

I hope you are doing well. We are Team Quantum Nexus, a group of third-year Bachelor of Science in Information Technology students from Camarines Sur Polytechnic Colleges, Nabua, Camarines Sur. Currently, we are working on our capstone project about Veterinary Clinic.

In connection to this, we would like to request a short interview with you to better understand the clinic's operations and needs. Your insights would be invaluable in ensuring our system effectively addresses key challenges and meets the specific needs of Kho Veterinary Clinic Rinconada.x

Please let us know your available schedule, and we will be happy to adjust accordingly. All information shared will be kept confidential and used solely for academic purposes.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

QUANTUM NEXUS

ANNE PSYLO

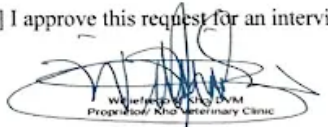
JEY KYLE E. PEREGRINO

MARY PONS RICA
E. RELATOIRES

Noted by: 
JOCELLE B. MONREAL, DIT
Capstone Project Subject Adviser/CaPSA

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

Approved:
[] I approve this request for an interview

Williefredo V. Kho DVM
Proprietor/ Kho Veterinary Clinic

DR. ADELMA A. MORALES
Resident Doctor

APPENDIX F

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: March 7, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and JOCELLE B MONREAL, DIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

JOCELLE B. MONREAL, DIT - Capstone Adviser

Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean

Name and Title

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Anne Psylocke S. Urbina, Mary Pons Rica E. Relatores, Jey Kyle E. Peregrino

which is entitled:

PetKho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in

information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR POLYTECHNIC
COLLEGES**


JOCELLE B. MONREAL, DIT
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: March 7, 2025

DATE: March 7, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: January 13, 2026

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and PHILIP ALGER M. SERRANO, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
PHILIP ALGER M. SERRANO, MIT - Consultant

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Anne Psylocke S. Urbina, Mary Pons Rica E. Relatores, Jey Kyle E. Peregrino

which is entitled:

PetKho Portal: A Smart veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
 - a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the

confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3,
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other,
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR
POLYTECHNIC COLLEGES**


PHILIP ALGER M. SERRANO, MIT
Expert


ROSEL G. ONESA, MIT
CSPC Representative

DATE: January 13, 2026

DATE: January 13, 2026

Non-Disclosure Agreement Form

EFFECTIVE DATE: MAY 09, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and MA. ALLAINE C. AGNA, LPT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

MA. ALLAINE C. AGNA, LPT - Grammarian
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean
Name and Title

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Anne Psylocke S. Urbina, Mary Pons Rica E. Relatores, Jey Kyle E. Peregrino

which is entitled:

PetKho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the

Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR POLYTECHNIC
COLLEGES**


MA. ALLIANE C. AGNA, LPT
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: MAY 9, 2025

DATE: MAY 9, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 16, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Dr. Ian P. Benitez hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

IAN P. BENITEZ, DIT – Panel Chair

Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean

Name and Title

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Anne Psylocke S. Urbina, Jey Kyle E. Peregrino, Mary Pons Rica E. Relatores

which is entitled:

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA

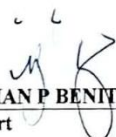
3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in



- confidential information
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
 10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3,
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other,
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
 11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
 12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
 13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
 14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR
POLYTECHNIC COLLEGES**


DR. IAN P. BENITEZ, DIT
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: January 13, 2026

DATE: January 13, 2026

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 16, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Kaela Marie N. Fortuno hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

KAELA MARIE N. FORTUNO, MIT – Panel Member

Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean

Name and Title

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Anne Psylocke S. Urbina, Jey Kyle E. Peregrino, Mary Pons Rica E. Relatores

which is entitled:

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in

confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3,
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR
POLYTECHNIC COLLEGES**


KAELA MARIE N. FORTUNO, MIT
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: January 13, 2026

DATE: January 13, 2026

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 16, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Vincent B. Cortez hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

VINCENT B. CORTEZ, MIT – Panel Member

Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - Dean

Name and Title

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Anne Psylocke S. Urbina, Jey Kyle E. Peregrino, Mary Pons Rica E. Relatores

which is entitled:

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in

confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3,
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other,
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:


VINCENT B. CORTEZ, MIT
Expert

DATE January 13, 2026

CAMARINES SUR
POLYTECHNIC COLLEGES


ROSEL O. ONESA, MIT
CSPC Representative

DATE January 13, 2026

APPENDIX G

JOINT AFFIDAVIT OF UNDERTAKING

Joint Affidavit of Undertaking

We, (1) JEY KYLE E. PEREGRINO, of legal age, single and a resident of LOURDES YOUNG, NABUA, CAMARINES SUR, (2) MARY PONS RICA E. RELATORES, of legal age, single and a resident of SAN FRANCISCO, NABUA, CAMARINES SUR, (3) ANNE PSYLOCKE S. URBINA, of legal age, single and a resident of SAN VICENTE, BATO, CAMARINES SUR, after having been sworn to in accordance with law, do hereby take oath and state :

- (i) That, we are officially enrolled for the thesis/capstone project on the topic titled PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA in the COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES.
- (ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY Degree are original and our own work, and is not plagiarized.
- (iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.
- (iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/ Institute.
- (v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this JAN 13 2026 day of _____ 2026 in

Nabua, Cam. Sur, Philippines.

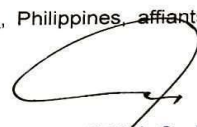
(1) JEY KYLE E. PEREGRINO
Affiant

(2) MARY PONS RICA E. RELATORES
Affiant

(3) ANNE PSYLOCKE S. URBINA
Affiant

SUBSCRIBED AND SWORN TO before me this _____ day of _____ at _____, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. : 302
Page No.: 62
Book No.: XIV
Series of 2026

JAN 13 2026

ATTY. EVELYN C. BERGADO
Notary Public Until December 31, 2027
Admin. Case No. IR-190
IBP No. 585856/01-04-2026/Camarines Sur
PTR No. 764182/01-05-2025 Nabua, Camarines Sur
MCLE Compliance No. VIII-0037003 valid until 04/14/2028
Attorney's Roll No. 89738
Cuba St., San Nicolas Nabua Camarines Sur

APPENDIX H

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	Quantum Nexus
Course Code	IT 3211
Subject adviser/ CaPSA	JOCELLE B. MONREAL, DIT

Name and Signature	Project Role	Email address and mobile#(s)
 Anne Psyche S. Urbina	Project Manager Documentation Writer QA Tester	anurbina@my.cspc.edu.ph 09938422994
 Jey Kyle E. Peregrino	Database Designer Software Engineer Programmer	jeperegrino@my.cspc.edu.ph 09913042008
 Mary Pons Rica E. Relatores	Network Designer UI Designer Database Designer	maryrelatores@my.cspc.edu.ph 09613678581

APPENDIX I

FINAL PROJECT TITLE FORM

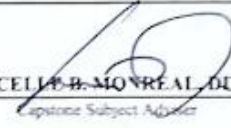
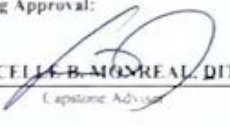
FINAL PROJECT TITLE FORM

Proponents/Researchers:

1. Jey Kyle E. Peregrino
2. Mary Pons Rica E. Relatores
3. Anne Psylocke S. Urbina

Proposed Thesis/ Capstone Project Title:

PetKho Portal: A Smart Veterinary Care and Monitoring System for Kho Veterinary Clinic Rinconada

Submitted by:  ANNE PSYLOCKE S. URBINA Project Head Date: <u>March 13, 2025</u>	Noted by:  JOCELYN B. MONREAL, DIT Capstone Subject Adviser Date: <u>March 13, 2025</u>
Recommending Approval:  JOCELYN B. MONREAL, DIT Capstone Adviser Date: <u>March 13, 2025</u>	Approved:  ROSEL O. ONESA, MIT OIC DEAN, CCS Date: <u>MAR 17 2025</u>

APPENDIX J
CAPSTONE HEARING FORM (TD, POD, FOD)

CAPSTONE PROJECT HEARING FORM

Date Filed: MAY 9, 2025 ☒ Title Proposal ☐ Pre-Oral ☐ Final Oral

Date of Hearing: MAY 16, 2025 Time: 8:00 - 10:00 AM Venue: CS LAB

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR
KHO VETERINARY CLINIC RINCONADA

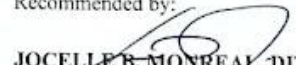
Proponent/s:

URBINA, ANNE PSYLOCKE S.

PEREGRINO, JEY KYLE E.

RELATORES, MARY PONS RICA F.

Recommended by:


JOCELLE B. MONREAL, DIT
Capstone Project Adviser

CERTIFICATION

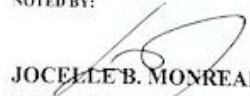
The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


KEALA MARIE N. FORTUNO, MIT
PANEL MEMBER 1


VINCENT B. CORTEZ, MIT
PANEL MEMBER 2


IAN P. BENITEZ, DIT
PANEL CHAIR

NOTED BY:


JOCELLE B. MONREAL, DIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC Dean, CCS

CAPSTONE PROJECT HEARING FORM

Date Filed: October 11, 2025 ☐ Title Proposal ☒ Pre-Oral ☐ Final Oral

Date of Hearing: October 16, 2025 Time: 3:00 – 5:00 PM Venue: OPEN LAB

Department: COLLEGE OF COMPUTER STUDIES (CCS)

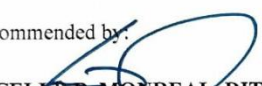
Research Title:

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA

Proponent/s:

URBINA, ANNE PSYLOCKE S. PEREGRINO, JEY KYLE E.
RELATORES, MARY PONS RICA E.

Recommended by:


JOCELLE B. MONREAL, DIT
Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


KAELA MARIE N. FORTUNO, MIT
PANEL MEMBER 1


VINCENT B. CORTEZ, MIT
PANEL MEMBER 2


IAN P. BENITEZ, DIT
PANEL CHAIR

NOTED BY:


JOCELLE B. MONREAL, DIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC Dean, CCS

CAPSTONE PROJECT HEARING FORM

Date Filed: December 04, 2025 ☐ Title Proposal ☐ Pre-Oral ☒ Final Oral

Date of Hearing: December 10, 2025 Time: 1:00 - 3:00 pm Venue: MC Lab

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR
KHO VETERINARY CLINIC RINCONADA

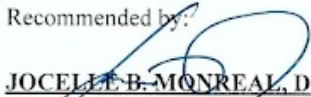
Proponent/s:

URBINA, ANNE PSYLOCKE S.

PEREGRINO, JEY KYLE E.

RELATORES, MARY PONS RICA E.

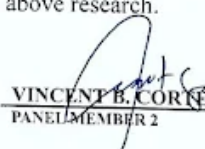
Recommended by:


JOCELLE B. MONREAL, DIT
Capstone Project Adviser

CERTIFICATION

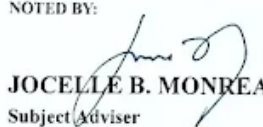
The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


KAELA MARIE N. FORTUNO, MIT
PANEL MEMBER 1


VINCENT B. CORTEZ, MIT
PANEL MEMBER 2


IAN P. BENTEZ, DIT
PANEL CHAIR

NOTED BY:


JOCELLE B. MONREAL, DIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC Dean, CCS

APPENDIX K

PANEL RSC (TD, POD, FOD)

Title : PETHKO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA
 Alias : TEAM QUANTUM NEXUS
 Date : May 16, 2025 ☒ TD ☐ POD ☐ FOD
 Secretary : ANTHONY L. LLABRES

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
		- Follow manuscript format - Fix page numbering	We have ensured that the entire document follows the required manuscript format, including consistent fonts, margins, headings, and citation styles. Additionally, we have fixed the page numbering to appear correctly throughout the manuscript for a professional and organized presentation.
	4	Objectives Of the Project - SO#2 should be: "Develop a component veterinary care and monitoring system that addresses the identified challenges by providing a centralized platform for: - All system modules" - SO#3 should be: "To test the management system based on black-box techniques using the ff: - Performance testing - Security testing - Use case testing - User Acceptance testing"	We have revised the specific objectives to ensure they clearly reflect the project goals and align with the system's scope and requirements. This includes refining SO#2 to emphasize a centralized platform for all system modules, specifying black-box testing techniques in SO#3, and clarifying the deployment target in SO#4 for Kho Veterinary Clinic Rinconada.

		Gap bridged by the Study - Discuss the edge of the study based on the RRLs - Cite specific RRL	We have discussed the edge of our study based on the reviewed literature and studies. Compared to existing systems, our study offers a more integrated approach by combining veterinary appointment scheduling, health monitoring, digital receipts, and referral functionalities in a single platform. For instance, unlike the system described by Santos et al. (2020), which focused solely on pet vaccination tracking, our system extends to include real-time referral and cashiering modules. Additionally, the study by Lee and Martinez (2019) emphasized user acceptance testing but lacked modules for internal clinic communication—an area our system addresses with its centralized referral system. These distinctions, supported by the cited RRLs, demonstrate the innovative and comprehensive nature of our system, giving it a clear advantage in addressing the operational needs of veterinary clinics like Kho Veterinary Clinic Rinconada.
		Notes All notes page should be in another page.	We have arranged all terms alphabetically to maintain clarity and

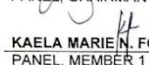
	3.3.2 Testing tools and framework - Each tool should be supported by RRL in chapter 2 - Discuss each tool used	We revised the framework and aligned it with the ISO standards to ensure a structured and recognized approach. However, the tools included in the framework are not supported by RRLS in Chapter 2.
	Notes Follow ACM style format	We have followed the ACM style format throughout the document to ensure consistency, professionalism, and adherence to recognized academic standards. This includes proper citation, formatting of headings, tables, figures, and references, all aligned with ACM guidelines.
ALL	- Revise all diagrams - Proper table and figure format. Do not screenshot tables	We have revised all the diagrams to improve clarity, accuracy, and alignment with standard conventions. Additionally, we ensured that every table and figure strictly follows the recommended formatting guidelines, with proper placement and concise discussions for each.

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN

6	Automate the cashiering process, including refunds, discounts, etc.	We have automated the cashiering process to efficiently handle transactions, including refunds, discounts, and other related operations. All these functionalities are integrated as part of the invoice system to ensure accurate and seamless financial management.
---	---	---

NOTED BY:


IAN P. BENITEZ, DIT
PANEL CHAIRMAN


KAELA MARIE N. FORTUNO, MIT
PANEL MEMBER 1


VINCENT B. CORTAZ, MIT
PANEL MEMBER 2

Title : PETHKO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC
RINCONADA

Alias : TEAM QUANTUM NEXUS

Date : October 27, 2025 ☐ TD ☒ POD ☐ FOD

Secretary : ANTHONY L. LLABRES

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
		- Use third person POV, do not use 'we' / 'our' - Validate questionnaire - Full revision in chapter 5	We use third person POV, we even validate questionnaire and also we revise the chapter 5.
	6	Objectives of the project Replace bullets to letters	The bullets are replaced into letters.
	8	Scope and limitations of the project Include SO4 in the discussion	The SO4 in the Scope and Limitation discussion is now included
	9	Project Dictionary - The definition of the terms should be in conceptual and operational definition. - Include authentication and authorization	The definition are now in conceptual and operational, and also authentication and authorization is added.
	14	Notes Follow ACM style format	The ACM format was followed
2		Related Literature and Studies Add RRL for testing	We added RRL for testing
4	60	Testing phase - Remove SDM - Discuss every testing used	The SDM is removed and for the testing phase we discuss every testing used.

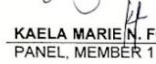
	68	Figure 7: ERD Provide a higher quality image to make it readable	We finally provide high quality image.
	70	Figure 8: context diagram Every 'ins' should have 'out'	It has now 'ins' and 'out'
	73	Figure 9: DFD - Follow proper format - Every 'ins' should have 'out' - Proper label per DFD	We fix the DFD and follow the proper format.
	97	Figure 28: notification management Figure label not the same as discussion	The figure label is now corrected
	105-107	Figure 36, 37, 38 Figure label not the same as discussion	The figure labels are now corrected
	111	Table 8: respondents Add row for total	The respondents has now a total number
	112	Include scaling for testing	Scaling is added to testing
	113	Table 9 3 decimal places for overall results	The overall results now is correct.
5	119	Summary - Summarize 1-4 - Remove table	We summarize the chapters 1 to 4 and the table is removed.
	120	Findings Findings per objective	The findings are now based on the Objective of the project.
	122	Recommendations - Recommendation for future researchers - Change bullets to numbers - Add introduction atleast one recommendation per objective	Recommendations for future researchers are added, the bullets are changed into numbers and it has an introduction for recommendation in every objective.
		Notes Follow ACM style format	ACM format is followed
ALL		- Revise all diagrams - Proper table and figure format. Do not screenshot tables	We revise all the diagrams.

--	--	--	--

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	Make the landing/home page responsive (slideshow of pets etc.)	The landing page is now responsive and has slideshow of pets.
	Registration of pets Breed – add dropdown of all breed type	We have added the dropdown of all breed types.
	Add/Display the availability of doctor in calendar	The availability of doctors is applied on the calendar.
	Do not reject appointments, instead reschedule and add notification/reason	We have replaced the reject appointment to reschedule appointment and can be viewed in the notification management and appointment calendar.
	Apply POS style in Billing	We applied the POS style for billing
	Application for Adoption - Automatic fill up of personal information - Restructure application form (use radio button for yes/no, etc.) - Add reason in rejecting application for adoption - When approving application, automatic send of date for interview if necessary	We applied all the needed changes for Application for Adoption
	Generation of reports	We created the reports for the overall transactions and tasks of the clinic.

NOTED BY:


IAN P. BENITEZ, DIT
PANEL, CHAIRMAN


KAELA MARIE N. FORTUNO, MIT
PANEL, MEMBER 1


VINCENT B. CORTEZ, MIT
PANEL, MEMBER 2

Title : PETHKO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC
RINCONADA
Alias : TEAM QUANTUM NEXUS
Date : DECEMBER 12, 2025 ☐ TD ☐ POD ☒ FOD
Secretary : ANTHONY L. LLABRES

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
		- Follow manuscript format - Proper ACM style format for all notes page (fullname of author) - All signatories in appendices must have signatures - No hanging pages	The manuscript was revised to strictly follow the prescribed manuscript format, all notes pages were updated to comply with the proper ACM style including the full name of the author, all signatories in the appendices were completed with their corresponding signatures, and the document layout was adjusted to ensure there are no hanging pages.
		Project Dictionary The definition of the terms should be in conceptual and operational definition. Add reference for conceptual definition	The Project Dictionary was added to the manuscript, with all key terms defined using both conceptual and operational definitions, and appropriate references were included to support each conceptual definition.
2		Proper citation of RRL's	All Review of Related Literature (RRL) sections were revised to ensure proper in-text citations and reference entries were formatted correctly and consistently according to the required citation style.

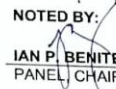
4		Figure 7: ERD - Provide a higher quality image to make it readable - Revise the diagram	Figure 7 (Entity Relationship Diagram) was revised and redesigned, and a higher-resolution image was provided to improve clarity, readability, and overall presentation.
		Figure 8: context diagram - Every 'ins' should have 'out' - Revise diagram - Use proper symbols/shapes - Highlight the main process only, do not include sub processes	Figure 8 (Context Diagram) was revised to ensure that every input has a corresponding output, proper standard symbols and shapes were used, and the diagram was simplified to highlight only the main process without including any sub-processes.
		Figure 9: DFD - Follow proper format - Every 'ins' should have 'out' - Use proper symbols - Revise all diagram lvls	Figure 9 (Data Flow Diagram) was revised across all levels to follow the proper DFD format, ensure that every input has a corresponding output, and apply the correct standard symbols consistently throughout the diagrams.
		Table of respondents Discuss the sampling technique used per respondents	The Table of Respondents was revised to include a discussion of the sampling technique used for each group of respondents, clearly explaining how participants were selected for the study.
		Table 21 use case testing Align testing based on interpretation	Table 21 (Use Case Testing) was revised to align all test cases and results based on the interpretation of the system requirements, ensuring consistency between expected outcomes and actual outcomes.
5		Findings - Discuss also the numerical results - Must be aligned/attuned to the SO'S	The Findings section was revised to include discussion of the numerical results, and all findings were aligned

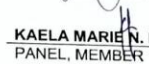
			and attuned to the corresponding Specific Objectives (SOs) of the study.
		Conclusion Must be aligned/attuned to findings	The Conclusion section was revised to ensure that all statements are aligned and attuned to the study's findings.
		Recommendations - Recommendation for future researchers - Change bullets to numbers - Add introduction atleast one recommendation per objective	The Recommendations section was revised to include an introduction, provide at least one recommendation per objective, specifically address future researchers, and the list format was changed from bullets to numbered items for clarity and organization.
		Appendix A relevant source code Limit / highlight only the main process	Appendix A (Relevant Source Code) was revised to highlight and include only the main processes, removing unnecessary or redundant code segments for clarity.
		Add label steps in user guide	The User Guide was revised to include clearly labeled steps for each procedure, improving readability and user comprehension.
ALL		Revise all diagrams	All diagrams in the manuscript were revised to ensure accuracy, clarity, proper formatting, and consistency with standard symbols and notations.

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	- Disable the previous dates in scheduling appointment and inputting doctors' availability - Appointment schedule date and time validation - Rescheduling date validation	The system was updated to disable past dates when scheduling appointments and inputting doctors' availability, and validation rules were implemented for appointment scheduling and rescheduling to ensure only valid future dates and times can be selected.
	Fix appointment bugs & process	All identified bugs in the appointment module were fixed, and the appointment scheduling and management processes were revised to ensure proper functionality, accuracy, and reliability.
	Apply/use modal for system responsiveness	Modals were applied throughout the system to improve responsiveness and user interaction, ensuring consistent display and functionality across different devices and screen sizes.
	- Add settings for adding VAT in payment billing - Include also VAT in invoice receipt	A VAT setting was added to the payment and billing module, and invoices were updated to include VAT calculations, ensuring accurate display of taxes in both billing and invoice receipts.
	Fix UI design in Adoption (showing of pet for adoption)	The UI design for the Adoption module was revised to improve the display and presentation of pets available for adoption, enhancing user experience and visual clarity.
	Remove age in adoption form	The age field was removed from the Adoption form to streamline data

		entry and simplify the application process.
	• Do not edit generated reports	he system was configured to prevent editing of generated reports, ensuring the integrity and accuracy of all report outputs.

NOTED BY:


IAN P. BENITEZ, DIT
PANEL CHAIRMAN


KAELA MARIE N. FORTUNO, MIT
PANEL, MEMBER 1


VINCENT B. CORTÉZ, MIT
PANEL, MEMBER 2

APPENDIX L

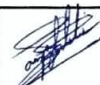
CONSULTATION LOGS FORM

CONSULTATION LOGS FORM

Capstone Project Title:	PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA						
Proponents:	HIRRINA ANNE PSYLOCKE S. PEREGRINO, JEY KYLE E. RELATORES, MARY PONS RICA E.						
Alias:	QUANTUM NEXUS						
Total # of Modules:	9			as approved by the CAPSA /TSA:			
PROTOTYPE							
	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
	10-17-2025	7	2				
	Remarks: - Report - Summary of Sales per week, month - photo and Pet Adoption Record to Owner's Pet Record -						
Deadline:							
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA						
Proponents:	Anne Psylocke S. Urbina, Jey Kyle E. Peregrino, Mary Pons Rica E. Relatores						
Alias:	Quantum Nexus						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
	12/10/28						
	Remarks: - Good to go Manuscript - System not yet uploaded to VPS or of date - Recommended for FOD						
Deadline:							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA						
Proponents:	URBINA, ANNE PSYLOCKE S. PEREGRINO, JEY KYLE E. RELATORES, MARY PONS RICA E.						
Alias:	QUANTUM NEXUS						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
							
	Remarks:						
	For bluebook-						
Deadline:							
	Remarks:						

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA						
Proponents:	URBINA, ANNE PSYLOCKE S. PEREGRINO, JEY KYLE E. RELATORES, MARY PONS RICA E.						
Alias:	QUANTUM NEXUS						
Total # of Modules:	9			as approved by the CAPSA /TSA:			
PROTOTYPE							
	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
	January 20, 2024	9					
	Remarks: - For 						
Deadline:							
	Remarks:						

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

APPENDIX M

SECRETARY CERTIFICATE



SECRETARY'S CERTIFICATION

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the capstone project entitled **"PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA"** prepared and submitted by Jey Kyle E. Peregrino, Mary Pons Rica E. Relatores and Anne Psylocke Urbina, and that the same have not been amended, modified, or obliterated.

Signed by: anthony

ANTHONY L. LLABRES
SECRETARY

12 / 03 / 2025
DATE SIGNED

APPENDIX N

GRAMMARIAN CERTIFICATE



GRAMMARIAN'S CERTIFICATION

This is to certify that the undersigned has reviewed and examined all the pages of the capstone project entitled **"PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA"** in accordance with the structural rules that govern research writing, particularly the proper composition of sentences, phrases, and words in the English language.

Signed by: 

MA. ALLANE C. AGNA
GRAMMARIAN

12-09-2025
DATE SIGNED

APPENDIX O

CERTIFICATE OF UTILIZATION



APPENDIX P

ACM FORMAT

PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA

JEY KYLE E. PEREGRINO

Camarines Sur Polytechnic Colleges, jeperegrino@my.cspc.edu.ph

MARY PONS RICA E. RELATORES

Camarines Sur Polytechnic Colleges, maryrelatores@my.cspc.edu.ph

ANNE PSYLOCKE S. URBINA

Camarines Sur Polytechnic Colleges, anurbina@my.cpc.edu.ph

The PetKho Portal is a smart online system created to help Kho Veterinary Clinic Rinconada improve its daily work and provide better service to pet owners. The clinic used to rely on handwritten records, walk-in appointments, and manual updates, which often caused delays, missing information, and extra workload for the staff. This project was developed to solve these problems by organizing all important tasks in one easy-to-use platform. The system includes features such as online appointment scheduling, pet record management, inventory tracking, billing and payment recording, referral handling, and automatic notifications. It was developed using the Agile approach, which allowed the team to build the system step by step and improve it based on feedback. Different tests such as performance testing, security testing, use case testing, and user acceptance testing were done to make sure the system works smoothly and safely. The results showed that the PetKho Portal helped reduce errors, save time, and make information easier to access. Pet owners also found it more convenient to book appointments and receive updates. Overall, the PetKho Portal provides a simple, modern, and reliable solution that supports the clinic in giving better and more organized veterinary.

CCS CONCEPTS • Information Systems – Web-based platform – Online appointment and service request system

Additional Keywords: Local Based Veterinary System, Automation of Pet Medical Health Record, Invoice Automation

1. INTRODUCTION

The PetKho Portal: A Smart Veterinary Care and Monitoring System was developed to address the operational challenges of Kho Veterinary Clinic Rinconada, which relied on manual processes for managing patient records, appointments, and client communication. The system aims to streamline workflows by providing a centralized, digital platform that automates recordkeeping, scheduling, billing, notifications, and multi-branch access, improving accuracy, efficiency, and service quality. By integrating these features, the portal enhances veterinary operations, fosters timely communication with pet owners, reduces administrative workload, and aligns the clinic with modern digital standards in veterinary care. The project also includes testing for performance, security, usability, and user acceptance to ensure a reliable and user-friendly system for both staff and clients.

2. BACKGROUND

In recent years, veterinary clinics have recognized the need to improve pet healthcare through technology. Many clinics have adopted digital systems to manage records, schedule appointments, and provide services more efficiently. Studies show that web-based platforms and electronic medical records (EMRs) help streamline operations by automating tasks such as record management, appointment scheduling, and health monitoring, resulting in more organized workflows and faster responses for pet owners (Afuolabi, 2023 [1]; Arifah Fasha & Muhammad Harith, 2023 [4]). These developments demonstrate that adopting digital tools can significantly enhance veterinary service delivery, but their success depends on careful planning, staff training, and system reliability (Beyer, 2023 [5]; Chua et al., 2024 [20]). By combining clinic efficiency with client-facing functionalities, the system aims to modernize veterinary operations, improve transparency, and strengthen communication between veterinarians and pet owners (Iskandar, 2023 [9]; Cabrera, 2023 [6]; S. S. Chua et al., 2024 [20]).

3. TECHNICAL BACKGROUND

3.1 Overview of Current Technologies to be Used in the System

The PetKho Portal was developed using modern front-end and back-end technologies to create a secure, efficient, and user-friendly platform. The front-end uses HTML5, CSS3, and Bootstrap for structure and responsive design, while JavaScript and jQuery handle interactive features. The back-end relies on PHP with the CodeIgniter framework, supported by MySQL and phpMyAdmin for database management, ensuring reliable storage and retrieval of pet records. The system is accessible on desktops, laptops, tablets, and smartphones through popular web browsers such as Chrome, Firefox, Edge, and Safari, ensuring cross-browser compatibility. This technology stack provides performance, security, and scalability, allowing real-time updates, safe handling of sensitive data, and supporting future enhancements for the veterinary clinic.

3.2 Resources

The PetKho Portal uses cloud-based servers for scalability and easy maintenance. It is accessible on desktops, laptops, tablets, and smartphones. The system is built with PHP, JavaScript, CodeIgniter, and MySQL, ensuring secure, efficient, and cost-effective development.

3.2.1 Hardware Specifications

The PetKho Portal uses a laptop with an Intel Core i5 or higher processor, 8GB RAM, 256GB SSD, and a 65W power supply, along with standard input and output devices like a keyboard, mouse, and monitor. This setup provides sufficient performance, portability, and reliability for development, testing, and general system use while running Windows 11 for a secure and modern environment.

3.2.2 Program Specifications

The PetKho Portal is developed using Windows 10 or 11 and MacOS 10.9 or later, with Visual Studio Code as the primary code editor and PHP v8.2.12 for backend development. Front-end technologies include HTML5, CSS3, Bootstrap, and JavaScript for responsive and user-friendly interfaces, while CodeIgniter serves as the backend framework. Database and server management are handled through XAMPP 8.1, MySQL, and phpMyAdmin 8.1, and the system supports major web browsers for cross-platform accessibility. Regular updates ensure security, stability, and performance, providing a reliable and scalable environment for the portal.

3.2.3 Program Specifications

Functional Requirements

The PetKho Portal includes core functional features such as customer and pet owner management, appointment and referral handling, pet records management, invoice and billing processing, stock and inventory management, adoption process facilitation, reporting and analytics, user and branch management, and automated notifications for appointments and updates. These features support veterinary operations and ensure efficient service delivery across all roles and branches.

Non-Functional Requirements

The system's non-functional requirements focus on overall quality and reliability. This includes user acceptance testing to confirm workflow support, use case testing to validate critical transactions, performance testing to ensure speed and stability under heavy usage, and security testing with strict role-based access control to protect staff and patient data. These requirements ensure the system operates efficiently, securely, and reliably.

3.3 Testing Plan

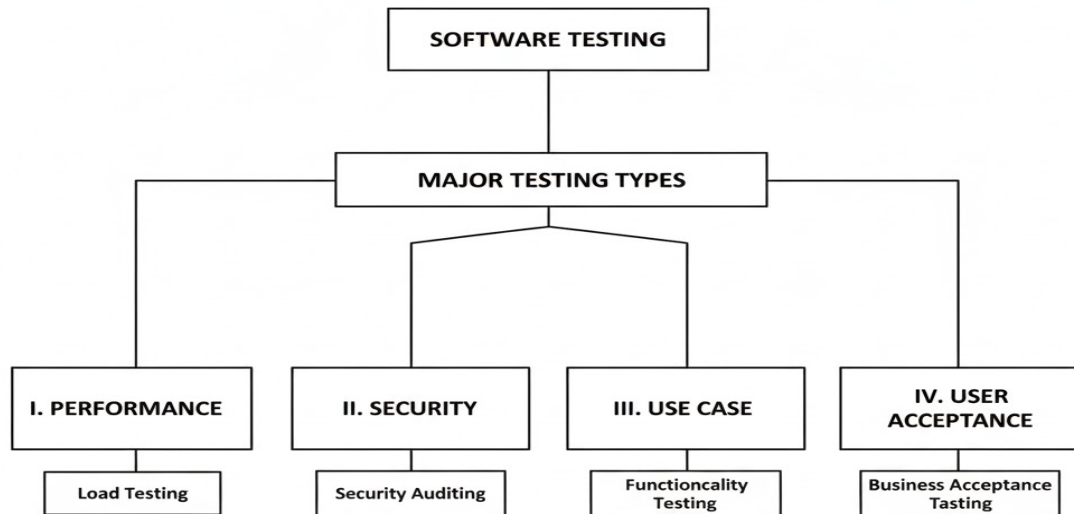


Figure 1: Hierarchy Diagram

The PetKho Portal will use an ISO-aligned questionnaire to ensure it meets user needs and system quality standards. Testing types include User Acceptance Testing (UAT) to verify usability and task completion, Security Testing to evaluate data protection and access controls, Performance Testing to assess response time and stability under varying loads, and Use Case Testing to confirm workflow accuracy and expected outputs. Each testing type follows ISO standards (ISO/IEC 25010, 9126, 27001).

4. Results and Discussion

4.1 Planning

The Planning phase defined the project's objectives, scope, and requirements for the PetKho Portal, analyzing the clinic's workflows, record management, and communication processes to identify inefficiencies and guide the system's development.

4.1.1 Convenient Sampling

Convenient sampling involved selecting participants based on availability, including clients, staff, and IT experts, to gather quick, practical feedback on the system's usability and performance.

4.1.2 Respondents

A total of 67 respondents participated, comprising pet owners from three branches, veterinary staff, the clinic owner, and IT experts, ensuring diverse perspectives on the system's effectiveness and functionality.

4.1.3 Scaling

Likert scales were used to measure performance, security, use case handling, and user acceptance, allowing testers to rate the system from 1 (Poor) to 4 (Excellent) for clear evaluation of usability and reliability.

4.1.4 Statistical Test

Weighted mean calculations analyzed feedback from groups of different sizes, providing an accurate overall assessment of system performance, usability, and user satisfaction.

4.1.5 Identified Problems

Interviews and observations revealed issues like manual patient records, inefficient scheduling, poor communication, uncoordinated referrals, lack of inventory tracking, and weak security, highlighting the need for a digital solution.

4.1.6 Proposed Solutions

The PetKho Portal addresses these problems with automated record management, scheduling, billing, referrals, inventory tracking, and secure communication, streamlining operations and enhancing service quality.

4.2 Design

The Design phase focused on creating a practical and user-friendly solution for Kho Veterinary Clinic Rinconada, defining the system structure, key features, and interfaces to integrate patient management, appointments, referrals, inventory, and communication into a single platform.

4.2.1 Entity Relationship Diagram (ERD):

The ERD illustrates the logical structure of the database, showing relationships among users, pets, appointments, adoptions, medical records, inventory, reports, and notifications, ensuring organized data management and smooth operation across all branches. It also identifies key entities, attributes, and constraints that help maintain data integrity, minimize redundancy, and support accurate, consistent, and efficient processing of information throughout the system.

4.2.2 Data Flow Diagram (DFD)

The DFD section illustrates how the Veterinary Management System manages and flows data between users and processes. The Context Diagram (Level 0) shows high-level interactions among the Veterinary Owner, branch staff, and pet owners, while Level 1 DFDs break down core processes like login and registration, appointment and referral management, pet records, invoices, adoptions, stock, reports, user/staff/branch management, and notifications. Together, these diagrams ensure data is handled efficiently, processes are coordinated, and system operations remain accurate, secure, and user-friendly across all branches.

4.3 Development

The development phase focused on deploying the PetKho Portal and making it fully operational at Kho Veterinary Clinic Rinconada. The system was installed, configured, and tested in the real clinic environment, with user accounts set up for staff, admins, and pet owners. Training was provided to ensure smooth navigation, and minor adjustments were made to maintain stable performance.

4.4. Testing

The testing phase of the PetKho Portal covered four main categories to ensure a reliable system. Performance Testing confirmed the portal is fast, stable, and responsive for all users. User Acceptance Testing (UAT) showed that pet owners and veterinarians can easily navigate and complete daily tasks. Use Case Testing verified that workflows function accurately and match real clinic operations. Security Testing ensured data protection and proper access control. All categories received the highest rating, “Excellent / Strongly Agree,” proving the portal is user-friendly, dependable, and ready for clinic use.

Overall Result

The overall test results for the Pet Owners’ side of the PetKho Portal, covering Performance Testing, User Acceptance Testing, and Use Case Testing, all received the highest rating of 4, “Excellent / Strongly Agree.” This shows that pet owners found the system fast, reliable, and easy to use. The portal efficiently supports daily tasks such as booking appointments, checking pet records, and receiving reminders, confirming it is ready for practical, everyday use.

Table 1. Overall Result

Category	Overall Results	Level of Functionality
Pet Owners	4.00	Strong Agree
Vet Owner/Resident Doctors	4.00	Strong Agree
IT Experts	4.00	Strong Agree
Total	4.00	Strong Agree

4.5 Deployment

The deployment phase of the PetKho Portal marked its transition to real-world use at Kho Veterinary Clinic Rinconada. The system was installed, configured, and made fully operational, including setting up servers, databases, and user accounts, while staff received training to navigate its features. Initial feedback helped refine minor adjustments, ensuring smooth adoption. The accompanying UML Deployment Diagram shows the system architecture, illustrating how client devices communicate with the web application server and databases via the Web API, and how core modules like user authentication, appointment management, pet records, notifications, inventory, billing, and referrals interact with the data storage layer.

5. Conclusion

The existing manual operations at Kho Veterinary Clinic caused inefficiencies, inconsistent records, scheduling conflicts, and communication delays, highlighting the need for an automated, centralized system. The PetKho Portal successfully addressed these issues by integrating key modules such as user authentication, appointment and pet record management, referrals, inventory, billing, and notifications, creating a unified platform that streamlined workflows and improved coordination with pet owners. Finally, deployment in the clinic proved the portal effective in real-world use, enhancing record accuracy, appointment management, communication.

6. Acknowledgement

We sincerely thank our friends, classmates, and loved ones for the happy memories, unwavering support, and encouraging words that lifted our spirits during difficult times and kept us motivated. We are especially grateful to Mrs. Agnes S. Urbina, Mr. and Mrs. Dolfo B. Peregrino, and Mrs. Maryann E. Relatores for their continuous financial, moral, and spiritual support, and for believing in our potential. Your sacrifices and encouragement inspired us to work harder and complete our Capstone Project. Above all, we thank our Almighty Father, our Lord and Savior, for granting us wisdom, knowledge, strength, and guidance. He has given us the determination and focus to overcome challenges and successfully pursue our studies.

REFERENCES

- [1] R. Arifah Fasha and M. Muhammad Harith. 2023. *An Online Scheduling Platform for Veterinary Appointments*. Jurnal Inteltek, 18, 2 (Aug. 2023). <https://doi.org/10.24191/ji.v18i2.22092>.
- [2] A. Benis. 2023. *One Digital Health Intervention for Monitoring Human and Animal Welfare in Smart Cities: Viewpoint and Use Case*. JMIR Medical Informatics, 11 (2023). <https://doi.org/10.2196/43871>
- [3] K. Beyer. 2023. *Unlocking the potential of ICT innovation in veterinary healthcare: The pathway to improve practices and business model*. Procedia Computer Science, 4775–4784 (2023)..
- [4] P. B. Bök and D. Micucci. 2024. *The future of human and animal digital health platforms*. Journal of Reliable Intelligent Environments (Sep. 2024). <https://doi.org/10.1007/s40860-024-00232-0>
- [5] M. Dayoub. 2023. *Applications of Smart Agriculture to Improve Animal Production: Opportunities, Challenges, Solutions, and Benefits*. 2023. <https://doi.org/10.20944/preprints202310.1229.v1>

APPENDIX S

CERTIFICATE OF PLAGIARISM CHECK



CERTIFICATION

Date of Release: January 27, 2026
Submission ID: 23367:126956774
Output Title: PETKHO PORTAL: A SMART VETERINARY CARE AND MONITORING SYSTEM FOR KHO VETERINARY CLINIC RINCONADA
Author(s): Anne Psylocke S. Urbina
Jey Kyle E. Peregrino
Mary Pons Rica E. Relatore
Program: Bachelor of Science in Information Technology

Authenticity Report:

Similarity Index Report: 6%

Internet Sources: 2%
Publications: 1%
Student Papers: 6%

- Interpretation: The similarity index report means that 6% of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: 29%

AI-generated only: 29%
AI-generated text that was AI-paraphrased: 0%

- Interpretation: The AI generated report means that 29% of the output indicates a *likely* AI-generated text and *likely* AI-paraphrased text, which is **within the acceptable AI usage** in the field of Engineering.


HAROLD JAN R. TERANO, Ph.D.
Director, Research & Development Services Office